



Cambridge IGCSE™ (9–1)

BIOLOGY

0970/11

Paper 1 Multiple Choice (Core)

May/June 2020

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

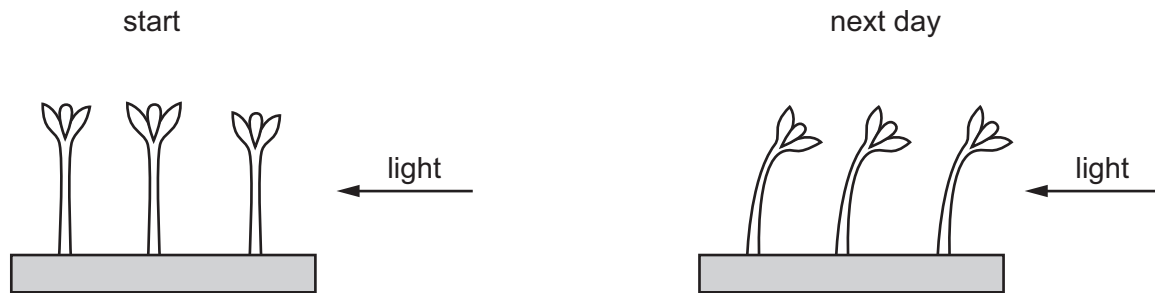
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
- Any rough working should be done on this question paper.

This document has **16** pages. Blank pages are indicated.



- 1 The diagram shows what happened in an experiment with plant seedlings.

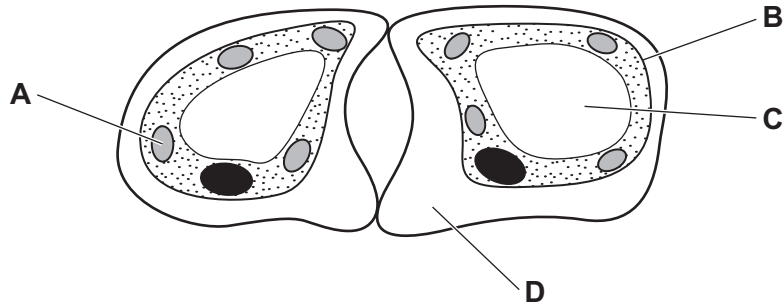


Which characteristic of living things made the seedlings grow towards the light?

- A excretion
 - B nutrition
 - C respiration
 - D sensitivity
- 2 A rat has the scientific name *Rattus rattus*.
- What do the two parts of this name refer to?
- A genus and species
 - B kingdom and genus
 - C kingdom and species
 - D variety and genus
- 3 An animal can swim, has a backbone and produces milk.
- To which group does it belong?
- A amphibians
 - B fish
 - C mammals
 - D reptiles

- 4 The diagram shows two guard cells.

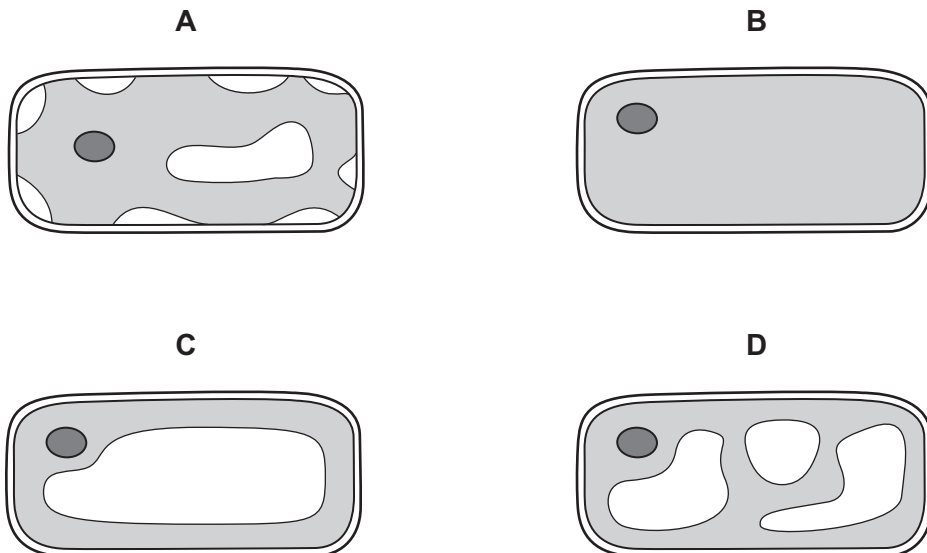
Which label shows the cell wall?



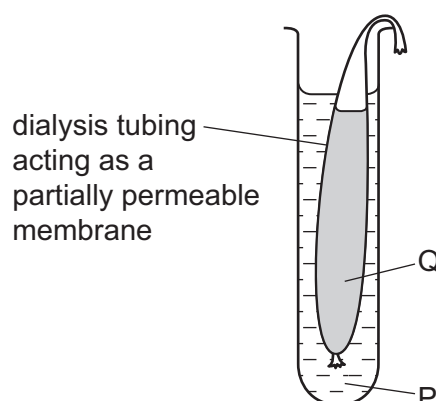
- 5 What is the correct order to describe an **increasing** level of organisation?

- A cell → organ → tissue → organ system
- B tissue → cell → organ → organ system
- C organ → tissue → cell → organ system
- D cell → tissue → organ → organ system

- 6 Which diagram shows the appearance of a plant cell several minutes after it has been placed in a concentrated solution of sugar?



- 7 The apparatus shown is used to demonstrate osmosis.



The mass of the dialysis tubing and contents was 11.2g at the start of the demonstration and 9.4g at the end.

Which solutions would cause this change in mass?

	solution P	solution Q
A	5% salt solution	10% salt solution
B	10% salt solution	5% salt solution
C	water	5% salt solution
D	water	10% salt solution

- 8 Which food-testing solution shows a positive result when it turns from blue to purple?

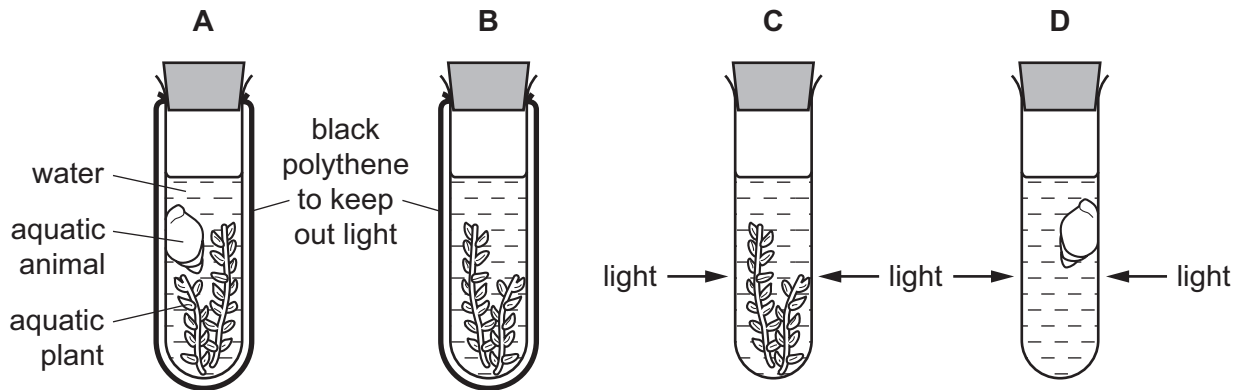
- A** Benedict's solution
- B** biuret solution
- C** ethanol
- D** iodine solution

- 9 What is the definition of an enzyme?

- A** a carbohydrate that acts as a catalyst
- B** a DNA molecule that acts as a catalyst
- C** a fat that acts as a catalyst
- D** a protein that acts as a catalyst

10 Four test-tubes are set up as shown.

Which test-tube contains the least carbon dioxide after one hour?



11 Which substance found in a healthy diet helps to prevent constipation?

- A fat
- B fibre
- C minerals
- D vitamins

12 The activity of amylase is measured in four parts of the alimentary canal.

Which two parts have the most amylase activity?

- A colon and duodenum
- B colon and stomach
- C mouth and duodenum
- D stomach and mouth

13 After passing through the root hair cells of a plant, what is the next tissue through which water passes?

- A cortex
- B epidermis
- C mesophyll
- D xylem

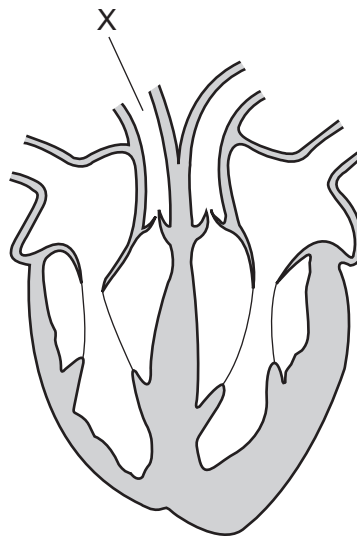
14 From which part of a leaf does most water evaporate during transpiration?

- A** the cuticle
- B** the guard cells
- C** the spongy mesophyll cells
- D** the xylem vessels

15 Which blood vessel carries blood to the muscle of the heart?

- A** coronary artery
- B** pulmonary artery
- C** renal vein
- D** vena cava

16 The diagram shows a section through the heart and its blood vessels.



Which row gives the vessel name and the direction of blood flow in vessel X?

	vessel name	direction of blood flow
A	aorta	towards the lungs
B	pulmonary artery	towards the lungs
C	pulmonary artery	away from the lungs
D	vena cava	away from the lungs

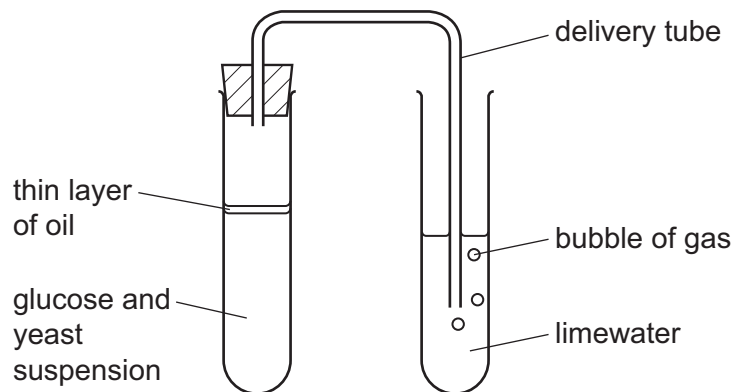
17 What is a chemical barrier to pathogens?

- A nose hairs
- B skin
- C stomach acid
- D white blood cells

18 What is the approximate percentage of oxygen contained in the air breathed out of the lungs?

- A 0%
- B 4%
- C 16%
- D 20%

19 The diagram shows an experiment to investigate the respiration of yeast. Oil prevents oxygen entering the glucose and yeast suspension.



If **no** oxygen is present in the glucose and yeast suspension, what will occur?

- A Ethanol will be produced and the limewater will stay clear.
- B Ethanol will be produced and the limewater will go cloudy.
- C Lactic acid will be produced and the limewater will stay clear.
- D Lactic acid will be produced and the limewater will go cloudy.

20 What happens during respiration in germinating seeds?

- A Seed temperature decreases.
- B Carbon dioxide is taken up by the seeds.
- C Glucose is used up by the seeds.
- D Oxygen is given out by the seeds.

21 Which row describes the urine produced by a person who is exercising on a hot day?

	concentration of urine	volume of urine
A	concentrated	large
B	concentrated	small
C	dilute	large
D	dilute	small

22 What is the function of the optic nerve?

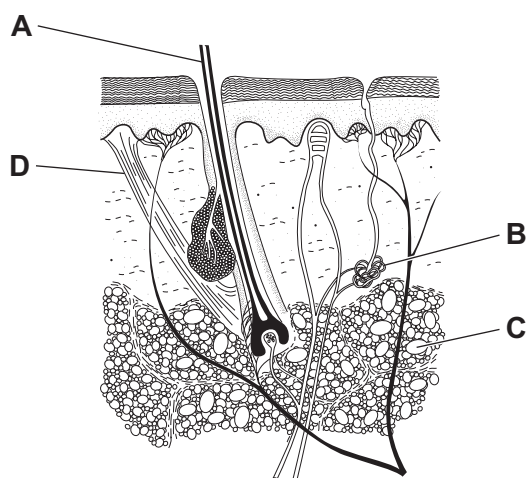
- A** to carry impulses from the brain to the retina
- B** to carry impulses from the retina to the brain
- C** to carry light from the brain to the retina
- D** to carry light from the retina to the brain

23 What causes the development of female secondary sexual characteristics?

- A** adrenaline production
- B** menstrual cycle starting
- C** oestrogen production
- D** ovulation occurring

24 The diagram shows a cross-section of the skin.

Which part produces a substance which cools the skin when it evaporates?



25 Gravitropism and phototropism are names given to specific plant growth responses.

How do shoots and roots normally respond?

	direction of growth of shoots		direction of growth of roots	
	light	gravity	light	gravity
A	towards	away	away	towards
B	away	towards	towards	away
C	towards	away	towards	away
D	away	towards	away	towards

26 Which type of infection could be treated successfully with antibiotics?

	infection caused by			
	bacteria	resistant bacteria	viruses	
A	✓	✓	✗	key ✓ = yes ✗ = no
B	✓	✗	✗	
C	✗	✓	✓	
D	✗	✗	✓	

27 What takes place during fertilisation?

- A** One gamete produces one zygote.
- B** One gamete produces two zygotes.
- C** Two gametes produce one zygote.
- D** Two gametes produce two zygotes.

28 What is the transfer of pollen from an anther to a stigma called?

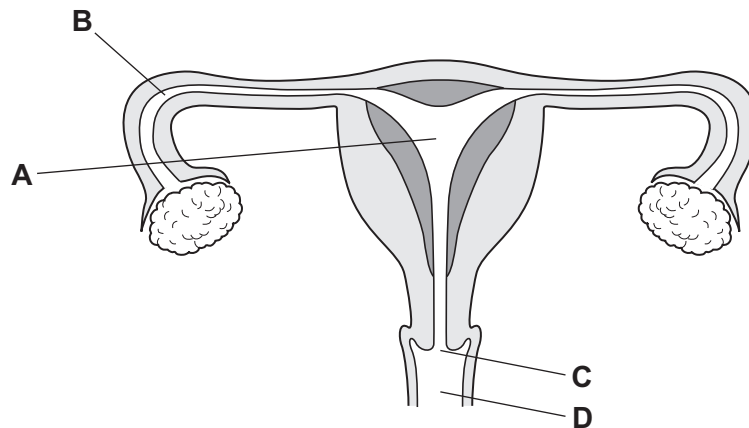
- A** fertilisation
- B** respiration
- C** meiosis
- D** pollination

29 What happens to the uterus lining during menstruation?

- A breaks down
- B new blood vessel growth
- C implantation of embryo
- D repair and thickening

30 The diagram of the female reproductive system shows where different methods of birth control can be used.

Where would an IUD be placed?



31 What is defined as a length of DNA that codes for a protein?

- A amino acid
- B chromosome
- C gene
- D nucleus

32 When damaged tissues are repaired cells undergo division by

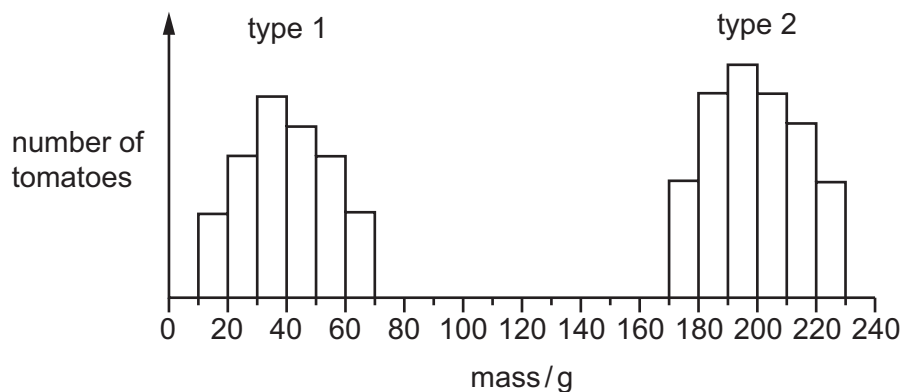
- A meiosis to produce genetically identical cells.
- B meiosis to produce genetically different cells.
- C mitosis to produce genetically identical cells.
- D mitosis to produce genetically different cells.

- 33** Pea plants produce either yellow or green seeds. Yellow (Y) is dominant to green (y).

What are the most likely phenotypes of the offspring of a cross between YY and Yy plants?

- A** 50% yellow and 50% green
- B** 75% yellow and 25% green
- C** 100% yellow
- D** 100% green

- 34** The graph shows the masses of two different types of tomato.



What can be concluded from the graph?

- A** Genes do not affect the mass of tomatoes.
- B** Type 1 tomatoes show continuous variation.
- C** Type 2 tomatoes are sometimes smaller than type 1 tomatoes.
- D** Type 2 tomatoes show discontinuous variation.

- 35** The diagram shows a simple food chain.

grass → gazelle → lion

The arrows show the energy flow through the food chain.

Where does the grass obtain its energy?

- A** oxygen from air
- B** water
- C** minerals in soil
- D** the Sun

- 36** The diagram shows a food chain in a rock pool.

seaweed → whelks → crabs → seagulls

What will happen if the number of secondary consumers increases?

There will be

- A** fewer crabs.
 - B** fewer seagulls.
 - C** fewer whelks.
 - D** less seaweed.
- 37** What is the role of anaerobic respiration in bread-making?
- A** to produce alcohol to flavour the bread
 - B** to produce gas to make the bread rise
 - C** to release enough energy to bake the bread
 - D** to release enough lactic acid to kill the yeast
- 38** A crop plant has been genetically modified to make it resistant to herbicides.
- Which is a possible disadvantage of introducing this new crop plant?
- A** Loss of weeds reduces competition.
 - B** Some weeds might become resistant to the herbicide.
 - C** The crop plant is unharmed and produces a higher yield.
 - D** The new gene will appear in new generations of the crop.
- 39** Which is an example of an undesirable effect of deforestation?
- A** decreased carbon dioxide concentration in the atmosphere
 - B** decreased soil erosion
 - C** increased flooding
 - D** the variety of organisms is maintained

40 Wheat is often grown in large areas as a monoculture.

What is a negative environmental impact of this method of farming?

- A** The variety of species in the area is reduced.
- B** Fewer types of pesticide are needed.
- C** Larger harvests can be produced.
- D** Specialist machinery can be used.

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BIOLOGY (9–1)

0970/11

Paper 1 Multiple Choice (Core)

May/June 2020

MARK SCHEME

Maximum Mark: 40

Published

Students did not sit exam papers in the June 2020 series due to the Covid-19 global pandemic.

This mark scheme is published to support teachers and students and should be read together with the question paper. It shows the requirements of the exam.

Mark schemes should usually be read together with the Principal Examiner Report for Teachers. However, because students did not sit exam papers, there is no Principal Examiner Report for Teachers for the June 2020 series.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the June 2020 series for most Cambridge IGCSE™ and Cambridge International A & AS Level components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	D	1
2	A	1
3	C	1
4	D	1
5	D	1
6	A	1
7	B	1
8	B	1
9	D	1
10	C	1
11	B	1
12	C	1
13	A	1
14	C	1
15	A	1
16	B	1
17	C	1
18	C	1
19	B	1
20	C	1
21	B	1
22	B	1
23	C	1
24	B	1
25	A	1
26	B	1
27	C	1
28	D	1

Question	Answer	Marks
29	A	1
30	A	1
31	C	1
32	C	1
33	C	1
34	B	1
35	D	1
36	C	1
37	B	1
38	B	1
39	C	1
40	A	1



Cambridge IGCSE™ (9–1)

BIOLOGY

0970/12

Paper 1 Multiple Choice (Core)

May/June 2020

45 minutes

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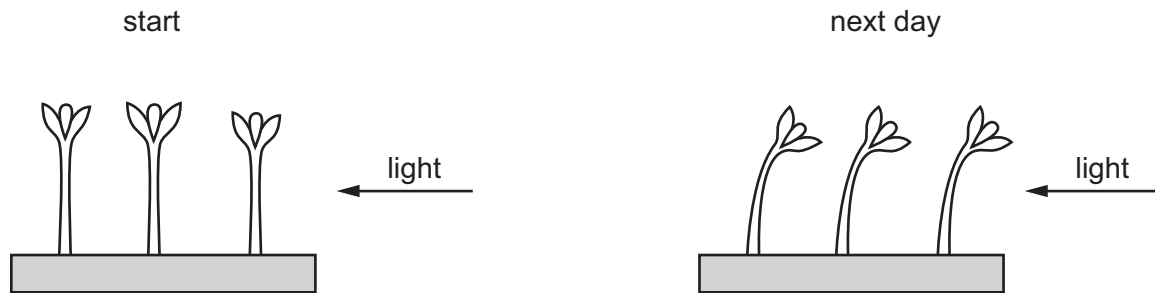
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- 1 The diagram shows what happened in an experiment with plant seedlings.



Which characteristic of living things made the seedlings grow towards the light?

- A excretion
 - B nutrition
 - C respiration
 - D sensitivity
- 2 A rat has the scientific name *Rattus rattus*.

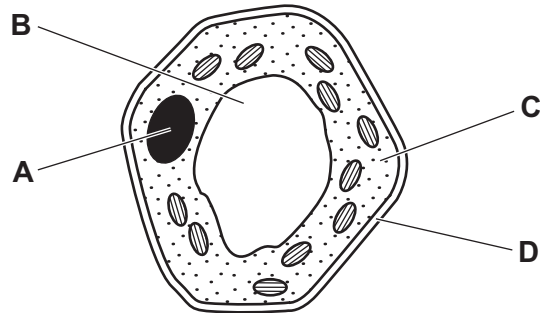
What do the two parts of this name refer to?

- A genus and species
 - B kingdom and genus
 - C kingdom and species
 - D variety and genus
- 3 Which list contains **only** arthropods?
- A amphibians, insects, myriapods
 - B arachnids, crustaceans, myriapods
 - C crustaceans, insects, fish
 - D insects, myriapods, reptiles

- 4 The diagram shows a plant cell.

A biologist wants to find out the number of chromosomes it contains.

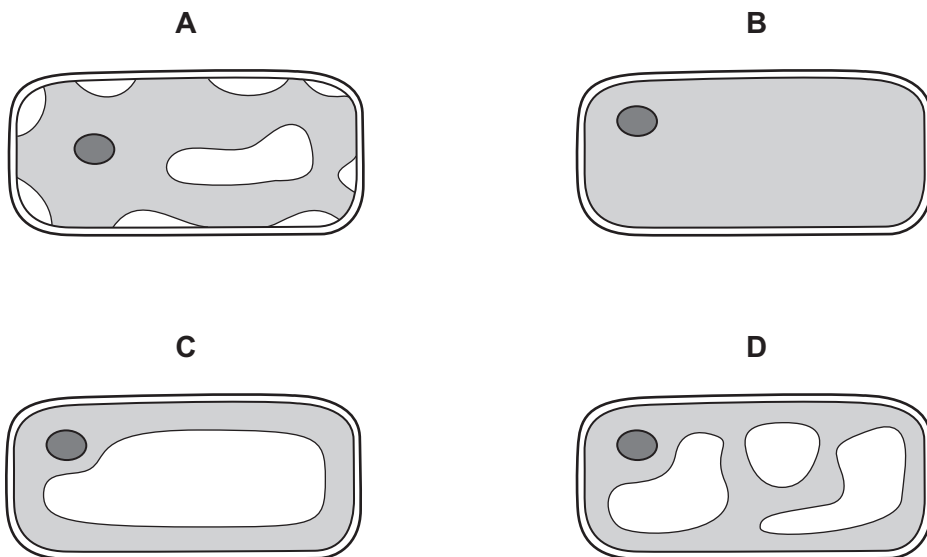
Which labelled part should be examined more closely?



- 5 What are leaves examples of?

- A cells
- B organs
- C organ systems
- D tissues

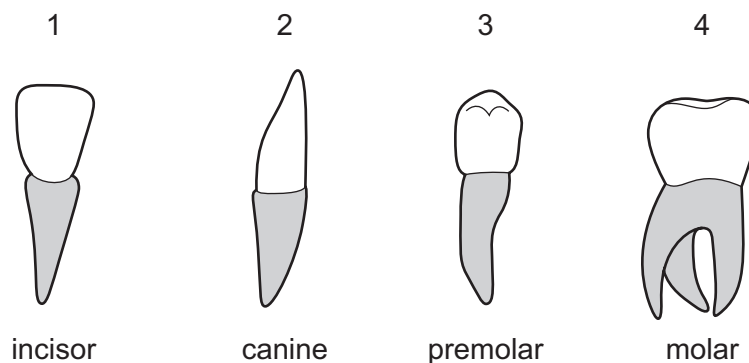
- 6 Which diagram shows the appearance of a plant cell several minutes after it has been placed in a concentrated solution of sugar?



- 7 Uncooked pieces of potato of identical size were placed in different liquids for one hour and then measured.

Which liquid will cause a decrease in the size of the piece of potato?

- A pure water
 - B sugar solution less concentrated than the potato cell contents
 - C sugar solution more concentrated than the potato cell contents
 - D sugar solution with the same concentration as the potato cell contents
- 8 Which food-testing solution shows a positive result when it turns from blue to purple?
- A Benedict's solution
 - B biuret solution
 - C ethanol
 - D iodine solution
- 9 The diagram shows the four types of human tooth.

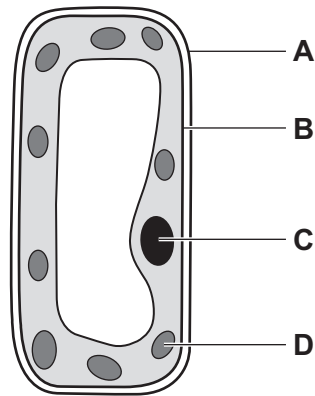


Which teeth are used for grinding food?

- A 1 and 2
- B 2 and 3
- C 3 and 4
- D 4 and 1

10 The diagram shows a plant cell.

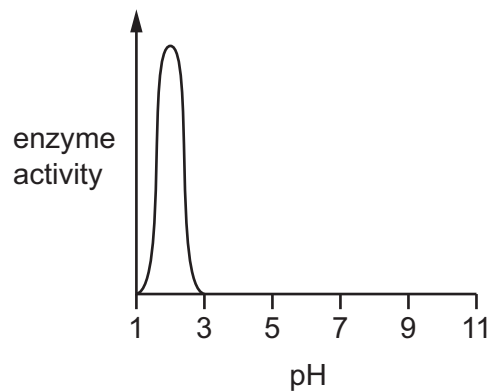
In which part of the cell does photosynthesis occur?



11 Which substance found in a healthy diet helps to prevent constipation?

- A fat
- B fibre
- C minerals
- D vitamins

12 The graph shows the effect of pH on the activity of an enzyme.

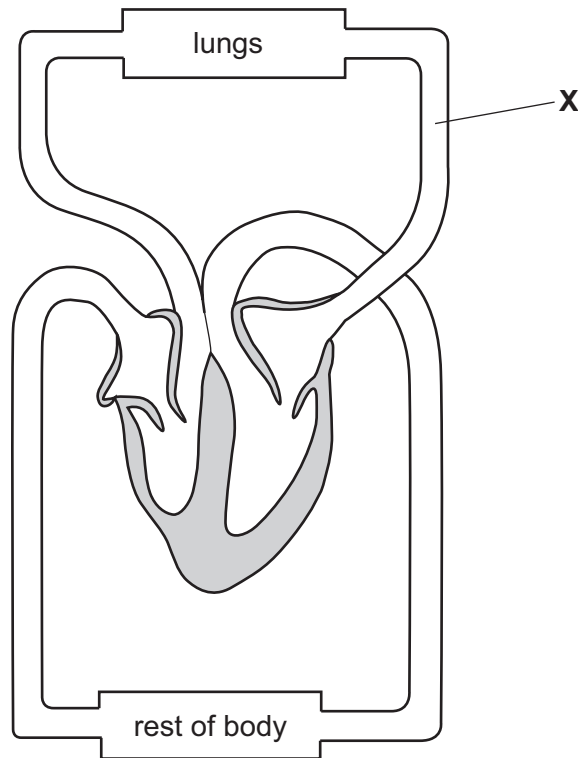


In which part of the alimentary canal would this enzyme be **most** active?

- A large intestine
- B mouth
- C small intestine
- D stomach

- 13** After passing through the root hair cells of a plant, what is the next tissue through which water passes?
- A** cortex
 - B** epidermis
 - C** mesophyll
 - D** xylem
- 14** From which part of a leaf does most water evaporate during transpiration?
- A** the cuticle
 - B** the guard cells
 - C** the spongy mesophyll cells
 - D** the xylem vessels
- 15** Which blood vessel carries blood to the muscle of the heart?
- A** coronary artery
 - B** pulmonary artery
 - C** renal vein
 - D** vena cava

16 The diagram shows the human heart and main blood vessels.



What is the blood vessel labelled **X**?

- A** aorta
- B** pulmonary artery
- C** pulmonary vein
- D** vena cava

17 What is a chemical barrier to pathogens?

- A** nose hairs
- B** skin
- C** stomach acid
- D** white blood cells

18 What is the site of gas exchange in humans?

- A** nose
- B** alveoli
- C** bronchus
- D** trachea

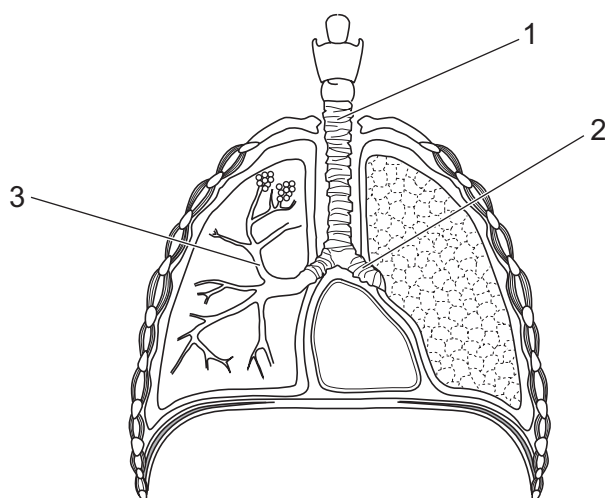
19 The substances listed are associated with aerobic respiration.

- 1 carbon dioxide
- 2 glucose
- 3 oxygen
- 4 water

Which substances are the products of aerobic respiration?

- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

20 The diagram shows the breathing system.

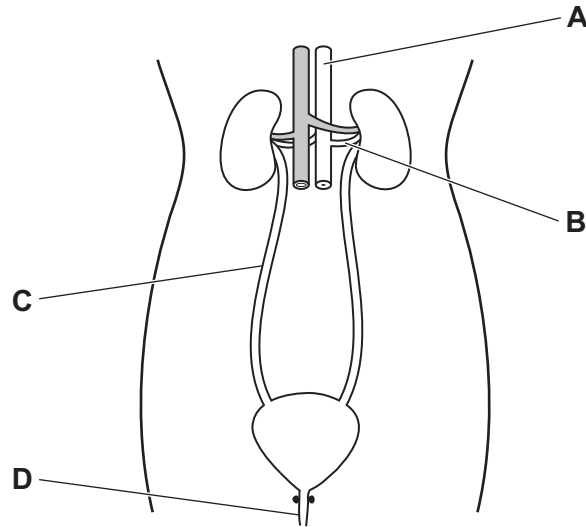


What are the labelled structures?

	1	2	3
A	bronchus	bronchiole	trachea
B	bronchiole	bronchus	trachea
C	trachea	bronchus	bronchiole
D	trachea	diaphragm	bronchus

21 The diagram shows the human excretory system.

Which labelled structure is the ureter?



22 Which structure transfers information to the central nervous system?

- A** effector
- B** motor neurone
- C** relay neurone
- D** sensory neurone

23 Examples of responses to hormones are listed.

- 1 increased breathing rate
- 2 increased pulse rate
- 3 growth of body hair
- 4 widened pupils

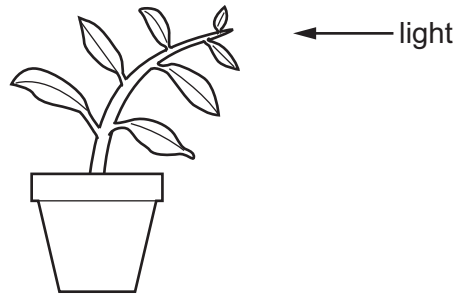
Which responses are caused by the hormone adrenaline?

- A** 1, 2, 3 and 4
- B** 1, 2 and 4 only
- C** 2, 3 and 4 only
- D** 3 and 4 only

24 Which part of the eye focuses light?

- A cornea
- B iris
- C lens
- D retina

25 The diagram shows a plant shoot growing towards the light.



Which response is shown by the shoot of the plant?

- A gravitropism
- B photosynthesis
- C phototropism
- D reflex

26 Which statement about antibiotic use in humans is correct?

- A Different types of antibiotic are used to remove antibodies from the body.
- B Antibiotics are injected in childhood to prevent diseases in adults.
- C Antibiotics are used to treat diseases caused by viruses.
- D Different types of antibiotic are used because bacteria can become resistant.

27 Which conditions are always required for the germination of seeds?

	condition			
	light	oxygen	suitable temperature	water
A	✓	✓	x	✓
B	✓	x	✓	x
C	x	✓	x	✓
D	x	✓	✓	✓

key

✓ = required

x = not required

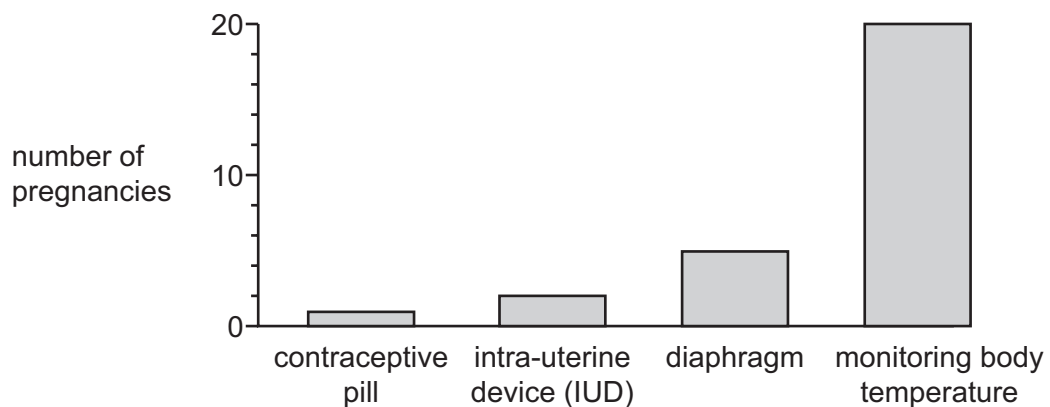
28 Where do fertilisation and pollination occur in a flower?

	fertilisation	pollination
A	carpel	ovule
B	ovary	stigma
C	stigma	anther
D	style	carpel

29 When do the following stages occur in a typical human menstrual cycle?

	bleeding	ovulation
A	days 13 – 14	days 6 – 25
B	days 1 – 4	days 27 – 28
C	days 6 – 25	days 1 – 4
D	days 1 – 4	days 13 – 14

30 The graph shows the number of pregnancies in 4 groups of 100 women. Each group used a different method of contraception.



The method of contraception which is the **most** effective is

- A** barrier.
- B** chemical.
- C** natural.
- D** surgical.

31 The photograph shows a tiger.



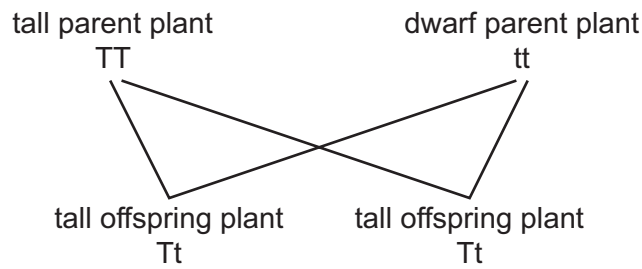
Which visible adaptive feature makes the tiger a good predator?

- A** large teeth
- B** long whiskers
- C** small ears
- D** thick fur

32 Which statement about the human sex chromosomes is correct?

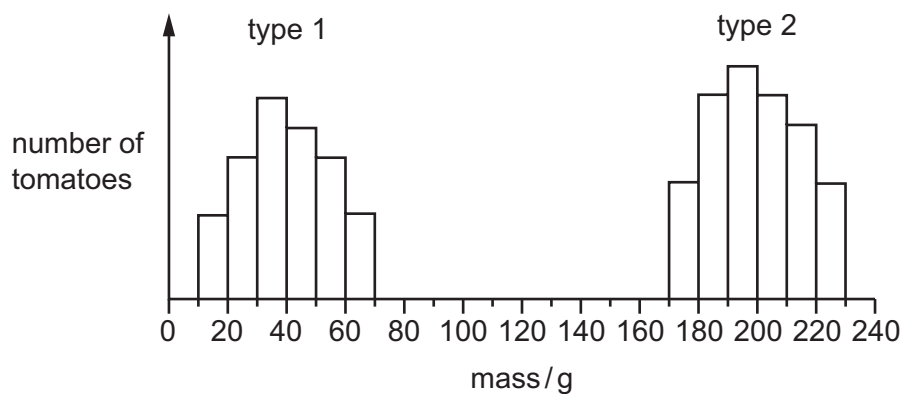
- A** Females have an X chromosome and a Y chromosome.
- B** Females have two Y chromosomes.
- C** Males and females have at least one X chromosome.
- D** Males and females have at least one Y chromosome.

- 33 The diagram shows the inheritance of height in pea plants.



Which plants have a heterozygous genotype?

- A both parent plants
 - B dwarf parent plant only
 - C both offspring plants
 - D tall parent plant only
- 34 The graph shows the masses of two different types of tomato.



What can be concluded from the graph?

- A Genes do not affect the mass of tomatoes.
 - B Type 1 tomatoes show continuous variation.
 - C Type 2 tomatoes are sometimes smaller than type 1 tomatoes.
 - D Type 2 tomatoes show discontinuous variation.
- 35 Which is an example of a population?
- A all the animals and plants living in a lake
 - B all the different herbivores in a forest
 - C all the mahogany trees growing in a forest
 - D all the species of animals in Africa

- 36** The diagram shows a food chain in a rock pool.

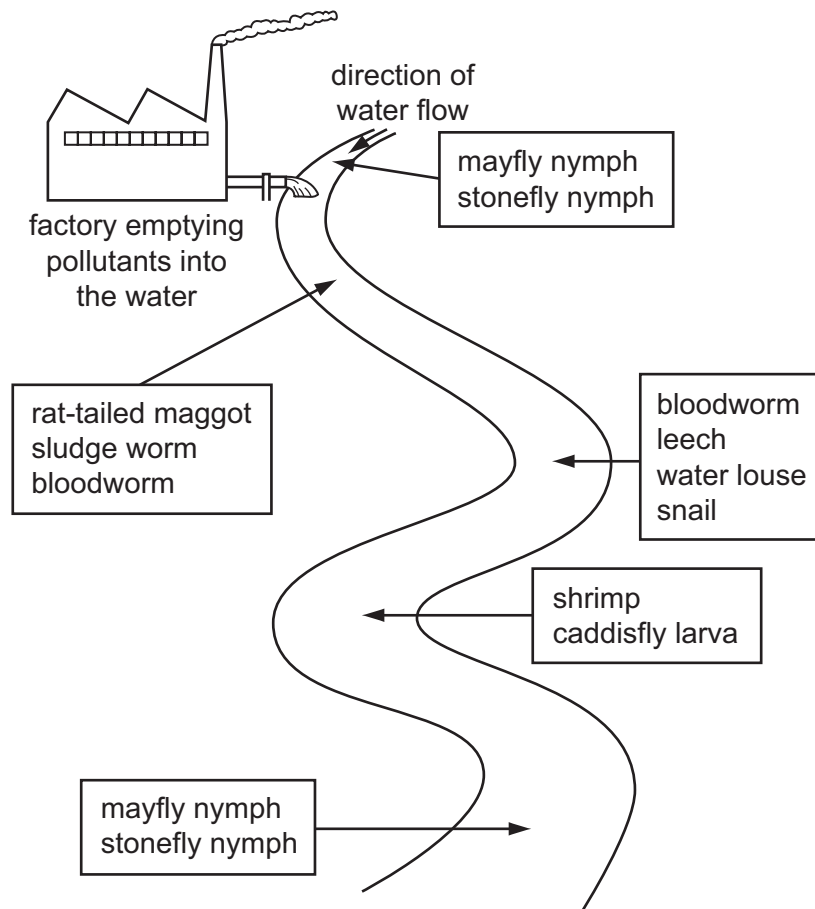
seaweed → whelks → crabs → seagulls

What will happen if the number of secondary consumers increases?

There will be

- A** fewer crabs.
 - B** fewer seagulls.
 - C** fewer whelks.
 - D** less seaweed.
- 37** What is the role of anaerobic respiration in bread-making?
- A** to produce alcohol to flavour the bread
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- 38** A crop plant has been genetically modified to make it resistant to herbicides.
- Which is a possible disadvantage of introducing this new crop plant?
- A** Loss of weeds reduces competition.
 - B** Some weeds might become resistant to the herbicide.
 - C** The crop plant is unharmed and produces a higher yield.
 - D** The new gene will appear in new generations of the crop.

- 39 The diagram shows the results of a survey on the types of animals found along a stretch of river near to a factory.

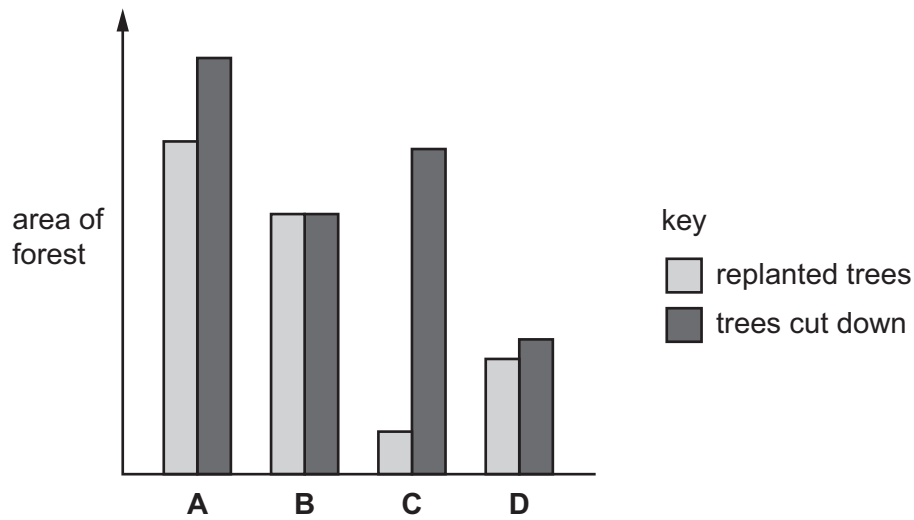


Which animal lives in the least polluted water?

- A bloodworm
- B caddisfly larva
- C leech
- D stonefly nymph

- 40 The graph shows four areas of forest that were cut down and then replanted with trees in one part of the world between 1995 and 2015.

Which area of forest was used most sustainably?



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Cambridge IGCSE™ (9–1)

BIOLOGY (9–1)

0970/12

Paper 1 Multiple Choice (Core)

May/June 2020

MARK SCHEME

Maximum Mark: 40

Published

Students did not sit exam papers in the June 2020 series due to the Covid-19 global pandemic.

This mark scheme is published to support teachers and students and should be read together with the question paper. It shows the requirements of the exam.

Mark schemes should usually be read together with the Principal Examiner Report for Teachers. However, because students did not sit exam papers, there is no Principal Examiner Report for Teachers for the June 2020 series.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the June 2020 series for most Cambridge IGCSE™ and Cambridge International A & AS Level components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	D	1
2	A	1
3	B	1
4	A	1
5	B	1
6	A	1
7	C	1
8	B	1
9	C	1
10	D	1
11	B	1
12	D	1
13	A	1
14	C	1
15	A	1
16	C	1
17	C	1
18	B	1
19	B	1
20	C	1
21	C	1
22	D	1
23	B	1
24	C	1
25	C	1
26	D	1
27	D	1
28	B	1

Question	Answer	Marks
29	D	1
30	B	1
31	A	1
32	C	1
33	C	1
34	B	1
35	C	1
36	C	1
37	B	1
38	B	1
39	D	1
40	B	1



Cambridge IGCSE™ (9–1)

BIOLOGY

0970/12

Paper 1 Multiple Choice (Core)

October/November 2020

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

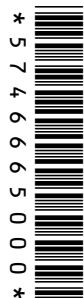
INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
- Any rough working should be done on this question paper.

This document has **16** pages. Blank pages are indicated.



1 Which process is carried out by all organisms?

- A growth
- B photosynthesis
- C sexual reproduction
- D transpiration

2 According to the binomial system, how should a human be named?

- A *Homo Sapiens*
- B *Homo sapiens*
- C *homo Sapiens*
- D *homo sapiens*

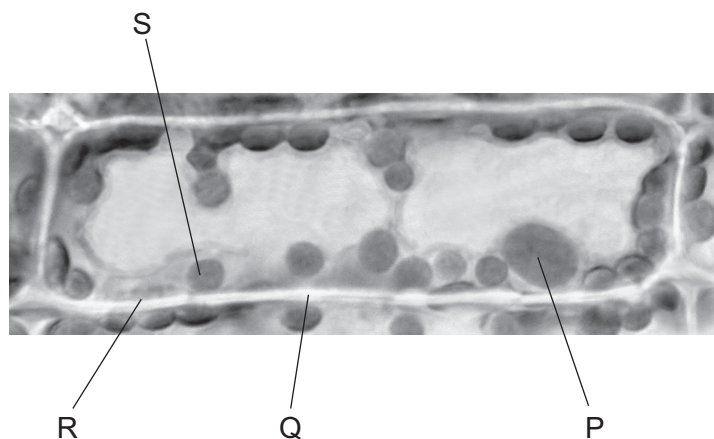
3 Some of the structures found in cells are listed:

- cytoplasm
- cell membrane
- cell wall
- chloroplast.

How many of these structures are found in the cells of **all** living organisms?

- A 1
- B 2
- C 3
- D 4

4 The photomicrograph shows a cell from a type of aquatic plant.



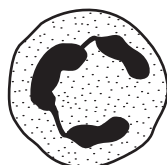
Which parts labelled on the photomicrograph indicate that this is a plant cell?

- A P and R
- B P and S
- C Q and R
- D Q and S

5 Which type of cell transports oxygen?

- A ciliated
- B nerve
- C palisade mesophyll
- D red blood

6 The cell shown in the diagram has been magnified 3000 times. The diagram is 21 mm wide.



What is the actual diameter of the cell?

- A 21 mm
- B $\frac{21\text{mm}}{3000}$
- C $21\text{ mm} \times 3000$
- D $\frac{3000}{21\text{mm}}$

7 What are features of osmosis?

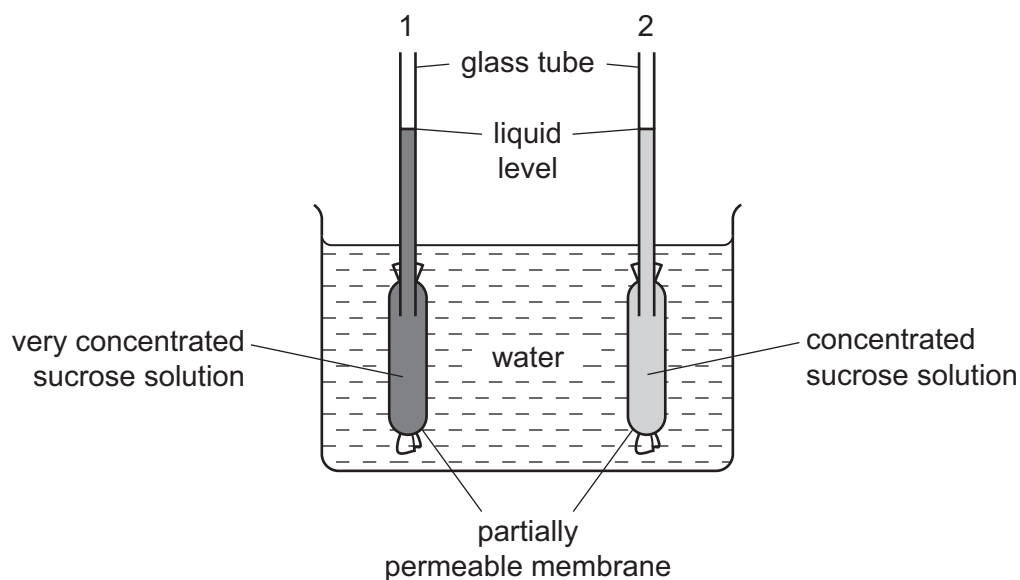
	diffusion is involved	requires cell walls	requires a partially permeable membrane
A	✓	x	✓
B	✓	x	x
C	x	✓	✓
D	x	✓	x

key

✓ = yes

x = no

- 8 The diagram shows apparatus which can be used to demonstrate osmosis.



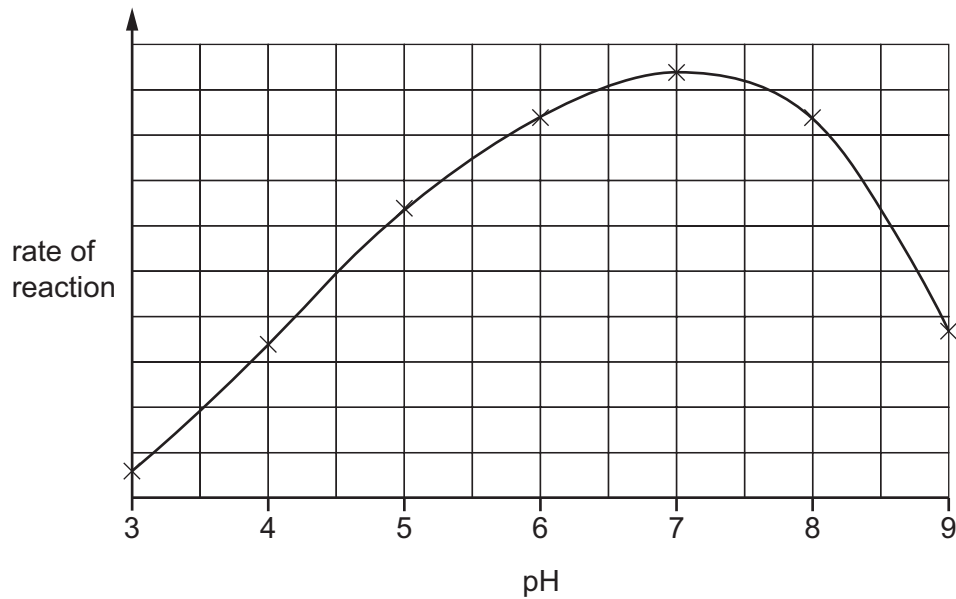
After one hour, what would happen to the liquid levels in the glass tubes?

	liquid level in tube 1	liquid level in tube 2
A	falls	falls
B	falls	rises
C	rises	falls
D	rises	rises

- 9 Which substances are made by linking together glucose molecules only?

- A** cellulose, glycogen and starch
- B** fats, cellulose and proteins
- C** proteins, oils and glycogen
- D** starch, fats and oils

10 The graph shows the effect of pH on the rate of reaction of an enzyme.



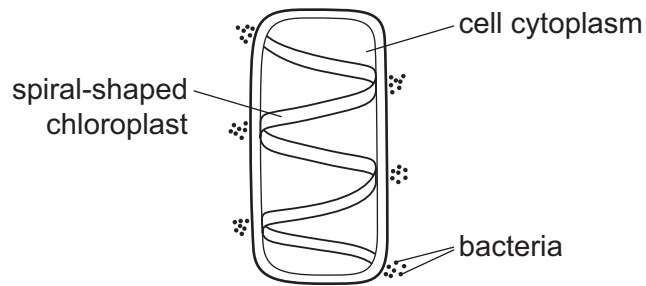
What does the graph show?

- A The enzyme is destroyed at pH 9.
- B The enzyme works best at pH 6.
- C The rate of reaction halves as the pH changes from pH 5 to pH 7.
- D The rate of reaction is the same at pH 5 and pH 8.5.

11 What kind of molecule is an enzyme?

- A fat
- B glucose
- C protein
- D starch

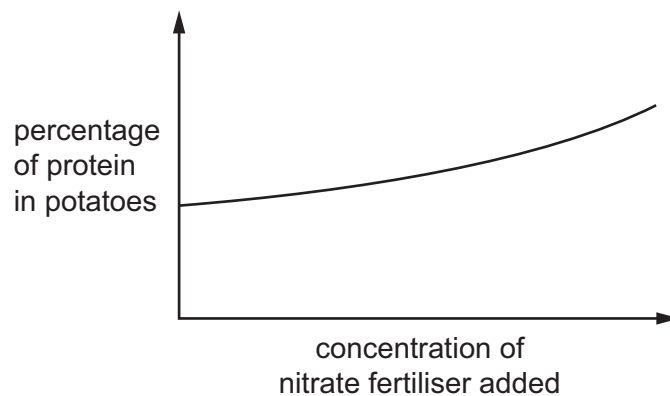
- 12 The diagram shows a cell with groups of bacteria around its edge.



The bacteria move to areas of high oxygen concentration.

Which process in the cell causes the bacteria to form these groups?

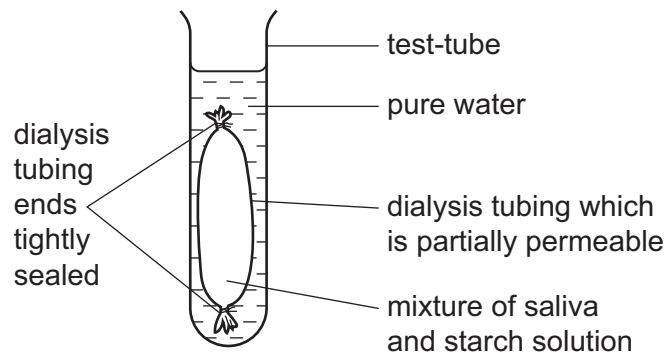
- A digestion
 - B photosynthesis
 - C reproduction
 - D respiration
- 13 The graph shows how the percentage of protein found in potatoes was affected by the concentration of nitrate fertiliser added to the crop when it was growing.



These results suggest that greater concentrations of nitrate fertiliser allow the potatoes to make more

- A amino acids.
- B cellulose.
- C lipid.
- D starch.

- 14 Which stage of nutrition takes place when food molecules become part of a body cell?
- A absorption
 - B assimilation
 - C digestion
 - D ingestion
- 15 The diagram shows the apparatus used in an experiment to demonstrate two processes that take place in the body.



After one hour, the water outside the dialysis tubing contained reducing sugars.

Which two processes are represented by this experiment?

- A absorption and digestion
 - B absorption and egestion
 - C digestion and egestion
 - D digestion and ingestion
- 16 Which type of organism causes tooth decay?
- A arachnid
 - B bacterium
 - C producer
 - D virus
- 17 In which order does water pass through these structures in a plant?
- A mesophyll → root hair → xylem
 - B mesophyll → xylem → root hair
 - C root hair → mesophyll → xylem
 - D root hair → xylem → mesophyll

18 What is a function of phloem?

- A** transports minerals to the roots
- B** transports starch to the roots
- C** transports sugar to the roots
- D** transports water to the roots

19 What ensures that blood flows in one direction in the human circulatory system?

- A** diffusion of carbon dioxide
- B** diffusion of oxygen
- C** thick walled arteries
- D** valves

20 Some functions of blood are listed.

- 1 antibody production
- 2 blood clotting
- 3 oxygen transport
- 4 phagocytosis

What are functions of white blood cells?

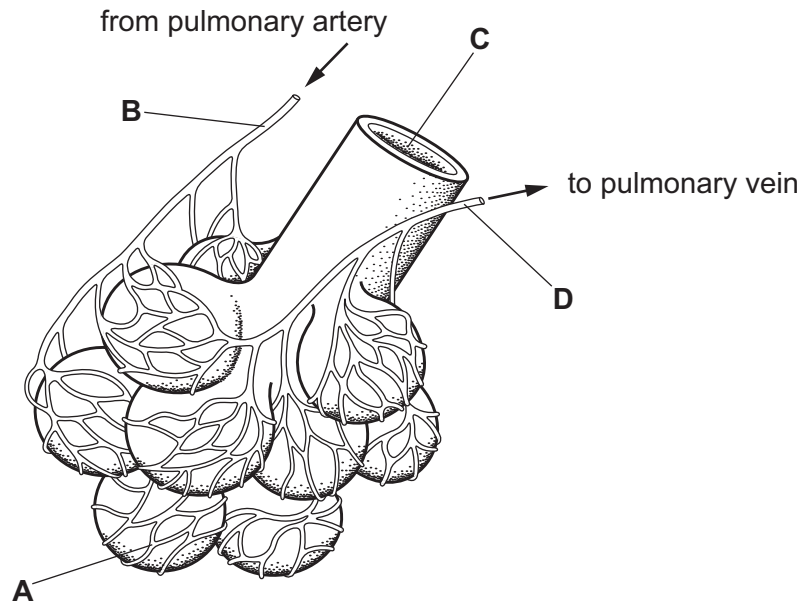
- A** 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

21 Which is an example of a transmissible disease?

- A** coronary heart disease due to eating high fat food
- B** cholera after drinking contaminated water
- C** liver damage due to drinking alcohol
- D** lung cancer due to inhaling tobacco smoke

22 The diagram shows some of the structures in a human lung.

Where is the carbon dioxide concentration highest?



23 What is produced during anaerobic respiration in muscles?

- A** carbon dioxide
- B** ethanol
- C** lactic acid
- D** water

24 The amount of urea in the blood increases as it passes through organ X.

What is organ X?

- A** heart
- B** kidney
- C** liver
- D** pancreas

25 Which structure contains relay neurones?

- A** gland
- B** muscle
- C** spinal cord
- D** synapse

- 26** A man injures his arm in an accident. Afterwards, he can move his hand but cannot feel objects touching his hand.

What could cause this?

- A** Both sensory and motor neurones are cut.
- B** Effectors are damaged.
- C** Motor neurones are cut.
- D** Sensory neurones are cut.

- 27** What are the effects of adrenaline on the human body?

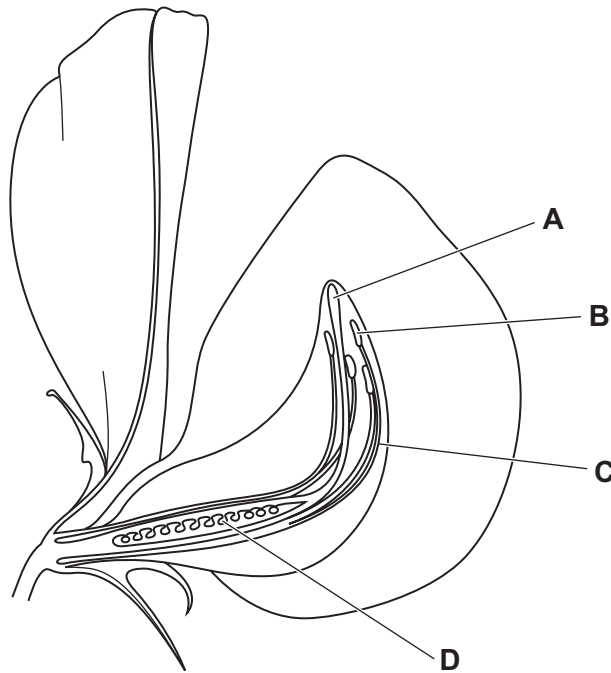
	breathing rate	pulse rate	size of pupils
A	decreased	decreased	decreased
B	decreased	decreased	increased
C	increased	increased	decreased
D	increased	increased	increased

- 28** Which is the correct definition of a drug?

- A** a substance that affects or modifies chemical reactions in the body
- B** a substance that can have negative social implications
- C** a substance that can affect reaction times and self-control
- D** a substance that can kill bacteria

29 The diagram shows a flower cut in half.

Which structure receives the pollen nucleus during fertilisation?



30 During childbirth, what must happen to the amniotic sac, cervix and the uterus wall?

	amniotic sac	cervix	uterus wall
A	breaks	contracts	contracts
B	breaks	dilates	contracts
C	expands	contracts	relaxes
D	expands	dilates	relaxes

31 What is a barrier method of birth control?

- A** contraceptive pill
- B** diaphragm
- C** contraceptive implant
- D** vasectomy

32 Which is a correct description of mitosis?

- A** It produces genetically identical cells called gametes.
- B** It produces genetically identical cells for growth and repair.
- C** It produces genetically different cells for growth.
- D** It produces genetically different cells called gametes.

33 Which term is used to describe alleles that are always expressed when they are present in the genotype?

- A** dominant
- B** heterozygous
- C** homozygous
- D** recessive

34 Some human phenotypes are listed.

- 1 body mass
- 2 foot size
- 3 height
- 4 sex

Which features are examples of continuous variation?

- A** 1, 2 and 3 **B** 1, 3 and 4 **C** 1 and 4 only **D** 2 and 3 only

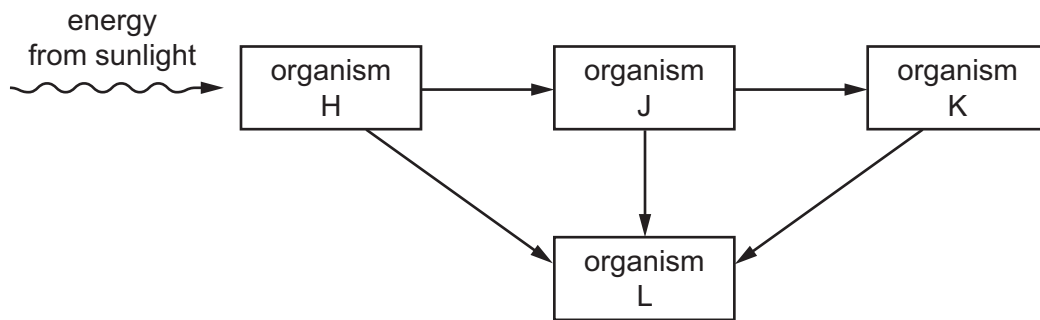
35 A food chain is shown.

potato plant → slug → hedgehog

By which process is energy transferred from the slug to the hedgehog?

- A** egestion
- B** excretion
- C** ingestion
- D** photosynthesis

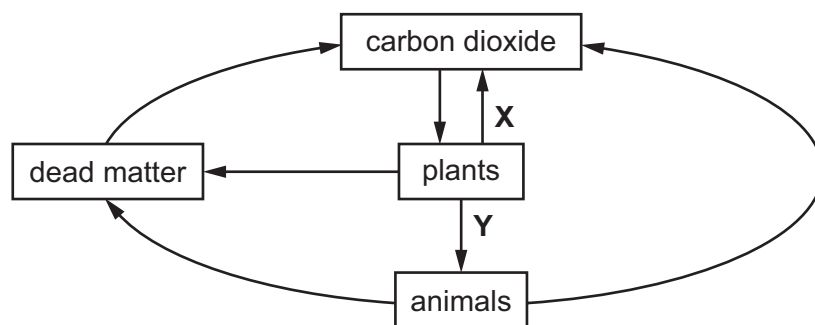
36 The diagram shows a food web.



What is organism L?

- A a carnivore
- B a decomposer
- C a herbivore
- D a producer

37 The diagram shows part of the carbon cycle.



What do arrows **X** and **Y** represent?

- A decay and respiration
- B feeding and photosynthesis
- C photosynthesis and respiration
- D respiration and feeding

38 What can be used to reduce competition between crop plants and weeds?

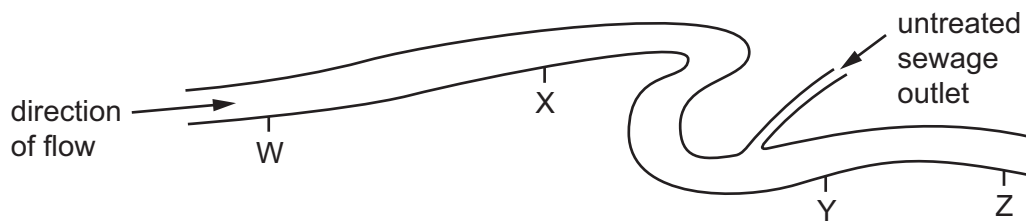
- A fertilisers
- B herbicides
- C insecticides
- D pollinators

- 39** Large areas of forest are cut down in order to clear the land for other uses.

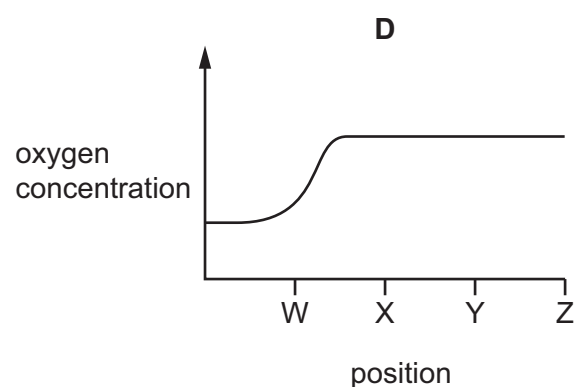
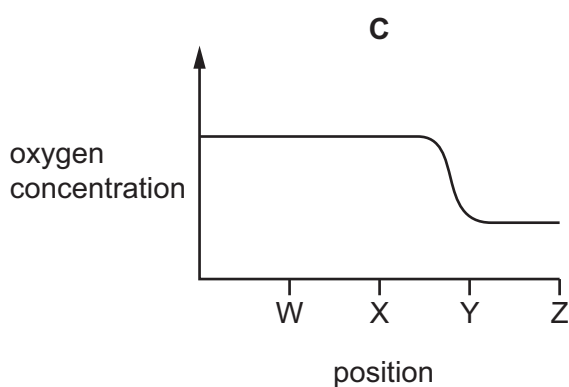
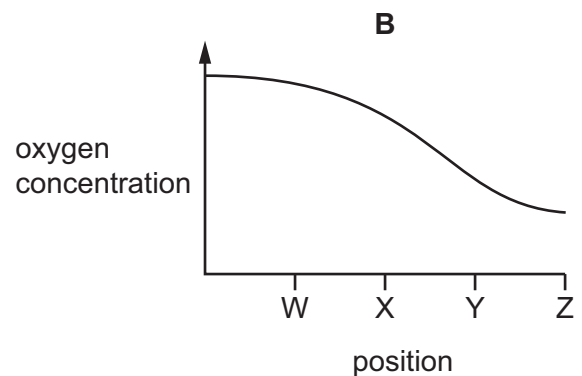
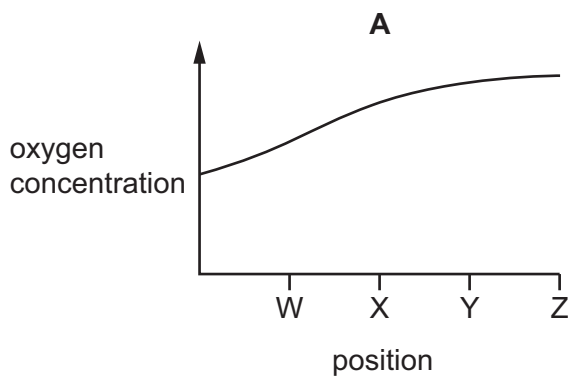
If the land is **not** replanted with trees or crops, what effect does this have on the atmosphere?

- A** Carbon dioxide remains the same.
- B** Carbon dioxide increases.
- C** Oxygen increases.
- D** Water vapour increases.

- 40** The diagram shows four positions on a river where water samples were taken.



Which graph shows oxygen concentrations in the river?



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Cambridge IGCSE™ (9–1)

BIOLOGY

0970/12

Paper 1 Multiple Choice (Core)

October/November 2020

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2020 series for most Cambridge IGCSE™, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	A	1
2	B	1
3	B	1
4	D	1
5	D	1
6	B	1
7	A	1
8	D	1
9	A	1
10	D	1
11	C	1
12	B	1
13	A	1
14	B	1
15	A	1
16	B	1
17	D	1
18	C	1
19	D	1
20	B	1
21	B	1
22	B	1
23	C	1
24	C	1
25	C	1
26	D	1
27	D	1
28	A	1

Question	Answer	Marks
29	D	1
30	B	1
31	B	1
32	B	1
33	A	1
34	A	1
35	C	1
36	B	1
37	D	1
38	B	1
39	B	1
40	C	1



Cambridge IGCSE™ (9–1)

BIOLOGY

0970/11

Paper 1 Multiple Choice (Core)

May/June 2021

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.



1 What is a characteristic of all living organisms?

- A breathing
- B circulation
- C egestion
- D sensitivity

2 The scientific names of some animals are listed.

- 1 *Camelus dromedarius*
- 2 *Camelus ferus*
- 3 *Equus ferus*
- 4 *Struthio camelus*

Which animals are in the same genus?

- A 1, 2 and 3 B 1, 2 and 4 C 1 and 2 only D 2 and 3 only

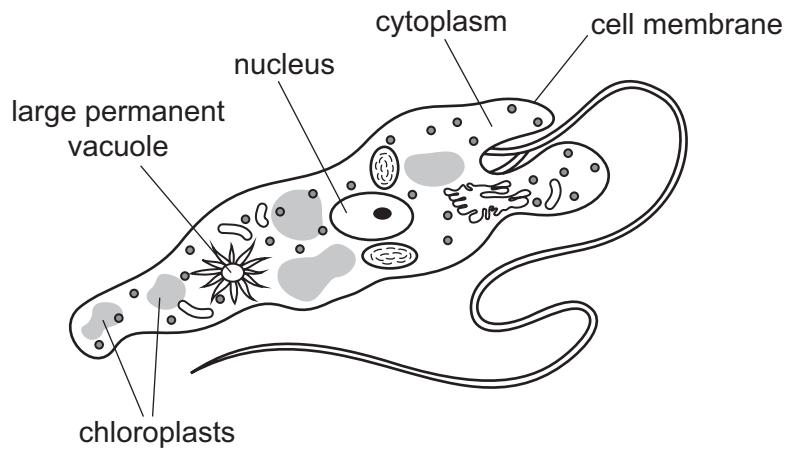
3 Scientists discover a new species of animal.

It has a segmented body with two pairs of legs on each segment.

To which group of animals does this new species belong?

- A arachnids
- B crustaceans
- C insects
- D myriapods

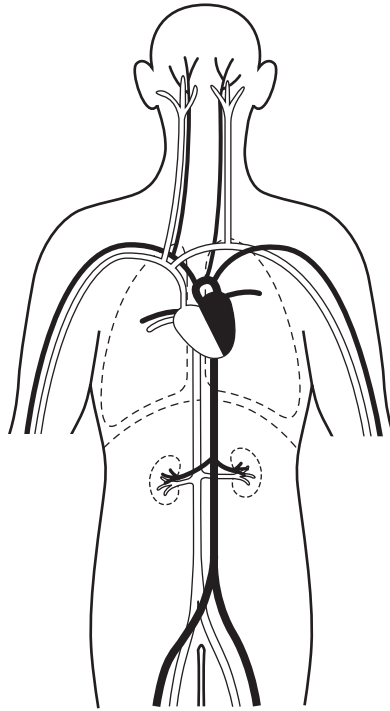
- 4 The diagram shows a single-celled organism called *Euglena*.



Which labelled structures would also be found in an animal cell?

- A cell membrane, chloroplast, nucleus
- B chloroplast, cytoplasm, nucleus
- C cell membrane, cytoplasm, nucleus
- D cell membrane, cytoplasm, large permanent vacuole

- 5 The diagram shows some of the blood vessels and other structures in the human body.



The blood vessels shown are all parts of the same

- A cell.
 - B organ.
 - C organ system.
 - D tissue.
- 6 A photograph shows a plant cell nucleus measuring 2 mm across.
- If the magnification of the cell is $\times 500$, what is the actual size of the nucleus?
- A 0.00002 mm B 0.004 mm C 0.04 mm D 250 mm
- 7 By which process do oxygen and carbon dioxide move between cells and capillaries?
- A breathing
 - B diffusion
 - C excretion
 - D respiration

- 8 A cylinder of potato tissue was placed in a beaker of very salty water. After one hour the mass of the potato cylinder had decreased.

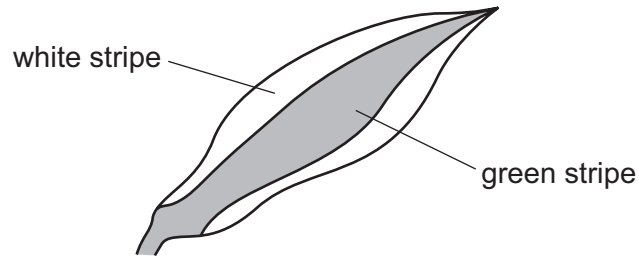
Why did this happen?

- A Salt entered the plant tissue by active transport.
 - B Salt left the plant tissue by osmosis.
 - C Water entered the plant tissue by active transport.
 - D Water left the plant tissue by osmosis.
- 9 Which element is found in proteins but **not** carbohydrates?
- A carbon
 - B hydrogen
 - C nitrogen
 - D oxygen
- 10 Starch is digested by amylase in the mouth, but it is not digested in the stomach.

What is the reason for this?

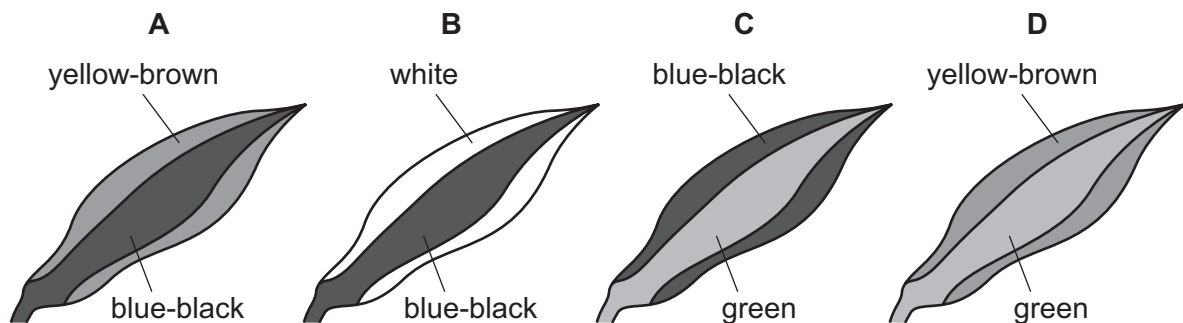
- A All starch digestion is completed in the mouth.
 - B The pH in the stomach is not suitable for the amylase to work.
 - C The starch does not stay in the stomach long enough to be digested.
 - D The temperature in the stomach is not suitable for the amylase to work.
- 11 Which statement describes a catalyst?
- A a substance that decreases the rate of a chemical reaction and is not changed by the reaction
 - B a substance that decreases the rate of a chemical reaction and is changed by the reaction
 - C a substance that increases the rate of a chemical reaction and is changed by the reaction
 - D a substance that increases the rate of a chemical reaction and is not changed by the reaction

- 12** A plant with striped leaves was kept in bright light for six hours.



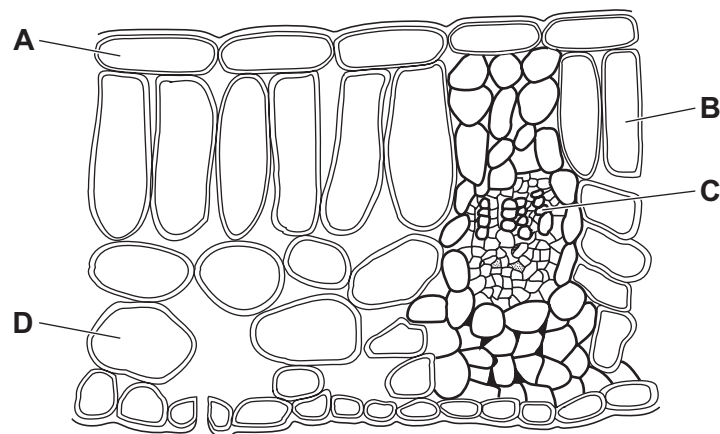
A leaf was taken from the plant and the chlorophyll was removed. The leaf was then tested for starch using iodine solution.

Which diagram shows the result of the test?



- 13** The diagram shows a section through the leaf of a plant.

Which label shows a palisade mesophyll cell?



- 14** In humans, where does most of the absorption of digested food take place?

- A** colon
- B** kidney
- C** liver
- D** small intestine

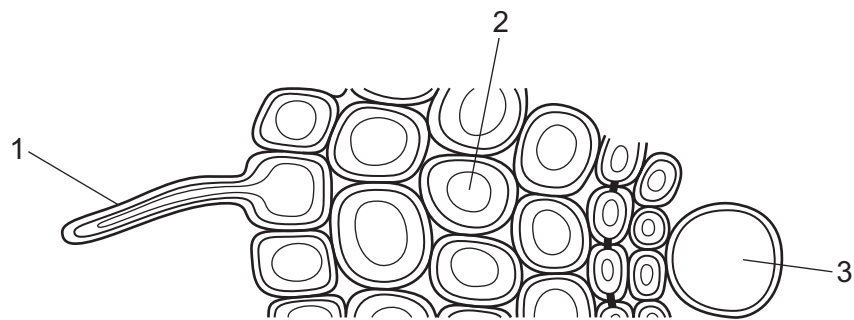
15 What are the products when proteins are broken down?

- A amino acids
- B fatty acids
- C glycerol
- D simple sugars

16 Where does **most** of the absorption of water take place in the alimentary canal?

- A colon
- B oesophagus
- C small intestine
- D stomach

17 The diagram shows part of a cross-section of a root.



What are cells 1, 2 and 3?

	1	2	3
A	root cortex cell	root hair cell	mesophyll cell
B	root hair cell	root cortex cell	xylem
C	root hair cell	root cortex cell	mesophyll cell
D	root cortex cell	root hair cell	xylem

- 18** A plant shoot is placed in a solution of dye.

The dye moves up the stem.

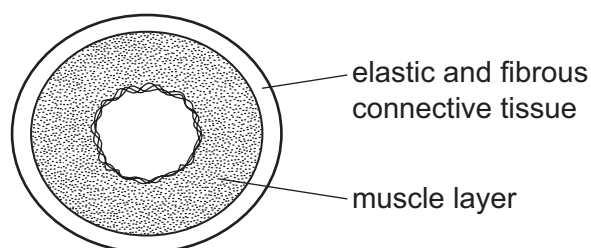
Under which conditions will the dye move most slowly?

	temperature	humidity
A	high	high
B	high	low
C	low	high
D	low	low

- 19** What are the main vessels carrying blood to and from the kidney?

	to kidney	from kidney
A	pulmonary artery	pulmonary vein
B	pulmonary vein	pulmonary artery
C	renal artery	renal vein
D	renal vein	renal artery

- 20** The diagram shows a cross-section through a human blood vessel.



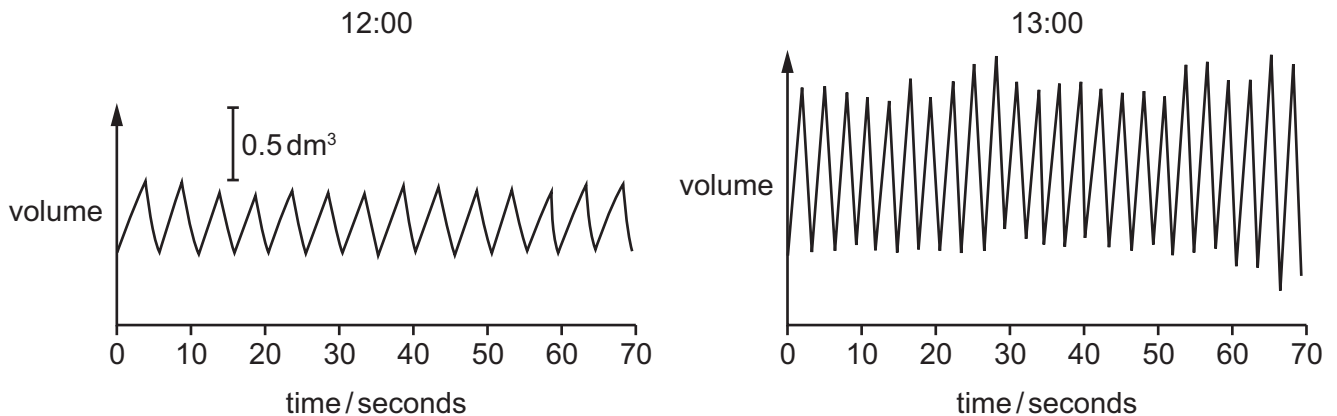
Which type of blood vessel does the diagram show?

- A** an artery
- B** a capillary
- C** a vein
- D** a ventricle

21 What is a function of some white blood cells?

- A** to carry glucose
- B** to carry oxygen
- C** to produce antibiotics
- D** to produce antibodies

22 The diagrams show the depth and rate of breathing in a person at 12:00 and 13:00.



What happens to the person's breathing between 12:00 and 13:00?

	depth of breathing	rate of breathing
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

23 The list shows some processes that take place in a human body.

- 1 production of new red blood cells
- 2 transmission of nerve impulses from the eyes to the brain
- 3 diffusion of gases into and out of the lungs

Which processes use energy released by respiration?

- A** 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

24 What is produced by anaerobic respiration in humans?

	alcohol	carbon dioxide	lactic acid	key ✓ = yes x = no
A	x	✓	✓	
B	✓	✓	x	
C	x	x	✓	
D	✓	x	x	

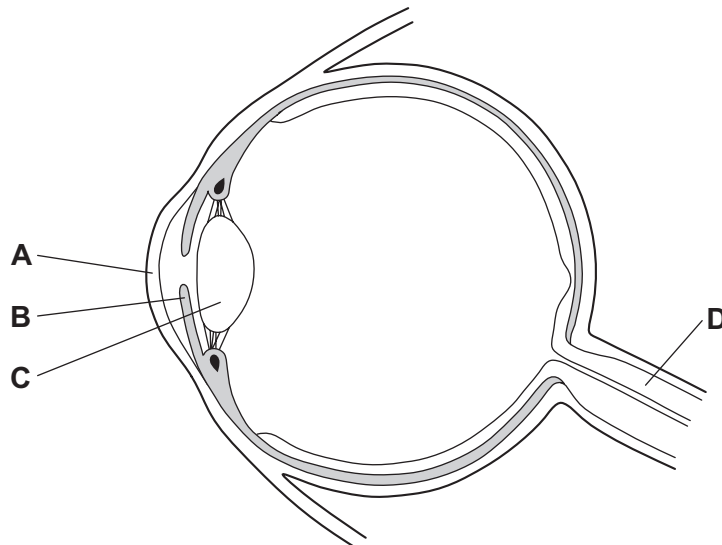
25 Which statement about urea is correct?

- A** Urea is formed from excess amino acids in the kidneys and excreted by the liver.
- B** Urea is formed from excess glucose in the liver and egested by the kidneys.
- C** Urea is formed from excess glucose in the kidneys and egested by the liver.
- D** Urea is formed from excess amino acids in the liver and excreted by the kidneys.

26 What is the sequence of neurones connecting a receptor to an effector in a reflex arc?

- A** motor → relay → sensory
- B** motor → sensory → relay
- C** sensory → motor → relay
- D** sensory → relay → motor

27 What controls how much light enters the eye?



28 Which row matches each hormone to its function?

	causes the growth of facial hair	reduces blood glucose concentration	repairs the lining of the uterus	widens the pupils
A	oestrogen	insulin	testosterone	adrenaline
B	oestrogen	insulin	insulin	testosterone
C	testosterone	adrenaline	insulin	oestrogen
D	testosterone	insulin	oestrogen	adrenaline

29 Scientists carried out a survey on the effect of giving up smoking on the risk of developing lung cancer. The results are shown in the graph.



The scientists made three conclusions:

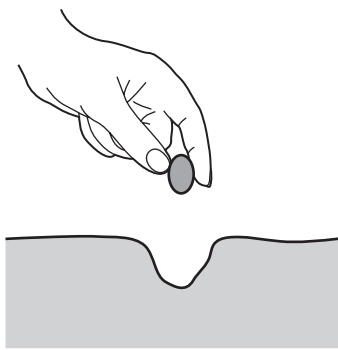
- 1 Stopping smoking reduces the risk of developing lung cancer.
- 2 Age increases the risk of lung cancer for smokers and non-smokers.
- 3 The earlier people stop smoking, the lower the risk.

Which conclusions are correct?

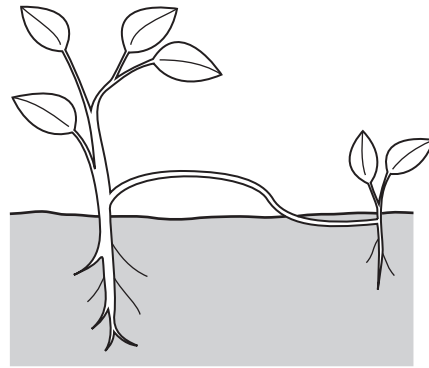
- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

30 A gardener wants to produce many genetically identical plants from a single plant.

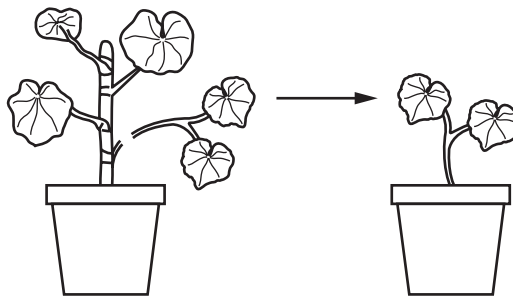
The diagram shows different methods of growing new plants.



1 planting seeds



2 runners



3 stem cuttings

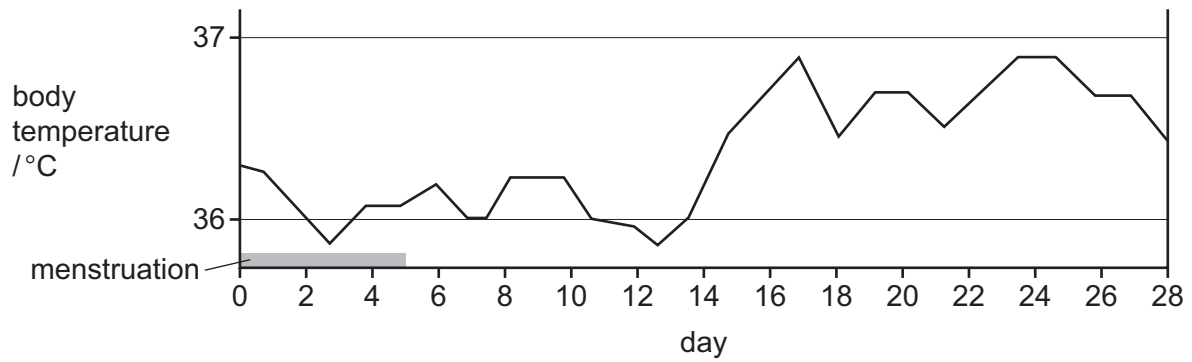
Which methods will produce plants that are genetically identical to the parent plant?

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

31 Which structure in the male reproductive system makes the fluid for sperm to swim in?

- A** penis
B scrotum
C prostate gland
D testis

32 The graph shows a woman's body temperature during the menstrual cycle.



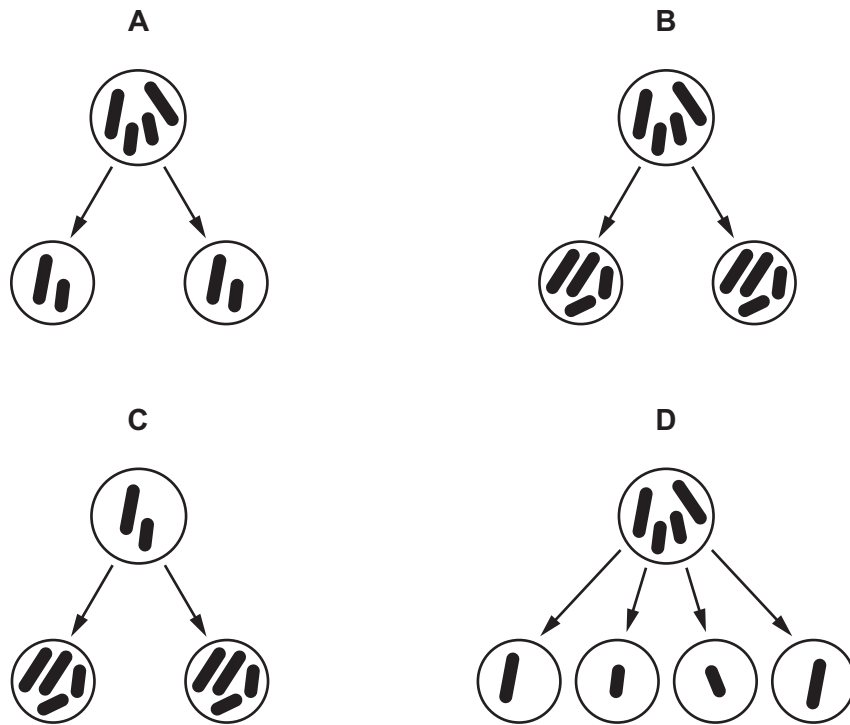
On which days is the woman **most** likely to become pregnant?

- A days 0–5
- B days 6–12
- C days 13–18
- D days 19–28

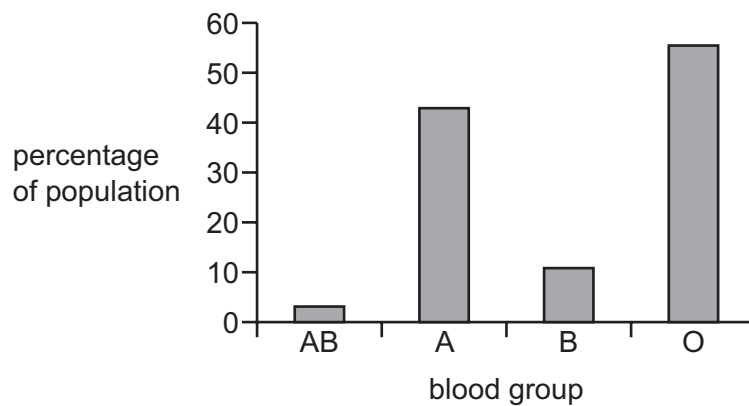
33 What is the transmission of genetic information from generation to generation called?

- A fertilisation
- B inheritance
- C meiosis
- D reproduction

34 Which diagram shows the results of the process of mitosis?



35 The graph shows the distribution of blood groups in one area.

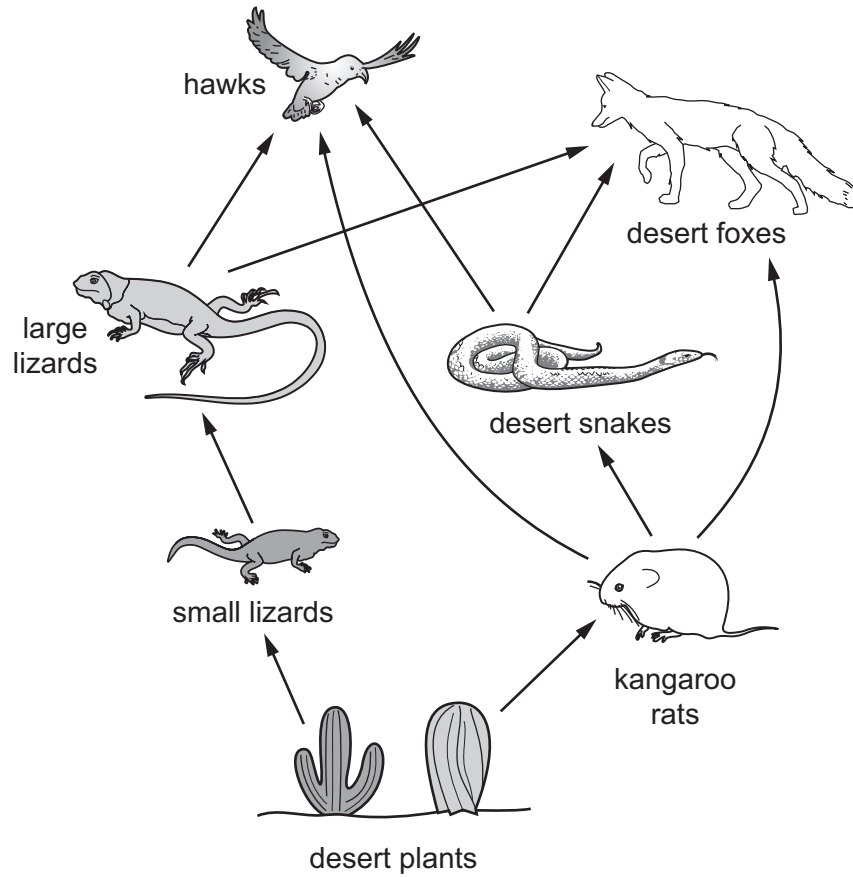


This is an example of discontinuous variation.

Which statement about discontinuous variation is correct?

- A** There is a range of genotypes between two extremes.
- B** There is a range of phenotypes between two extremes.
- C** There are intermediates between the phenotypes.
- D** There are no intermediates between the phenotypes.

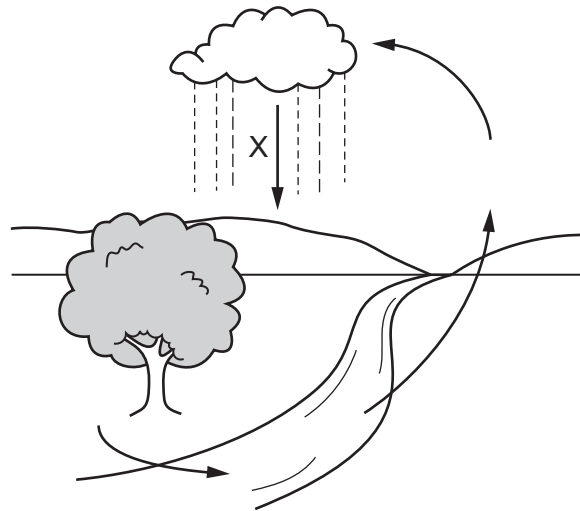
36 The diagram shows a food web in a desert.



Which organisms are **all** secondary consumers?

- A desert foxes, hawks, large lizards and desert snakes
- B desert foxes and hawks only
- C kangaroo rats and small lizards
- D large lizards and desert snakes only

37 The diagram shows stages in the water cycle.



Which process is represented by the letter X?

- A condensation
 - B evaporation
 - C precipitation
 - D transpiration
- 38 What is an example of a population?
- A all the arthropods in a pond
 - B all the crocodiles in a river
 - C all the plants in a wood
 - D all the zebras that lived from 1990–2010
- 39 What is a use of pectinase in the food industry?
- A to extract juice from fruit
 - B to make biofuels
 - C to make biological washing powders
 - D to make bread rise

40 Which factors are believed to have contributed to the rise in sea levels?

- A** pollution of the air by carbon dioxide only
- B** pollution of the air by carbon dioxide and methane
- C** pollution of the sea by plastics only
- D** pollution of the sea by insecticides and nuclear fall-out

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Cambridge IGCSE™ (9–1)

BIOLOGY (9–1)

0970/11

Paper 1 Multiple Choice (Core)

May/June 2021

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2021 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	D	1
2	C	1
3	D	1
4	C	1
5	C	1
6	B	1
7	B	1
8	D	1
9	C	1
10	B	1
11	D	1
12	A	1
13	B	1
14	D	1
15	A	1
16	C	1
17	B	1
18	C	1
19	C	1
20	A	1
21	D	1
22	D	1
23	A	1
24	C	1
25	D	1
26	D	1
27	B	1
28	D	1

Question	Answer	Marks
29	A	1
30	D	1
31	C	1
32	C	1
33	B	1
34	B	1
35	D	1
36	A	1
37	C	1
38	B	1
39	A	1
40	B	1



Cambridge IGCSE™ (9–1)

BIOLOGY

0970/12

Paper 1 Multiple Choice (Core)

October/November 2021

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
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- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

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- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.



- 1 All living things can remove toxic materials and other substances that are in excess of requirements.

What is this process called?

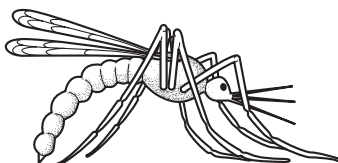
- A egestion
- B excretion
- C nutrition
- D respiration

- 2 The table shows features of some animals.

Which animal is a mammal?

	gills	hair	scales	wings	
A	✓	x	x	x	key ✓ = present x = absent
B	✓	x	✓	x	
C	x	✓	x	✓	
D	x	x	✓	✓	

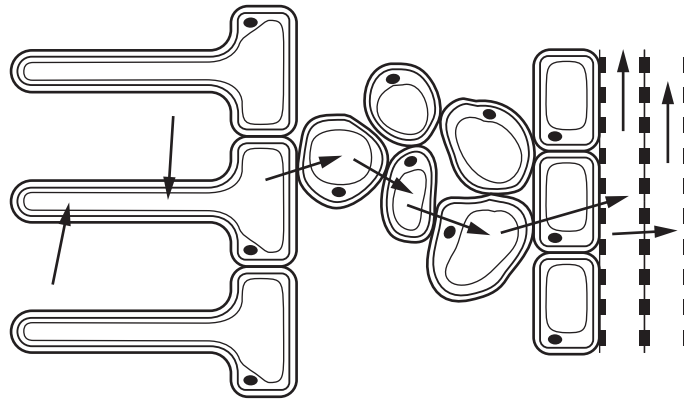
- 3 The diagram shows an animal.



What is the animal?

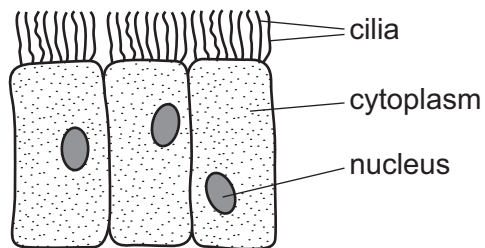
- 1 animal with a vertebral column vertebrate
- animal with an exoskeleton go to 2
- 2 no distinct head, thorax and abdomen **A**
- distinct head, thorax and abdomen go to 3
- 3 eye occupies less than one third of the head **B**
- eye occupies more than one third of the head go to 4
- 4 wings are wider than they are long **C**
- wings are longer than they are wide **D**

- 4 The diagram shows some tissues in part of a plant root. The arrows represent the movement of water into and across the plant root tissues.



How many different tissues are shown?

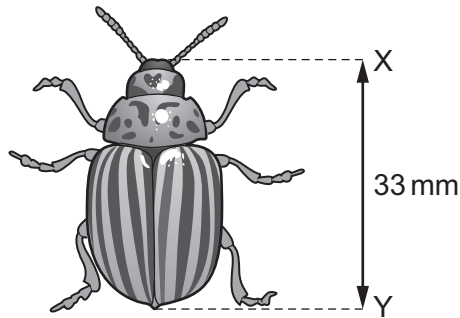
- A 2 B 4 C 5 D 7
- 5 The diagram shows some cells.



Where are these cells found?

- A alimentary canal
B blood
C bronchus
D plant roots

- 6 The diagram shows a beetle.



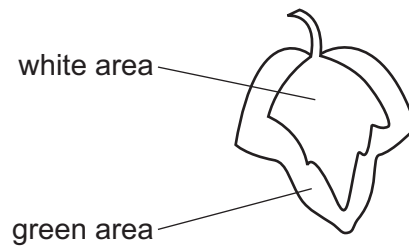
The actual size of the beetle between X and Y is 15 mm.

What is the magnification of the image?

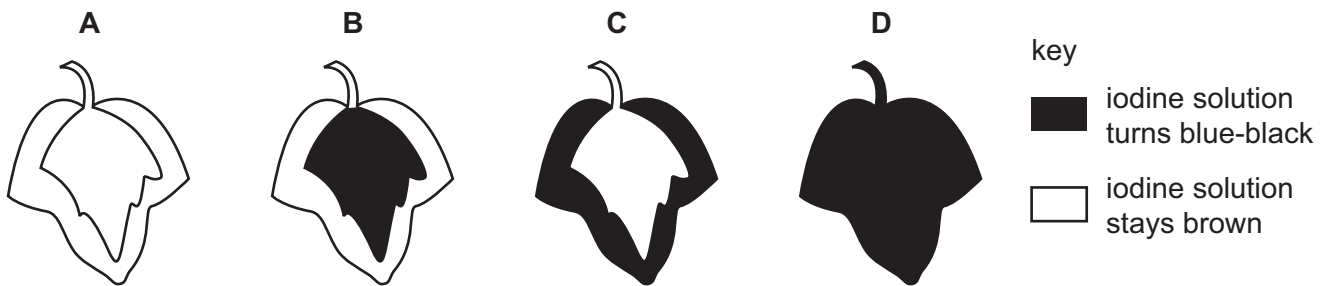
- A** $\times 0.45$ **B** $\times 2.2$ **C** $\times 48$ **D** $\times 495$
- 7 Which statement about diffusion is correct?
- A** The process requires energy from respiration.
B The particles must cross a cell membrane.
C The net movement of particles is up a concentration gradient.
D The process involves the random movement of particles.
- 8 The cells in a plant root have a higher concentration of magnesium ions than in the surrounding soil.
- Which process will move the magnesium ions into the root?
- A** active transport
B diffusion
C osmosis
D transpiration

- 9 In a photosynthesis experiment, a plant is left in bright sunlight for several hours. A leaf is then removed from the plant and tested for starch, using iodine solution.

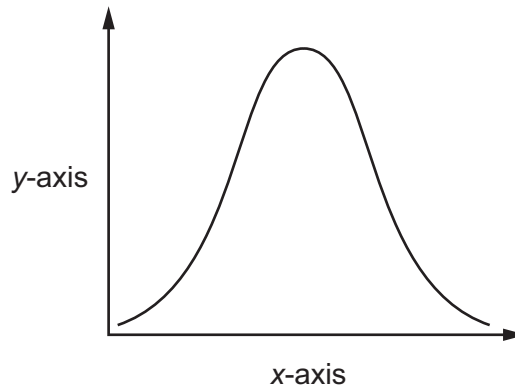
The diagram shows the leaf from the plant that was used in the experiment.



Which diagram shows the result of the experiment?



- 10 An experiment was carried out to investigate the effect of pH on enzyme action. The graph shows the results.

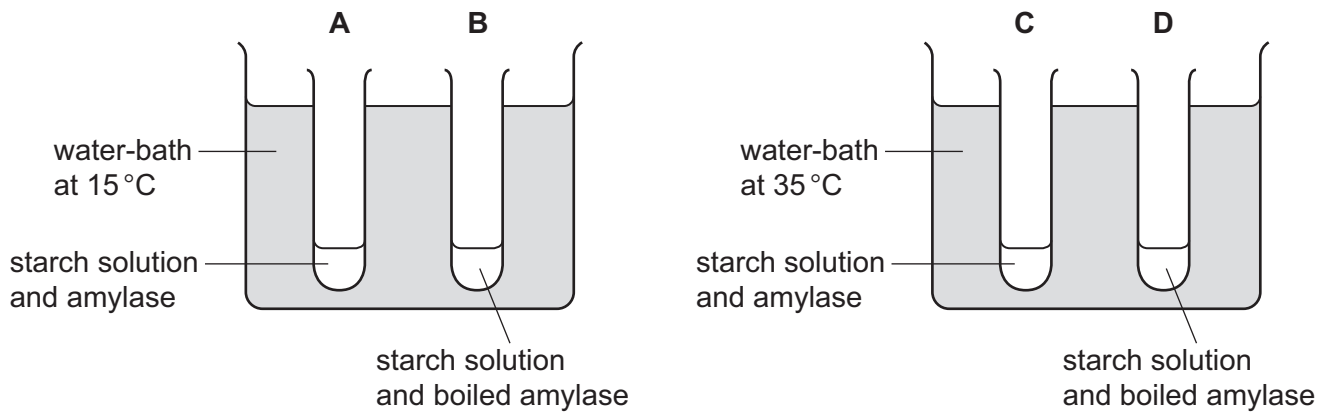


What are the labels for the x-axis and the y-axis?

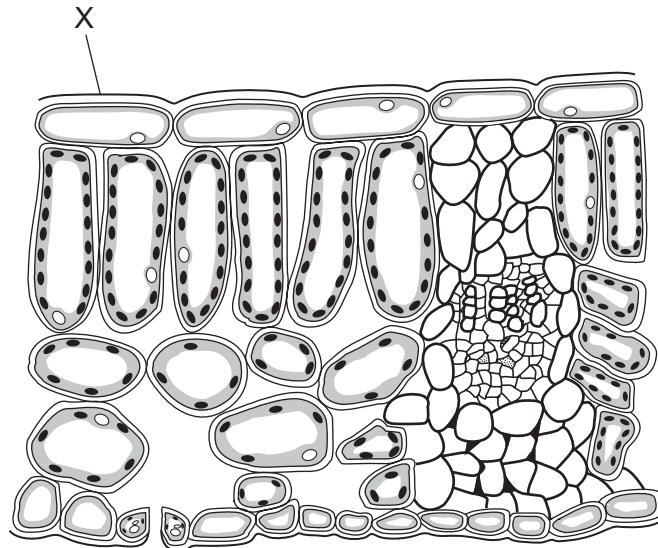
	x-axis	y-axis
A	pH	rate of reaction
B	pH	time
C	rate of reaction	pH
D	time	pH

11 Four test-tubes were set up as shown in the diagram.

In which test-tube is the starch digested most quickly?



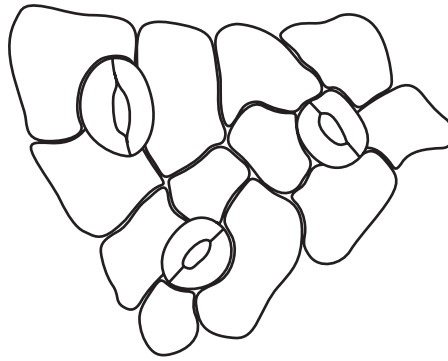
12 The diagram shows a cross-section through a leaf.



What is the structure labelled X?

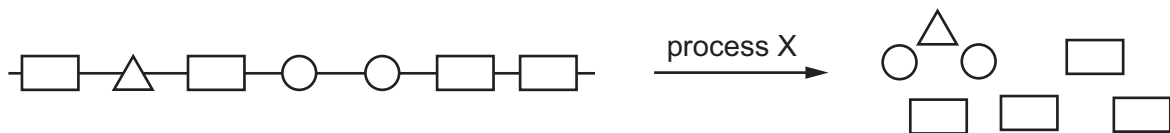
- A cuticle
- B palisade mesophyll
- C spongy mesophyll
- D stomata

- 13 The diagram shows the surface view of part of the epidermis of a leaf.



How many guard cells are present?

- A** 0 **B** 3 **C** 6 **D** 12
- 14 What is the dietary importance of carbohydrates?
- A** to promote healthy bones and teeth
- B** to make fats
- C** to provide energy for the body
- D** to make proteins
- 15 The diagram shows a large food molecule changing into smaller molecules.

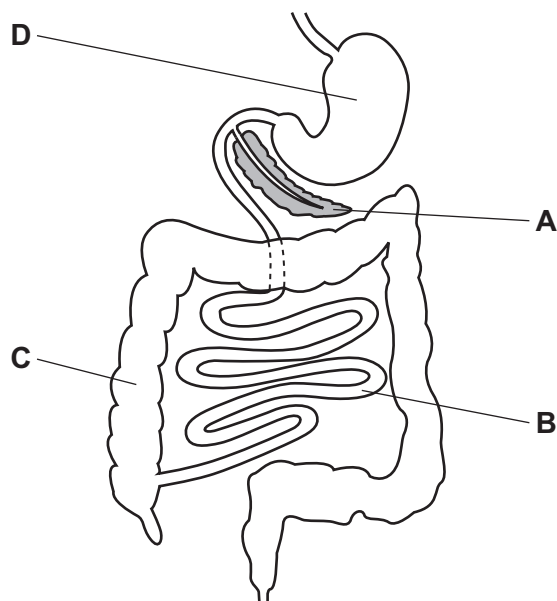


What is process X?

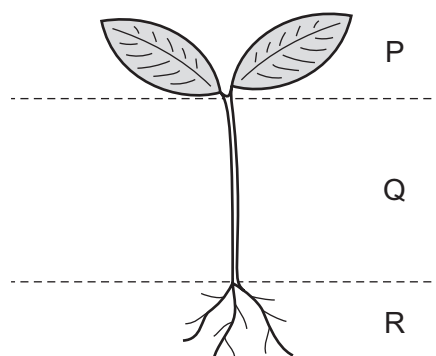
- A** absorption
- B** chewing
- C** digestion
- D** secretion

16 The diagram shows part of the alimentary canal and associated organs.

Where does most water absorption take place?



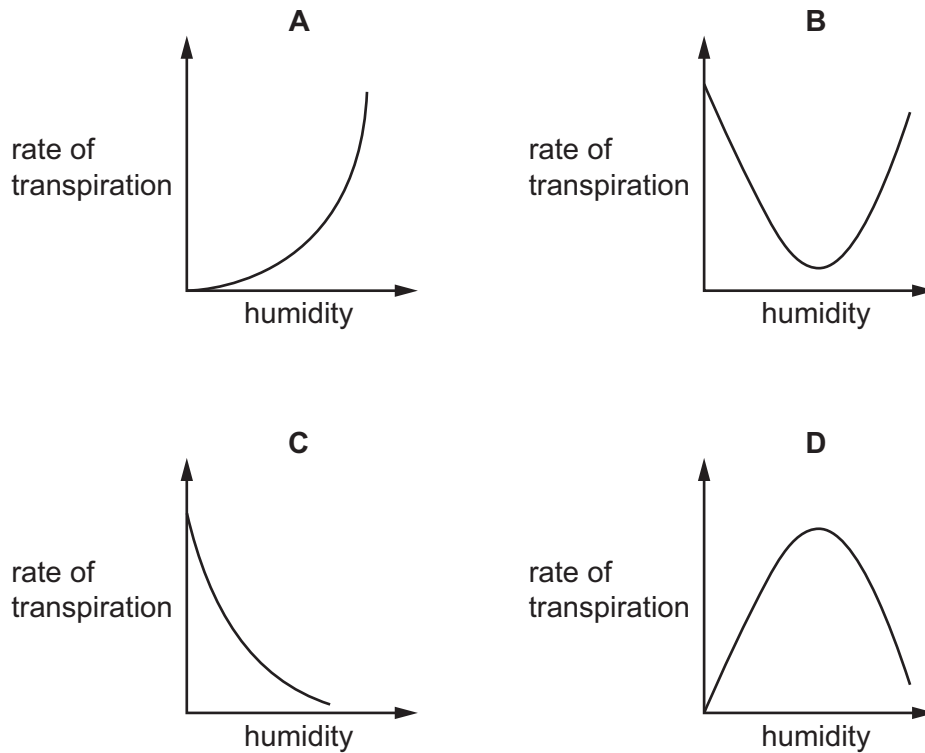
17 The diagram shows a young plant.



In which parts of the plant are xylem vessels found?

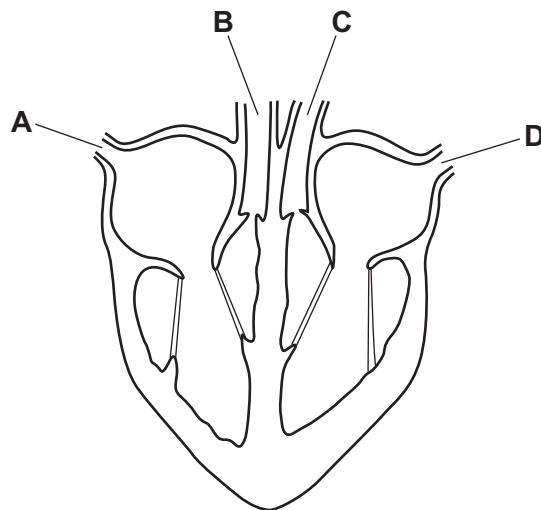
- A** P, Q and R **B** P and Q only **C** Q and R only **D** Q only

18 Which graph shows the effect of increasing humidity on the rate of transpiration?



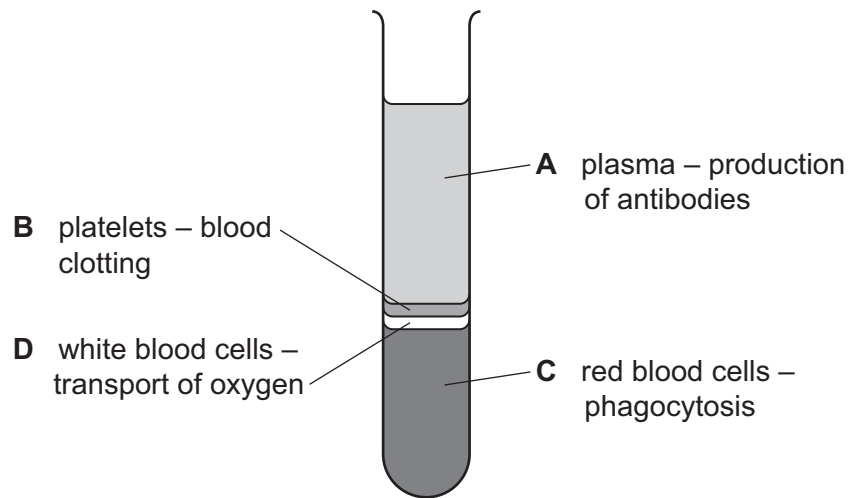
19 The diagram shows a section through a mammalian heart.

Which part carries blood directly from the lungs?



20 The diagram shows the four components of blood separated into layers.

Which component is labelled with the correct function?



21 Some features that help to defend the body against pathogens are listed.

- 1 mucus
- 2 skin
- 3 stomach acid
- 4 phagocytosis

Which features can prevent pathogens entering body tissues?

- A** 1, 2, 3 and 4
- B** 1, 2 and 3 only
- C** 2 and 3 only
- D** 4 only

22 Which shows the correct order of structures that air travels through when a human breathes in?

- A** alveoli → trachea → bronchioles → bronchi
- B** trachea → bronchi → bronchioles → alveoli
- C** alveoli → bronchioles → bronchi → trachea
- D** trachea → bronchioles → bronchi → alveoli

23 Which chemicals are needed to release energy in aerobic respiration?

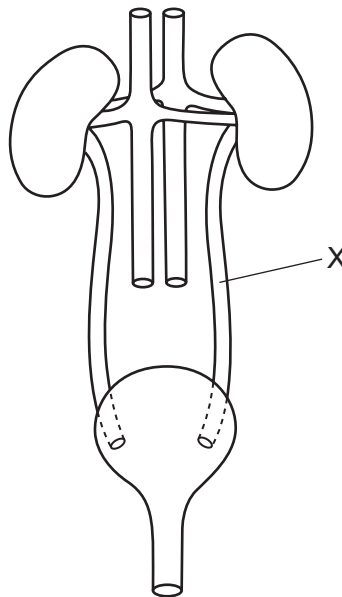
- A** carbon dioxide and glucose
- B** carbon dioxide and water
- C** oxygen and glucose
- D** oxygen and water

24 Yeast is an organism used in the production of biofuels.

Which statement describes why yeast is used for biofuel production?

- A** Yeast respires aerobically to produce carbon dioxide.
- B** Yeast respires aerobically to produce carbon dioxide and ethanol.
- C** Yeast respires anaerobically to produce carbon dioxide.
- D** Yeast respires anaerobically to produce carbon dioxide and ethanol.

25 The diagram shows part of the excretory system of a female mammal.



What is tube X?

- A** ureter
- B** urethra
- C** uterus
- D** vein

26 The structures listed are part of the nervous system.

- 1 brain
- 2 effector
- 3 receptor
- 4 relay neurone

In a reflex arc, which structures are connected to a motor neurone by synapses?

- A** 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

27 A student investigated the effect of changing light intensity on pupil diameter in the eye.

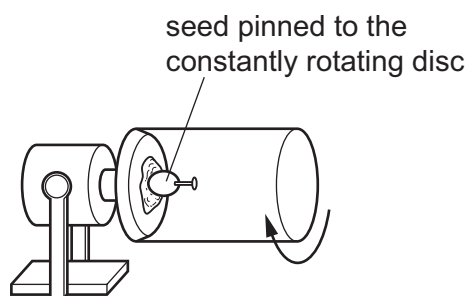
The table shows the results.

starting pupil diameter / mm	final pupil diameter / mm
8	2

Which explanation for the student's results is correct?

- A** A reduced light intensity reduced the pupil diameter by 75%.
- B** A reduced light intensity reduced the pupil diameter by 25%.
- C** An increased light intensity reduced the pupil diameter by 75%.
- D** An increased light intensity reduced the pupil diameter by 25%.

28 A seed is placed and grown on a rotating disc, as shown.



Which diagram shows the appearance of the seedling shoot after seven days?



29 Which substance can be used to treat a bacterial infection?

- A adrenaline
- B antibiotics
- C antigens
- D insulin

30 Fertilisation is defined as the fusion of theX..... of two gametes to form aY..... .

Which row completes the sentence?

	X	Y
A	alleles	sex cell
B	alleles	zygote
C	nuclei	sex cell
D	nuclei	zygote

31 What is the correct description of pollination?

- A Pollen grains are transferred from the anther to the ovary.
- B Pollen grains are transferred from the anther to the stigma.
- C Pollen grains are transferred from the ovary to the stigma.
- D Pollen grains are transferred from the stamen to the anther.

32 HIV is a sexually transmitted infection (STI).

Which methods could be used to reduce the spread of HIV?

- 1 breast-feed babies
- 2 education about how HIV is spread
- 3 provide easy access to condoms
- 4 provide needle exchange schemes for drug users

- A 1, 2 and 3 B 1, 2 and 4 C 1, 3 and 4 D 2, 3 and 4

33 Which of these determine the sex of humans?

- A X and Y genes
- B X and Y alleles
- C X and Y chromosomes
- D X and Y genotypes

- 34** Part of a tree branch is broken off and planted in the soil. A new tree grows from the branch.

Which row is correct?

	the cells in the new tree are formed by		the cells in the original tree and in the new tree will be genetically	
	meiosis	mitosis	different	identical
A	✓	✗	✓	✗
B	✗	✓	✗	✓
C	✗	✓	✓	✗
D	✓	✗	✗	✓

key

✓ = correct

✗ = not correct

- 35** A species of insect usually has pale-coloured wings. This helps to camouflage them on pale-coloured tree trunks. A few of the insects in this species have darker coloured wings.

After a number of years the tree trunks become darker in colour due to environmental changes. The insects with dark-coloured wings become more common than insects with pale-coloured wings in this species.

Which process causes this change in the proportion of insects with dark-coloured wings?

- A** biotechnology
- B** conservation
- C** natural selection
- D** selective breeding

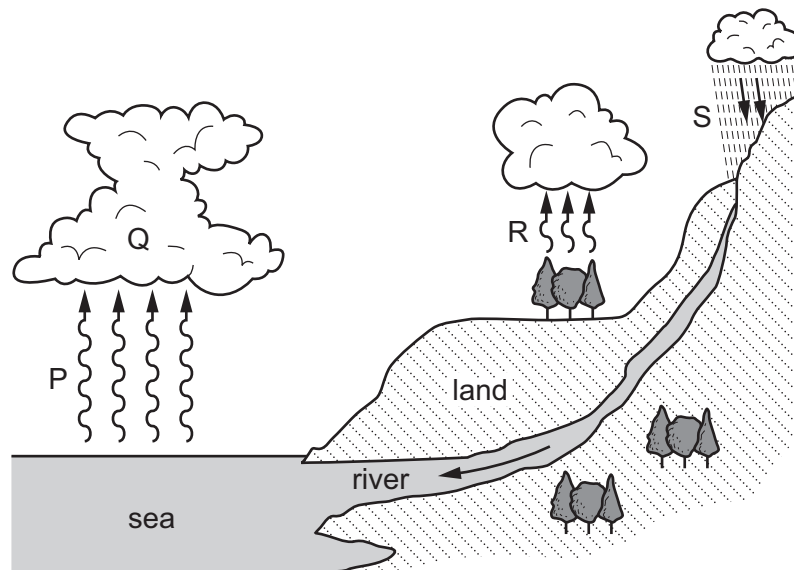
- 36** The diagram shows a food chain.

grass → grasshopper → frog → snake → hawk

What is the tertiary consumer in this food chain?

- A** frog
- B** grasshopper
- C** hawk
- D** snake

37 The diagram shows the water cycle.



The table shows the processes involved in the water cycle.

Which row is correct?

	condensation	evaporation	precipitation	transpiration
A	P	Q	R	S
B	R	S	Q	P
C	S	R	P	Q
D	Q	P	S	R

38 Which process releases carbon dioxide into the atmosphere?

- A** decomposition
- B** fossilisation
- C** photosynthesis
- D** transpiration

39 What is the useful product of anaerobic respiration in the manufacture of bread?

- A** carbon dioxide
- B** ethanol
- C** lactic acid
- D** oxygen

- 40** What are reasons why species become endangered?
- A** habitat destruction, hunting and introduced species
 - B** habitat protection, introduced species and monitoring
 - C** habitat destruction, hunting and captive breeding programmes
 - D** hunting, pollution and seed banks

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Cambridge IGCSE™ (9–1)

BIOLOGY

0970/12

Paper 1 Multiple Choice (Core)

October/November 2021

MARK SCHEME

Maximum Mark: 40

Published

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Question	Answer	Marks
1	B	1
2	C	1
3	B	1
4	B	1
5	C	1
6	B	1
7	D	1
8	A	1
9	C	1
10	A	1
11	C	1
12	A	1
13	C	1
14	C	1
15	C	1
16	B	1
17	A	1
18	C	1
19	D	1
20	B	1
21	B	1
22	B	1
23	C	1
24	D	1
25	A	1
26	C	1
27	C	1
28	A	1

Question	Answer	Marks
29	B	1
30	D	1
31	B	1
32	D	1
33	C	1
34	B	1
35	C	1
36	D	1
37	D	1
38	A	1
39	A	1
40	A	1



Cambridge IGCSE™ (9–1)

BIOLOGY

0970/11

Paper 1 Multiple Choice (Core)

May/June 2022

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

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INFORMATION

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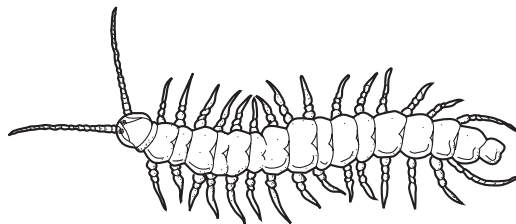
- 1 All living organisms release energy from nutrient molecules within their cells.

What is the name of this characteristic?

- A growth
 - B nutrition
 - C respiration
 - D sensitivity
- 2 The horse, *Equus ferus*, and the donkey, *Equus asinus*, are able to interbreed. The offspring they produce is called a mule.

Which statement is correct?

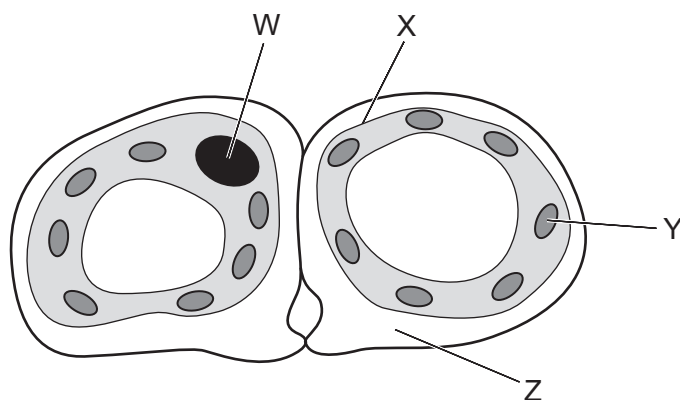
- A The horse and the donkey are the same genus; the mule is infertile.
 - B The horse and the donkey are the same species; the mule is infertile.
 - C The horse and the donkey are the same genus; the mule is fertile.
 - D The horse and the donkey are the same species; the mule is fertile.
- 3 The diagram shows an organism.



Which type of arthropod is this organism?

- A myriapod
- B insect
- C crustacean
- D arachnid

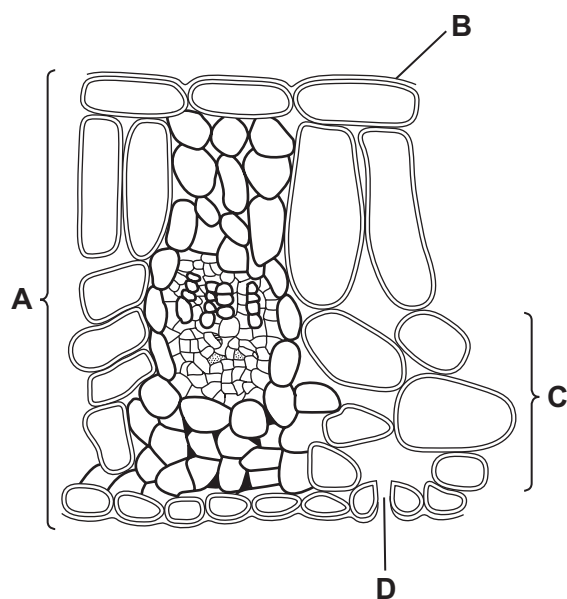
- 4 The diagram shows a cross-section through two guard cells of a leaf.



Which labelled structures would also be found in an animal cell?

- A** W and X **B** X and Y **C** Y and Z **D** Z and W
- 5 The diagram shows a cross-section through a leaf.

Which label shows a tissue?



- 6 A student found four different worms in a sample of soil and drew diagrams of them. The diagrams were drawn with different magnifications.

Which worm was the longest?

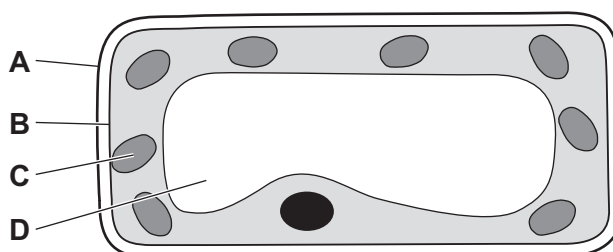
	length of diagram / mm	magnification
A	60	$\times 3$
B	70	$\times 1$
C	100	$\times 2$
D	120	$\times 5$

- 7 Which statement describes the net movement of particles during diffusion?

- A** from higher to lower concentration down a concentration gradient
- B** from higher to lower concentration against a concentration gradient
- C** from lower to higher concentration down a concentration gradient
- D** from lower to higher concentration against a concentration gradient

- 8 The diagram shows a section through a mesophyll cell of a leaf.

Which part is partially permeable?



- 9 What are the smaller basic units of starch and glycogen molecules?

	starch	glycogen
A	amino acids	fatty acids and glycerol
B	amino acids	glucose
C	glucose	fatty acids and glycerol
D	glucose	glucose

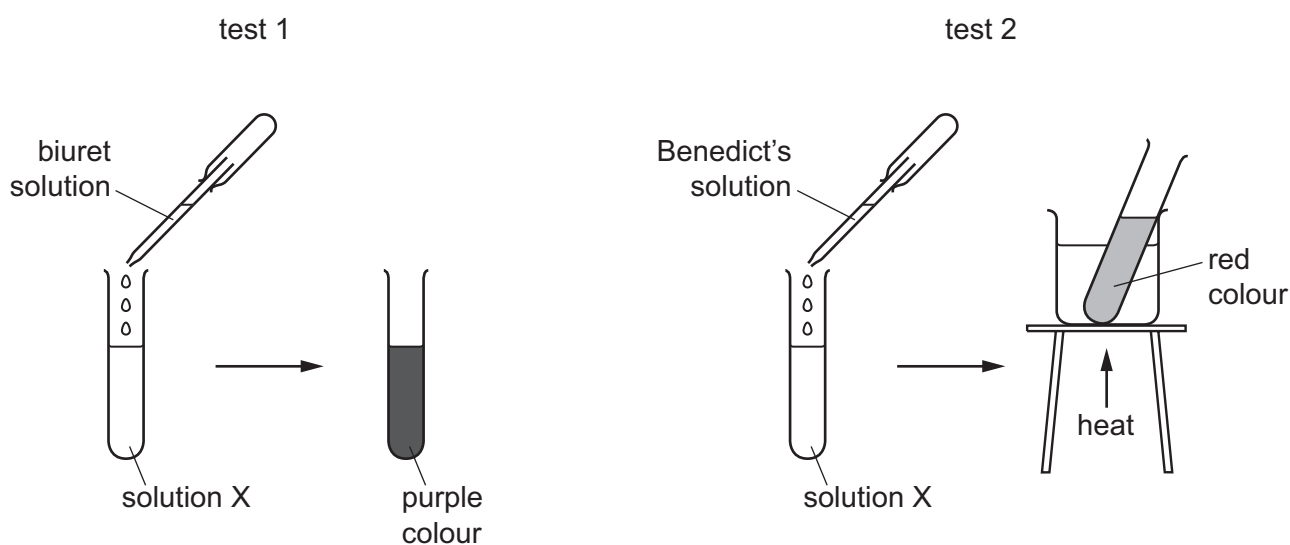
10 Some terms used to describe enzyme-controlled reactions are listed.

- 1 catalyst
- 2 product
- 3 protein
- 4 substrate

Which terms describe an enzyme?

- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

11 The diagram shows two food tests being carried out on solution X.



Which nutrients are present in solution X?

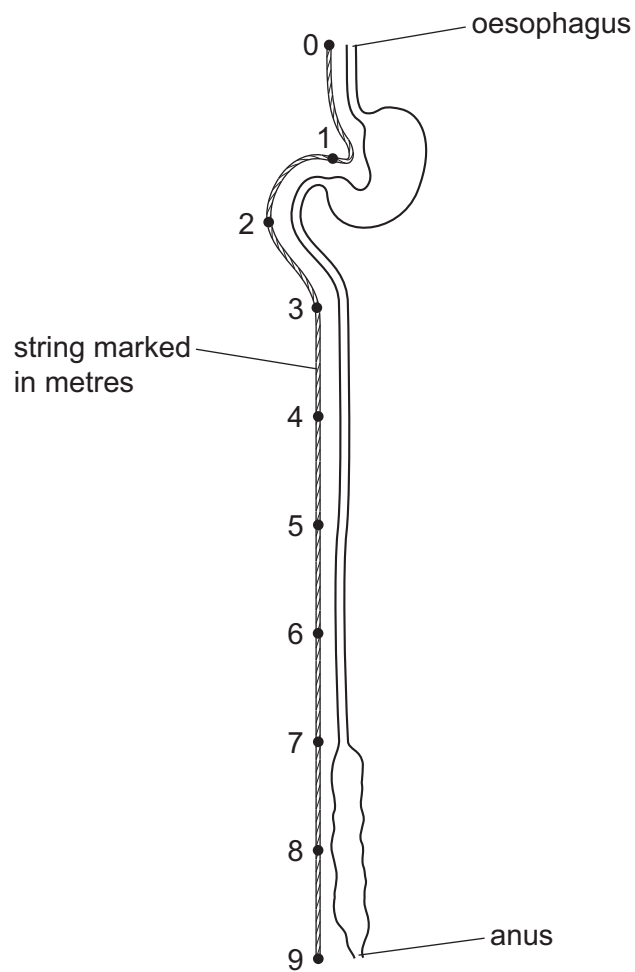
- A** protein and starch
B protein and sugar
C starch and fat
D starch and sugar

12 Which chemical is a product of photosynthesis that moves out of a green leaf through its stomata?

- A** carbon dioxide
B glucose
C oxygen
D water

- 13** Which tissues are found in vascular bundles?
- A** palisade mesophyll and phloem
 - B** palisade mesophyll and spongy mesophyll
 - C** phloem and xylem
 - D** spongy mesophyll and xylem
- 14** What is the dietary importance of fibre in humans?
- A** for growth and repair of cells
 - B** to help food move through the alimentary canal
 - C** to maintain strong bones and teeth
 - D** to provide energy
- 15** In which part of the alimentary canal does ingestion take place?
- A** anus
 - B** liver
 - C** mouth
 - D** stomach

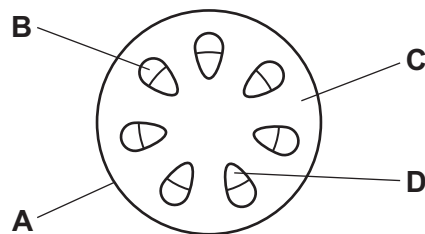
- 16** The diagram shows the human alimentary canal, with a string marked in metres beside it.



How long is the small intestine?

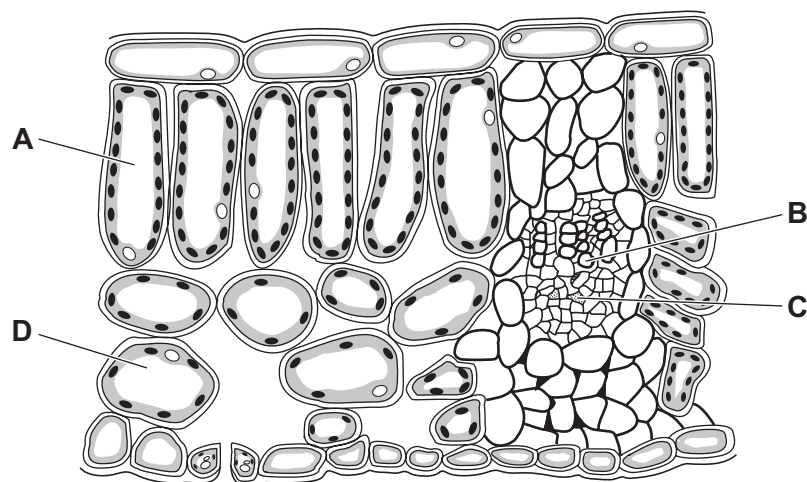
- A** 2m **B** 6m **C** 8m **D** 9m
- 17** A non-woody plant stem is placed in water containing a red dye. It is left for one day and then a section of the stem is examined using a microscope.

Which tissue is stained red?

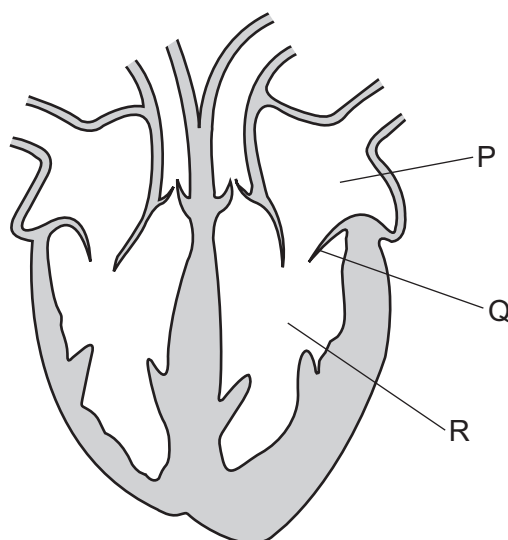


18 The diagram shows a cross-section through a leaf.

From which cell will most water evaporate during transpiration?



19 The diagram shows a cross-section through the heart of a mammal.



Which row correctly identifies P, Q and R?

	P	Q	R
A	valve	right atrium	right ventricle
B	left atrium	valve	left ventricle
C	left ventricle	left atrium	valve
D	right atrium	valve	right ventricle

20 Which process is slower than normal in a person with very few platelets?

- A** antibody formation
- B** blood clotting
- C** oxygen transport
- D** phagocytosis

21 The body has several defence mechanisms to protect against disease.

Which defence mechanism is a chemical barrier?

- A** hairs in the nose
- B** mucus
- C** skin
- D** white blood cells

22 A person lives in a climate where the air is very dry.

Which row correctly compares inspired air with expired air for this person?

	inspired air	expired air
A	higher carbon dioxide concentration	higher oxygen concentration
B	lower water vapour concentration	higher carbon dioxide concentration
C	lower oxygen concentration	higher water vapour concentration
D	lower carbon dioxide concentration	lower water vapour concentration

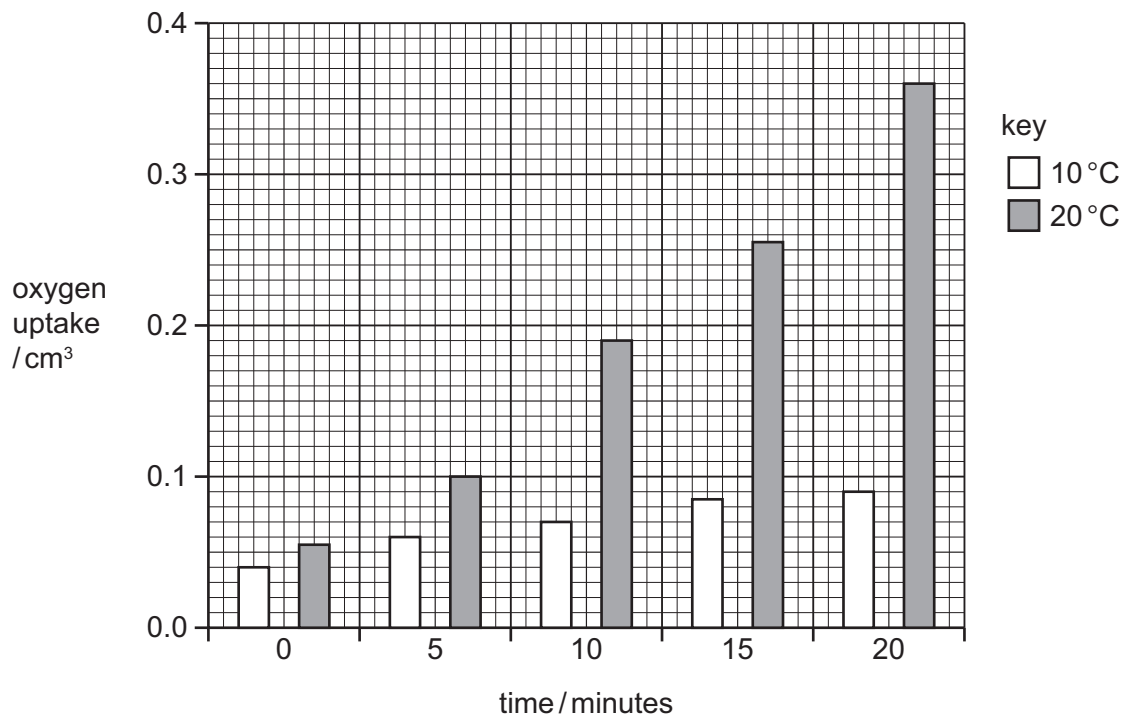
23 Substances involved in aerobic respiration are listed.

- 1 carbon dioxide
- 2 glucose
- 3 oxygen
- 4 water

Which substances are used during aerobic respiration?

- A** 1 and 3
- B** 1 and 4
- C** 2 and 3
- D** 2 and 4

24 The diagram shows the oxygen uptake by germinating seeds at two different temperatures.



At which time is the oxygen uptake at 20 °C four times greater than at 10 °C?

- A** 5 minutes
- B** 10 minutes
- C** 15 minutes
- D** 20 minutes

25 A person produced 1.75 dm³ of urine on a cool day.

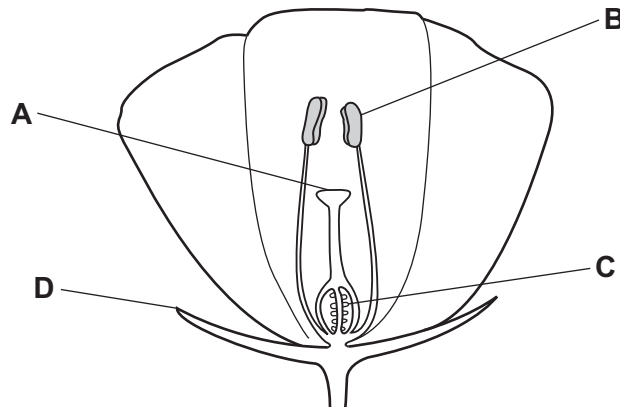
How will the volume and concentration of urine change on a hot day, if fluid intake is the same as on the cool day?

	volume	concentration
A	increases	decreases
B	increases	increases
C	decreases	decreases
D	decreases	increases

- 26 Which hormone triggers the 'fight or flight' response?
- A testosterone
 - B oestrogen
 - C adrenaline
 - D insulin
- 27 Which organ detects and coordinates the response to changes in internal body temperature?
- A brain
 - B heart
 - C liver
 - D skin
- 28 Which statement describes the plant response known as phototropism?
- A All parts of a plant grow towards light.
 - B All parts of a plant grow away from light.
 - C Plant shoots grow towards light.
 - D Plant roots grow towards light.
- 29 Why is excessive consumption of alcohol discouraged before someone operates machinery or drives a vehicle?
- A Alcohol decreases reaction times.
 - B Alcohol increases reaction times.
 - C Alcohol is a stimulant drug.
 - D Alcohol is an addictive drug.
- 30 What is an example of sexual reproduction?
- A one bacterial cell dividing to produce two daughter bacterial cells
 - B one banana plant being divided into two banana plants
 - C one pollen grain nucleus fusing with one ovule nucleus in a flower
 - D one yeast cell producing buds which separate to become new yeast cells

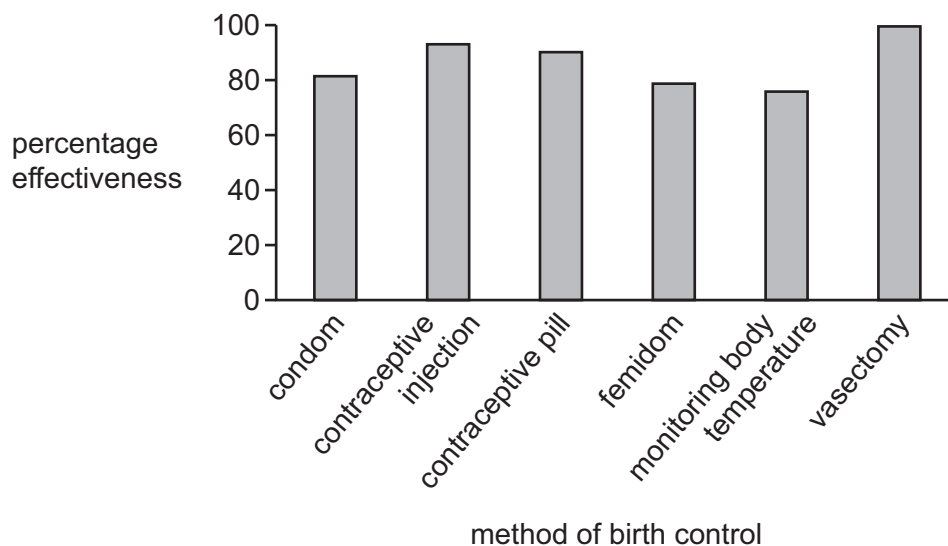
31 The diagram shows a cross-section of an insect-pollinated flower.

Which label is the stigma?



32 The effectiveness of different methods of birth control were studied in one country.

The graph shows the results of the study.



Which type of birth control is the most effective in this study?

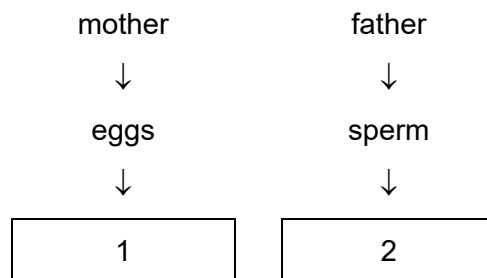
- A barrier
- B chemical
- C natural
- D surgical

- 33** Beak shape in birds is controlled by genes passed from the parent birds to their offspring.

What is this an example of?

- A** selection
- B** fertilisation
- C** mitosis
- D** inheritance

- 34** A student used a diagram to explain the inheritance of sex in humans.



Which statements should she use to complete boxes 1 and 2?

	box 1	box 2
A	all have an X chromosome	half have an X chromosome and half have a Y chromosome
B	all have a Y chromosome	half have an X chromosome and half have a Y chromosome
C	half have an X chromosome and half have a Y chromosome	all have an X chromosome
D	half have an X chromosome and half have a Y chromosome	all have a Y chromosome

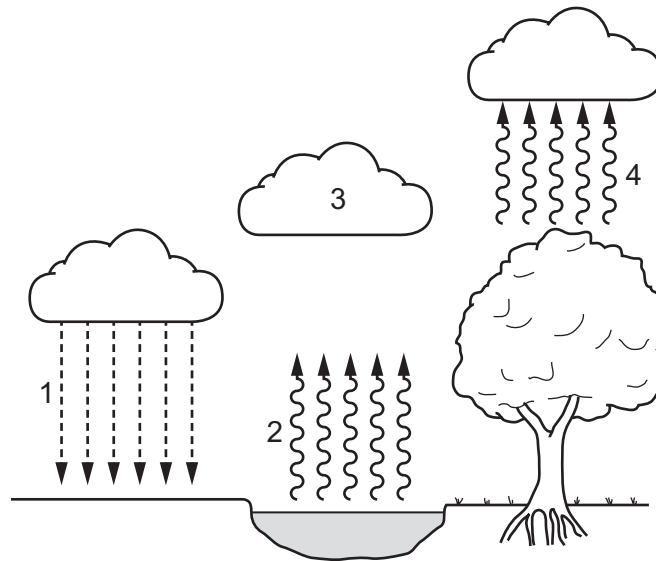
- 35** Which statement about selective breeding is correct?

- A** It does not involve humans.
- B** It involves a struggle for survival.
- C** It always involves only one parent.
- D** It involves parents that possess desirable features.

36 Which process uses the principal source of energy input to biological systems?

- A** ingestion
- B** decomposition
- C** photosynthesis
- D** respiration

37 The diagram shows the water cycle. Four different processes are numbered.



Which row identifies the processes correctly?

	condensation	evaporation	precipitation	transpiration
A	1	4	3	2
B	2	1	4	3
C	3	2	1	4
D	4	3	2	1

38 A herd of red deer live in a forest that contains snakes and a large variety of birds.

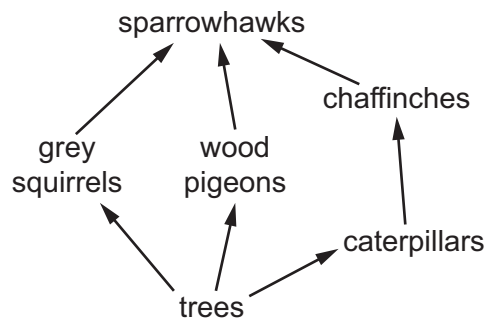
Which group of organisms is an example of a population?

- A** all the animals in the forest
- B** all the red deer in the forest
- C** all the organisms in the forest
- D** all the plants in the forest

39 Which process makes use of a genetically engineered organism?

- A using bacteria to produce insulin
- B using enzymes in biological washing powders
- C using pectinase in fruit juice production
- D using yeast to produce ethanol

40 The food web shows the feeding relationships in a woodland.



If all the chaffinches in the food web die, which effect would this have?

- A The amount of damage to trees will increase.
- B The food supply for grey squirrels will increase.
- C The number of wood pigeons will increase.
- D The population of caterpillars will decrease.

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Cambridge IGCSE™ (9–1)

BIOLOGY (9–1)

0970/11

Paper 1 Multiple Choice (Core)

May/June 2022

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2022 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	C	1
2	A	1
3	A	1
4	A	1
5	C	1
6	B	1
7	A	1
8	B	1
9	D	1
10	A	1
11	B	1
12	C	1
13	C	1
14	B	1
15	C	1
16	B	1
17	D	1
18	D	1
19	B	1
20	B	1
21	B	1
22	B	1
23	C	1
24	D	1
25	D	1
26	C	1
27	A	1
28	C	1

Question	Answer	Marks
29	B	1
30	C	1
31	A	1
32	D	1
33	D	1
34	A	1
35	D	1
36	C	1
37	C	1
38	B	1
39	A	1
40	A	1



Cambridge IGCSE™ (9–1)

BIOLOGY

0970/12

Paper 1 Multiple Choice (Core)

May/June 2022

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.



- 1 All living organisms release energy from nutrient molecules within their cells.

What is the name of this characteristic?

- A growth
- B nutrition
- C respiration
- D sensitivity

- 2 Which levels of classification are shown in the scientific name of an organism, using the binominal system?

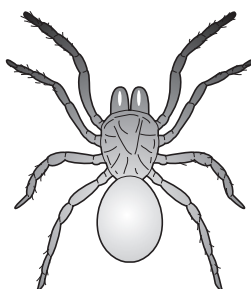
	genus	kingdom	species
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✓
D	✗	✓	✓

key

✓ = yes

✗ = no

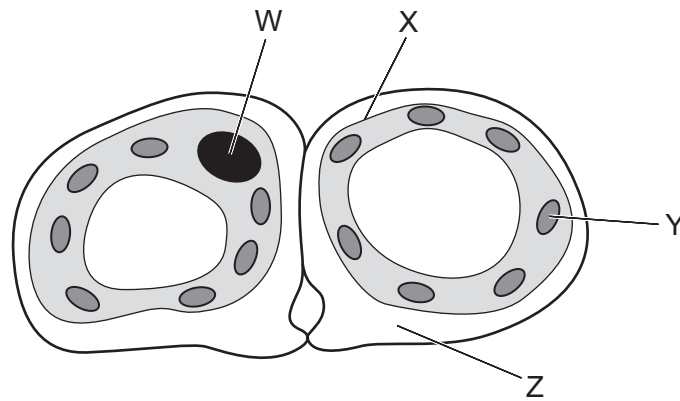
- 3 The diagram shows an arthropod that lives on land.



To which group of arthropods does this animal belong?

- A arachnids
- B crustaceans
- C insects
- D myriapods

- 4 The diagram shows a cross-section through two guard cells of a leaf.



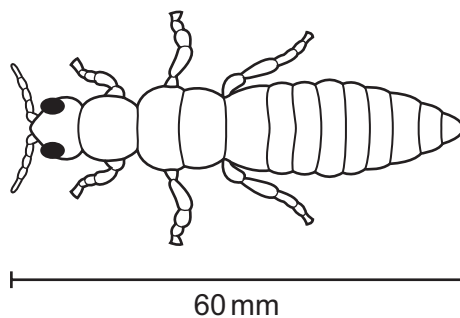
Which labelled structures would also be found in an animal cell?

- A** W and X **B** X and Y **C** Y and Z **D** Z and W
- 5 The table shows some structures found in the human body.

Which row shows the correct level of organisation, from the simplest structure to the most complex?

	simple → complex			
A	red blood cell	heart	heart muscle	circulatory system
B	heart muscle	circulatory system	red blood cell	heart
C	red blood cell	heart muscle	heart	circulatory system
D	heart muscle	red blood cell	circulatory system	heart

- 6 The diagram shows an animal. The actual length of this animal is 0.6 mm.



What is the magnification of the diagram?

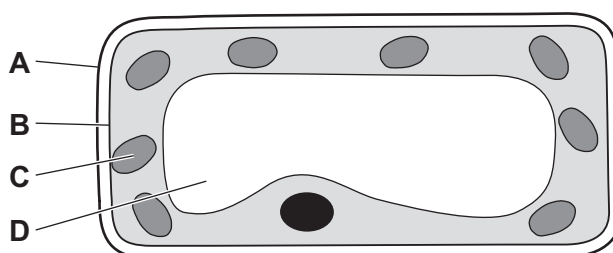
- A** $\times 10$ **B** $\times 100$ **C** $\times 1000$ **D** $\times 10\,000$

7 By which process does oxygen move from a region of higher concentration in the alveoli to a region of lower concentration in the blood?

- A osmosis
- B breathing
- C diffusion
- D active transport

8 The diagram shows a section through a mesophyll cell of a leaf.

Which part is partially permeable?



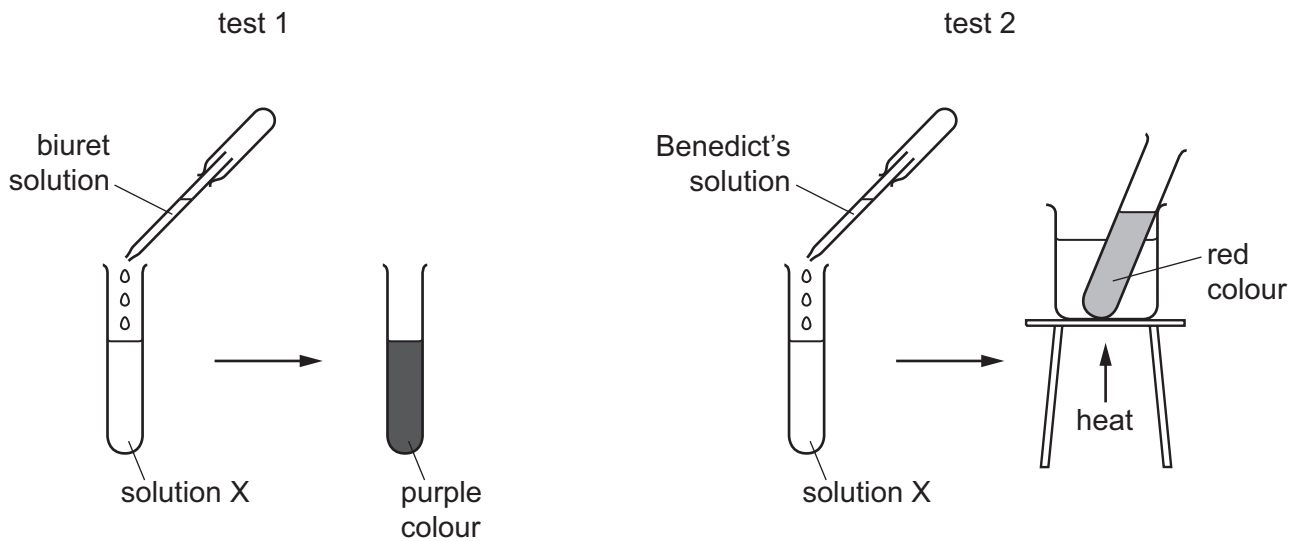
9 What are the smaller basic units of starch and glycogen molecules?

	starch	glycogen
A	amino acids	fatty acids and glycerol
B	amino acids	glucose
C	glucose	fatty acids and glycerol
D	glucose	glucose

10 Which type of molecule are enzymes made of?

- A carbohydrates
- B fats
- C proteins
- D vitamins

11 The diagram shows two food tests being carried out on solution X.



Which nutrients are present in solution X?

- A protein and starch
 - B protein and sugar
 - C starch and fat
 - D starch and sugar
- 12 Which chemical is a product of photosynthesis that moves out of a green leaf through its stomata?
- A carbon dioxide
 - B glucose
 - C oxygen
 - D water
- 13 In a leaf, water moves from the surface of a mesophyll cell and then out of the leaf into the atmosphere.

What is the correct order of the processes involved?

- A active transport → osmosis
- B diffusion → evaporation
- C evaporation → diffusion
- D osmosis → active transport

14 The table shows some different components in a balanced diet.

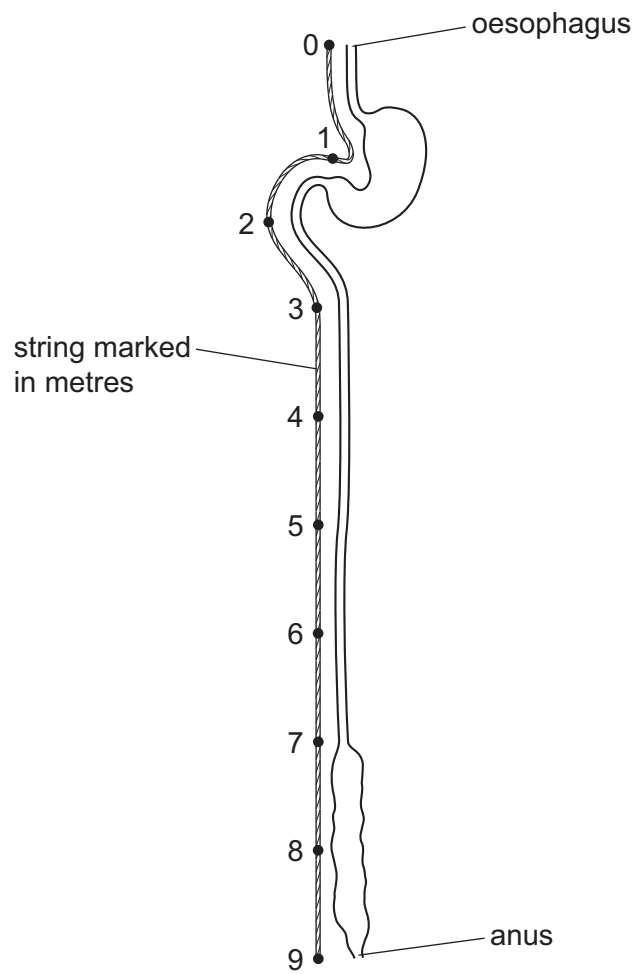
Which row correctly matches the components to the reasons why they are needed?

	vitamin D	water	fats
A	insulation	bone growth	to make haemoglobin
B	bone growth	transport	insulation
C	transport	to make haemoglobin	bone growth
D	to make haemoglobin	insulation	transport

15 Which process involves the removal of material from the rectum?

- A** egestion
- B** ingestion
- C** chemical digestion
- D** mechanical digestion

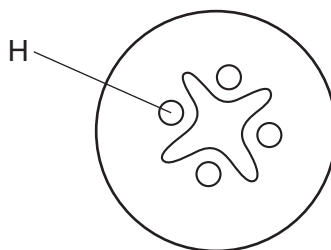
- 16 The diagram shows the human alimentary canal, with a string marked in metres beside it.



How long is the small intestine?

- A** 2m **B** 6m **C** 8m **D** 9m

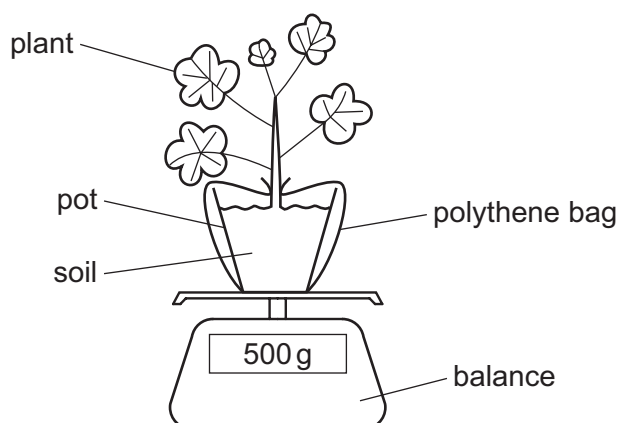
- 17 The diagram shows a cross-section of part of a plant.



Which part of the plant is shown and what is the tissue labelled H?

	plant part	tissue H
A	root	phloem
B	root	xylem
C	stem	phloem
D	stem	xylem

- 18 The apparatus shown can be used to investigate the effect of temperature and humidity on the rate of transpiration.



Which set of conditions would give the highest rate of transpiration and the greatest decrease in mass?

	temperature	humidity
A	cool	high
B	cool	low
C	warm	high
D	warm	low

19 Which row correctly links the blood vessels with their functions?

	carries blood to the right atrium from the body	carries blood from the right ventricle to the lungs	carries blood from the lungs to the left atrium	carries blood from the left ventricle to the body
A	vena cava	pulmonary artery	pulmonary vein	aorta
B	aorta	pulmonary vein	pulmonary artery	vena cava
C	aorta	pulmonary artery	pulmonary vein	vena cava
D	vena cava	pulmonary vein	pulmonary artery	aorta

20 Which process is slower than normal in a person with very few platelets?

- A** antibody formation
- B** blood clotting
- C** oxygen transport
- D** phagocytosis

21 Which disease is caused by a pathogen?

- A** cholera
- B** coronary heart disease
- C** lung cancer
- D** scurvy

22 The photographs show a positive test for the presence of carbon dioxide gas.



test-tube at the start



test-tube at the end

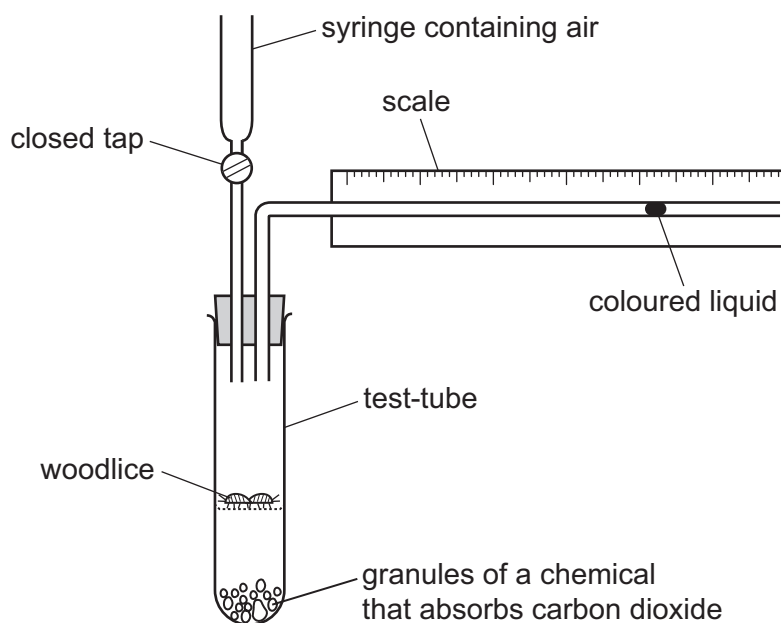
Which reagent is being used in the test-tube?

- A** Benedict's
- B** biuret
- C** DCPIP
- D** limewater

23 What is the word equation for anaerobic respiration in yeast?

- A** glucose → alcohol
- B** glucose → alcohol + carbon dioxide
- C** glucose → lactic acid
- D** glucose → lactic acid + carbon dioxide

24 A student used this apparatus to investigate the rate of aerobic respiration in woodlice.



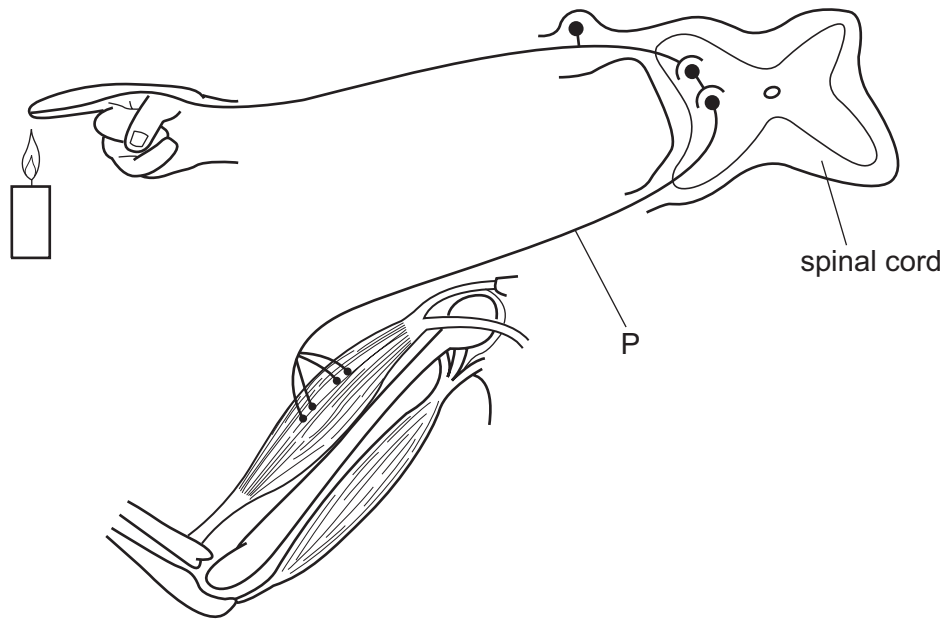
Which statement describes and explains the movement of the coloured liquid when the woodlice are respiring?

- A** The coloured liquid moves towards the test-tube because the woodlice are using carbon dioxide.
- B** The coloured liquid moves towards the test-tube because the woodlice are using oxygen.
- C** The coloured liquid moves away from the test-tube because the woodlice are using carbon dioxide.
- D** The coloured liquid moves away from the test-tube because the woodlice are using oxygen.

25 Which organ excretes urea?

- A** kidney
- B** liver
- C** lungs
- D** pancreas

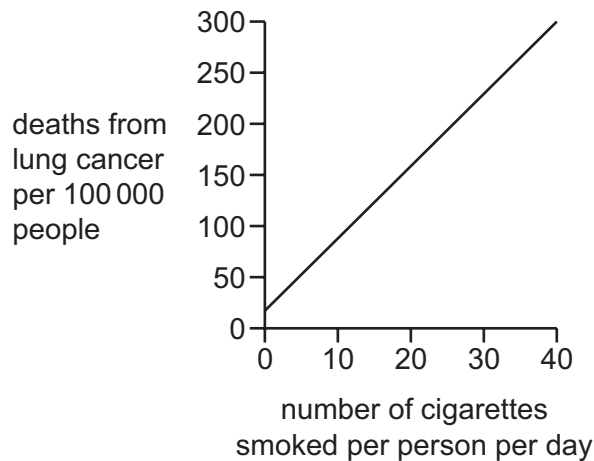
26 The diagram shows a simple reflex arc.



What is the function of the part labelled P?

- A passing impulses to the central nervous system
 - B passing impulses to the effector
 - C passing impulses to the receptor
 - D passing impulses to the sensory neurone
- 27 Which endocrine gland produces insulin?
- A adrenal gland
 - B ovary
 - C pancreas
 - D testes
- 28 Where in the body are the blood temperature receptors?
- A brain
 - B liver
 - C muscles
 - D skin

- 29** The graph shows the relationship between deaths from lung cancer and the number of cigarettes smoked per day. Cigarettes contain tobacco.



Which statement is a possible conclusion for the data shown in the graph?

- A** A person will die of lung cancer only if they smoke.
- B** A person's health is only affected if they smoke more than 10 cigarettes per day.
- C** A person will have less chance of dying of lung cancer if they smoke fewer cigarettes.
- D** If a person smokes more than 40 cigarettes per day, their risk of dying from lung cancer is the same as a person who smokes 40 cigarettes per day.

- 30** The statements refer to reproduction.

- 1 A zygote is formed.
- 2 Offspring are genetically identical to the parent.
- 3 Two gamete nuclei fuse together.

Which statements refer to sexual reproduction?

- A** 1, 2 and 3 **B** 1 and 3 only **C** 1 only **D** 2 and 3 only

- 31** What is the name of the ball of cells that forms soon after fertilisation in humans?

- A** embryo
- B** ovum
- C** placenta
- D** fetus

32 The diagram shows the inheritance of sex in humans.

		male gametes	
		X	Y
female gametes	X	XX	XY
	X	XX	XY

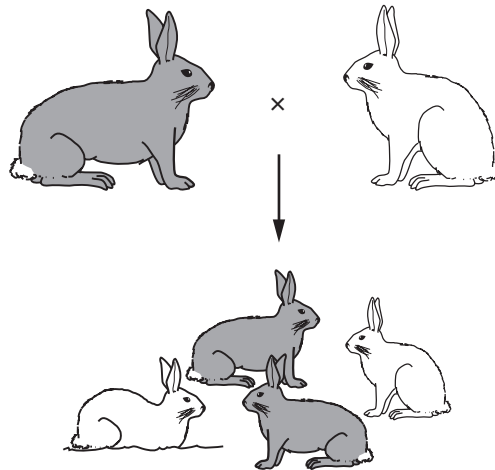
A man and a woman have one female child.

What is the chance of the couple's next child being female?

- A** 25% **B** 50% **C** 75% **D** 100%

33 In rabbits, the allele for dark fur, R, is dominant to the allele for white fur, r.

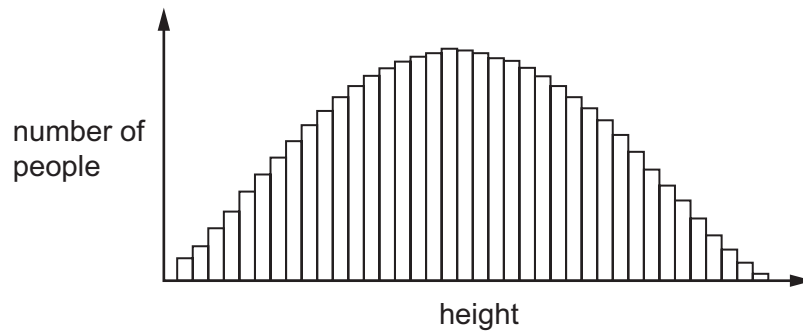
The diagram shows a cross between a rabbit with dark fur and a rabbit with white fur.



What are the genotypes of the parents?

- A** Rr and rr **B** RR and rr **C** RR and Rr **D** R and r

34 The graph shows the different heights of people in a human population.



Which row describes the variation shown by the graph?

	type of variation	has intermediate phenotypes
A	continuous	no
B	continuous	yes
C	discontinuous	no
D	discontinuous	yes

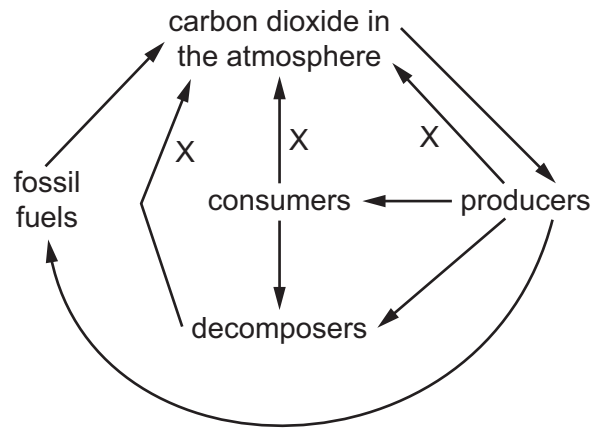
35 Which statement about selective breeding is correct?

- A** It does not involve humans.
- B** It involves a struggle for survival.
- C** It always involves only one parent.
- D** It involves parents that possess desirable features.

36 Which process uses the principal source of energy input to biological systems?

- A** ingestion
- B** decomposition
- C** photosynthesis
- D** respiration

37 The diagram shows part of the carbon cycle.



Which process is shown by the arrows labelled X?

- A** combustion
- B** fossilisation
- C** photosynthesis
- D** respiration

38 A herd of red deer live in a forest that contains snakes and a large variety of birds.

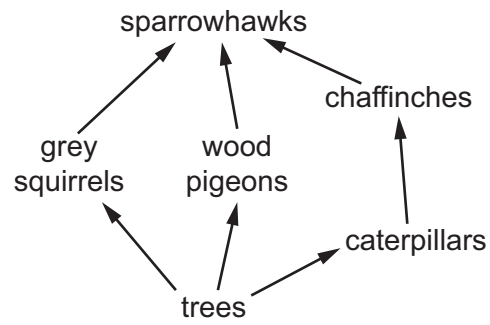
Which group of organisms is an example of a population?

- A** all the animals in the forest
- B** all the red deer in the forest
- C** all the organisms in the forest
- D** all the plants in the forest

39 Which process makes use of a genetically engineered organism?

- A** using bacteria to produce insulin
- B** using enzymes in biological washing powders
- C** using pectinase in fruit juice production
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If all the chaffinches in the food web die, which effect would this have?

- A The amount of damage to trees will increase.
- B The food supply for grey squirrels will increase.
- C The number of wood pigeons will increase.
- D The population of caterpillars will decrease.

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Cambridge IGCSE™ (9–1)

BIOLOGY (9–1)

0970/12

Paper 1 Multiple Choice (Core)

May/June 2022

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2022 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	C	1
2	C	1
3	A	1
4	A	1
5	C	1
6	B	1
7	C	1
8	B	1
9	D	1
10	C	1
11	B	1
12	C	1
13	C	1
14	B	1
15	A	1
16	B	1
17	A	1
18	D	1
19	A	1
20	B	1
21	A	1
22	D	1
23	B	1
24	B	1
25	A	1
26	B	1
27	C	1
28	A	1

Question	Answer	Marks
29	C	1
30	B	1
31	A	1
32	B	1
33	A	1
34	B	1
35	D	1
36	C	1
37	D	1
38	B	1
39	A	1
40	A	1



Cambridge IGCSE™ (9–1)

BIOLOGY

0970/12

Paper 1 Multiple Choice (Core)

October/November 2022

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.



1 Which process provides an organism with the raw materials needed for tissue repair?

- A excretion
- B growth
- C nutrition
- D respiration

2 Which name is given to a group of individuals that can reproduce to produce fertile offspring?

- A a genus
- B a kingdom
- C a species
- D an organ system

3 An animal has four legs, hair and a tail.

To which group of vertebrates does it belong?

- A amphibians
- B birds
- C mammals
- D reptiles

4 Root hair cells are found on plant roots.

Which feature is present in a root hair cell but **not** in a sperm cell?

- A cell membrane
- B cell wall
- C chloroplasts
- D cytoplasm

5 The cells listed have specialised structures that allow them to carry out their functions.

- 1 ciliated cell
- 2 nerve cell
- 3 root cortex cell
- 4 sperm cell

Which cells have structures that can move?

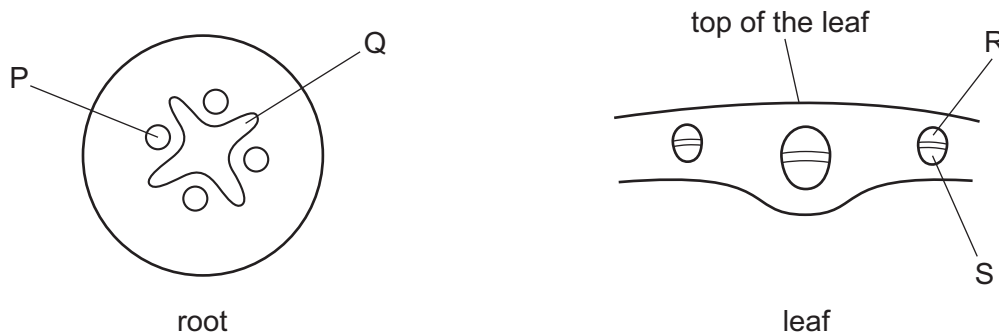
- A 1 and 2
- B 1 and 4
- C 2 and 3
- D 3 and 4

- 6 A cell was viewed under a microscope. The actual length of the cell is 0.025 mm. The magnification of the image is $\times 400$.

What is the length of the cell in the image?

- A 10 mm B 100 mm C 1600 mm D 16 000 mm

- 7 The roots of a plant were placed in a solution of red dye. After 24 hours, a section of root and a section of leaf were cut from the plant.



In which tissues will the red dye be visible?

- A P and R B P and S C Q and R D Q and S

- 8 Which row shows features of osmosis in cells?

	requires a cell membrane	requires a cell wall	water can move into the cell	water can move out of the cell	key ✓ = yes ✗ = no
A	✓	✓	✓	✓	
B	✓	✗	✓	✓	
C	✓	✓	✗	✓	
D	✗	✓	✓	✗	

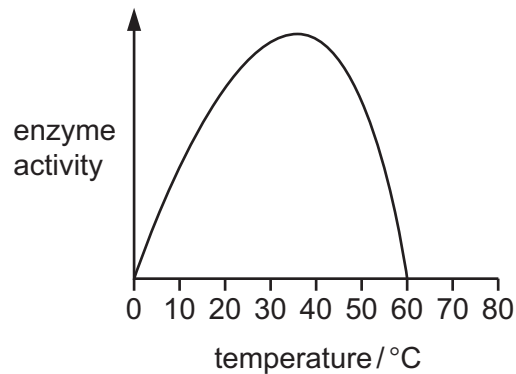
- 9 Which chemical is used to test for the presence of protein in a food sample?

- A Benedict's solution
B biuret solution
C DCPIP
D iodine solution

10 Which statement about enzymes is correct?

- A They are made of carbohydrates.
- B Their activity is unaffected by pH.
- C They are used up during the reaction.
- D They have a complementary shape to their substrate.

11 The graph shows the effect of temperature on the activity of an enzyme.



At which temperature is the enzyme most active?

- A 15°C B 25°C C 35°C D 60°C

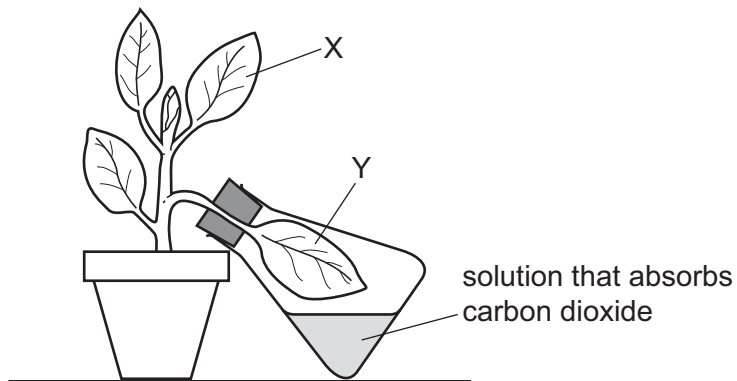
12 The substances listed are found in the leaf of a plant.

Which substance is obtained from the soil?

- A carbon dioxide
- B chlorophyll
- C glucose
- D mineral ions

- 13** The diagram shows an experiment to find out if carbon dioxide is necessary for photosynthesis.

The plant is kept in the dark for 48 hours to remove any starch. Then it is set up in the light, as shown.

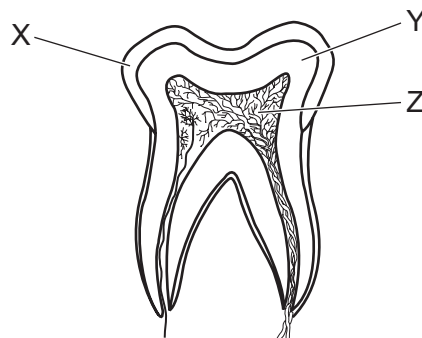


After 10 hours, leaves X and Y were tested for starch.

What is the expected result?

	presence of starch		
	in leaf X	in leaf Y	
A	✓	✓	key ✓ = present x = absent
B	x	x	
C	✓	x	
D	x	✓	

- 14** The diagram shows a cross-section of a human tooth.



Which row identifies the labels in the diagram?

	X	Y	Z
A	dentine	enamel	pulp cavity
B	dentine	pulp cavity	enamel
C	enamel	pulp cavity	dentine
D	enamel	dentine	pulp cavity

15 Which statement is correct?

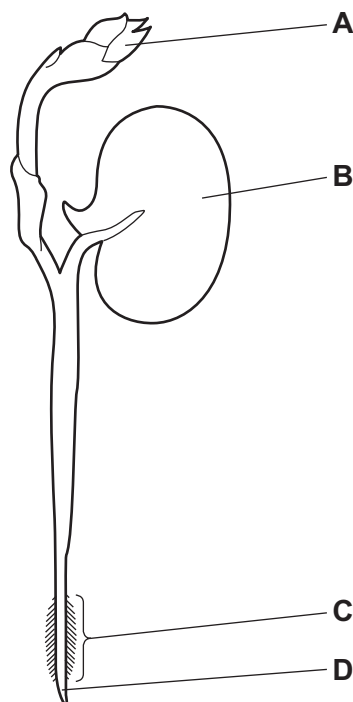
- A Amylase breaks down glucose to starch.
- B Amylase is secreted into the mouth and small intestine.
- C Lipase breaks down glycerol to fatty acids.
- D Protease is secreted into the oesophagus and stomach.

16 Where is most water absorbed in the alimentary canal?

- A large intestine
- B oesophagus
- C small intestine
- D stomach

17 The diagram shows a bean seedling soon after it has germinated.

Where is most water absorbed?

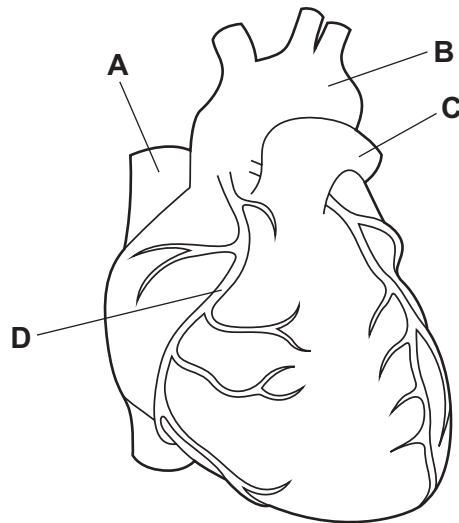


18 Which process occurs during transpiration?

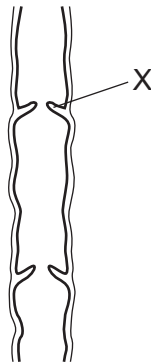
- A evaporation of water from the xylem
- B loss of water by osmosis from the guard cells
- C movement of water vapour through the spongy mesophyll by active transport
- D movement of water vapour through the stomata by diffusion

19 The diagram shows the outside of a human heart.

Which structure is a coronary artery?



20 The diagram shows a section of a human vein.



What is the function of the part labelled X?

- A to make sure the blood flows to the heart
- B to make sure the blood flows to the kidneys
- C to make sure the blood flows to the brain
- D to make sure the blood flows to the lungs

21 *Campylobacter* is a bacterium that can cause food poisoning.

Which word describes *Campylobacter*?

- A antibody
- B disease
- C pathogen
- D symptom

22 What is the approximate percentage of oxygen in expired air?

- A** 0.04% **B** 4% **C** 16% **D** 21%

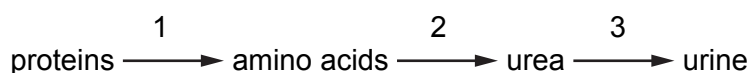
23 Which process could continue without energy from respiration?

- A** active transport
B growth
C osmosis
D protein synthesis

24 Which row shows the products of anaerobic respiration in humans and yeast?

	products in humans	products in yeast
A	lactic acid only	alcohol only
B	lactic acid only	alcohol and carbon dioxide
C	lactic acid and carbon dioxide	alcohol only
D	lactic acid and carbon dioxide	alcohol and carbon dioxide

25 A person eats foods containing proteins. Some of the amino acids in the proteins are converted to urea and excreted from the body.



Which row identifies the sites of processes 1, 2 and 3?

	site of process 1	site of process 2	site of process 3
A	stomach	liver	kidney
B	liver	kidney	bladder
C	stomach	kidney	liver
D	liver	stomach	kidney

26 Which part of the eye refracts light?

- A** cornea
B iris
C pupil
D retina

27 Which hormone causes an increase in breathing rate, an increase in heart rate and the widening of pupils?

- A** insulin
- B** adrenaline
- C** oestrogen
- D** testosterone

28 Which row about tropic responses is correct?

	gravitropism	phototropism
A	root grows away from gravity	shoot grows away from light source
B	root grows away from gravity	shoot grows towards light source
C	root grows towards gravity	shoot grows away from light source
D	root grows towards gravity	shoot grows towards light source

29 What is the definition of a drug?

- A** a substance produced by a gland that is carried in the blood and alters the activity of target organs
- B** a substance produced by white blood cells that is carried in the blood and destroys bacteria and viruses
- C** a substance taken into the body that modifies or affects chemical reactions carried out in the body
- D** a substance that increases the rate of chemical reactions in the body and is not changed by the reaction

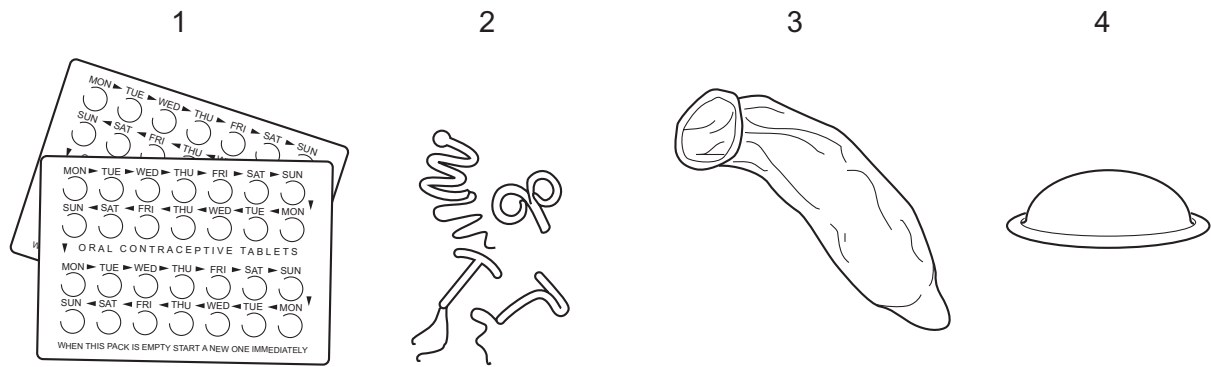
30 What occurs during fertilisation?

- A** fusion of two gamete nuclei
- B** fusion of two zygote nuclei
- C** splitting of a gamete nucleus
- D** splitting of a zygote nucleus

31 Which hormone causes the development of secondary sexual characteristics in human males?

- A** adrenaline
- B** insulin
- C** progesterone
- D** testosterone

32 The diagrams show four methods of birth control.



not to scale

What are barrier methods of birth control?

	1	2	3	4
A	✓	✓	✗	✗
B	✗	✓	✓	✗
C	✗	✗	✓	✓
D	✗	✓	✗	✓

key

✓ = yes

✗ = no

33 In one species of plant, the allele for red-coloured fruit is dominant and is represented by the letter R. The allele for white-coloured fruit is recessive and is represented by the letter r.

Two plants that are heterozygous for fruit colour are crossed.

What are the possible genotypes of the offspring plants from this cross?

A RR, Rr and rr

B all Rr

C all rr

D Rr and rr only

34 Which phenotype shows discontinuous variation in humans?

A foot length

B height

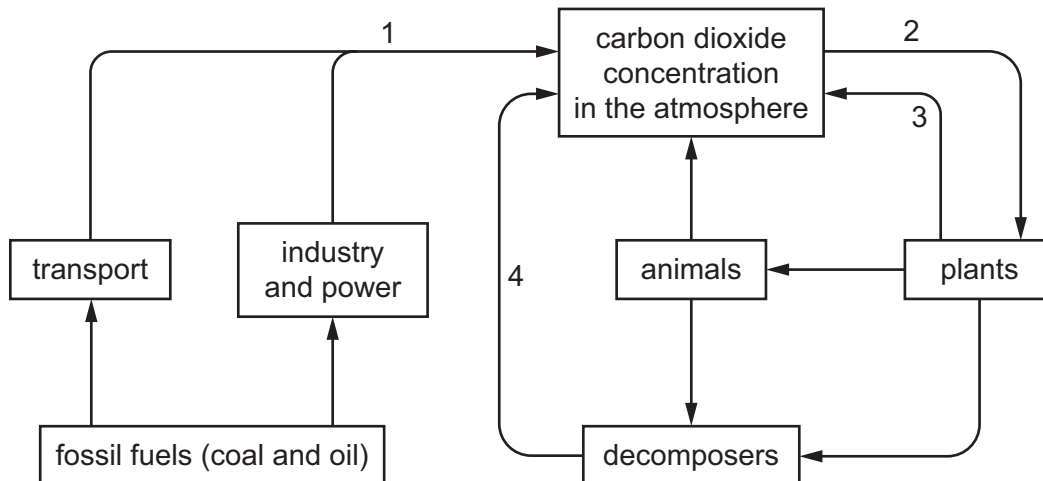
C sex

D weight

35 What can increase the genetic variation in a species?

- A growth
- B malnutrition
- C mitosis
- D mutation

36 The diagram shows part of the carbon cycle.



Which processes reduce or increase carbon dioxide concentration in the atmosphere?

	reduces	increases
A	1	4
B	2	3
C	3	2
D	4	1

37 The diagram shows a food chain.

rose bush → aphids → ladybirds → spiders

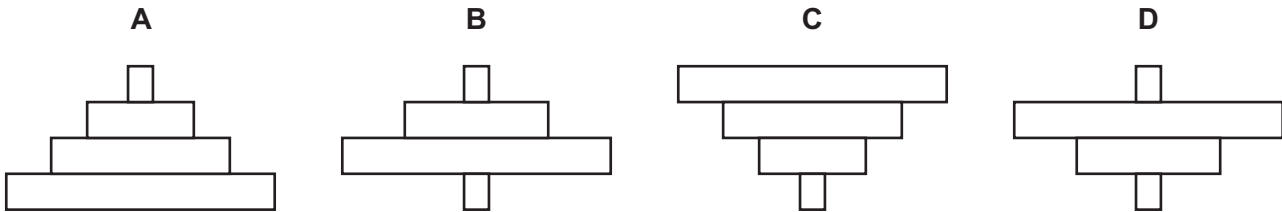
A disease reduced the number of ladybirds.

What would be the likely effect of this on the numbers of aphids and spiders?

	aphids	spiders
A	decrease	decrease
B	decrease	increase
C	increase	decrease
D	increase	increase

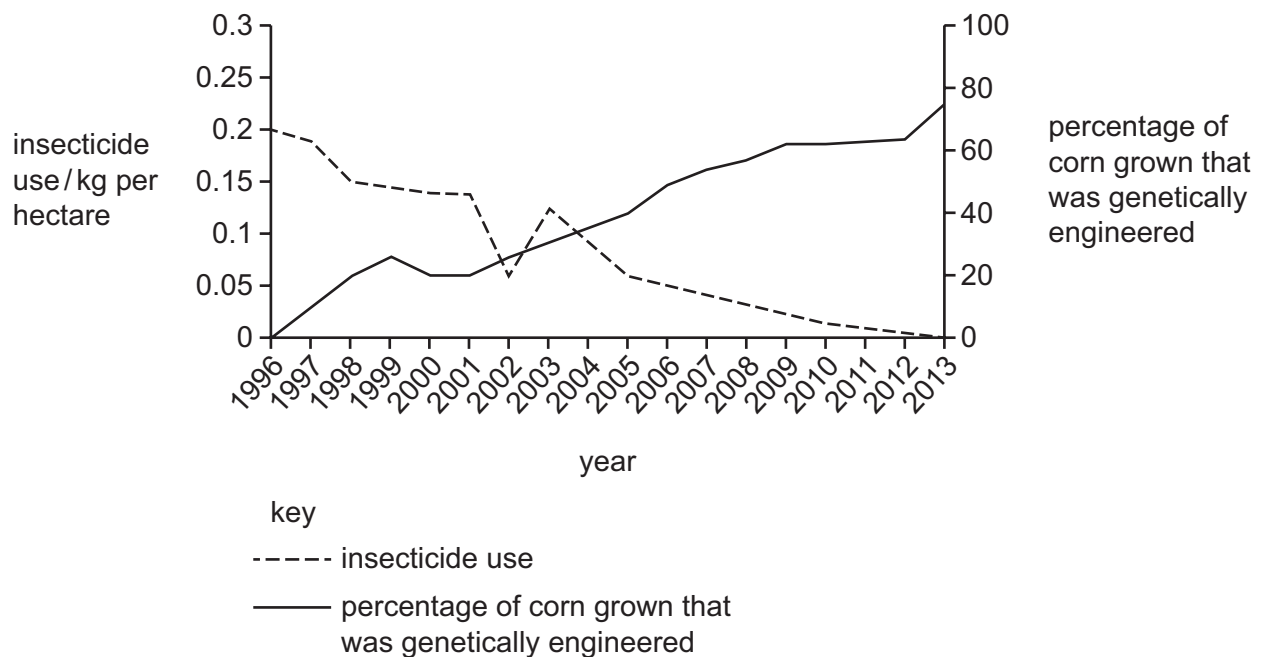
38 The block at the base of each pyramid represents the number of producers.

Which pyramid of numbers has more tertiary consumers than secondary consumers?



39 Corn that has been genetically engineered to be resistant to pests was developed in the 1990s.

The graph shows the change in the percentage of corn grown that was genetically engineered and the change in the mass of chemical insecticides used.



Which statement describes the data shown in the graph between 1996 and 2013?

- A** The percentage of corn grown that was genetically engineered increased and the use of insecticides decreased.
- B** The percentage of corn grown that was genetically engineered increased and the use of insecticides increased.
- C** The percentage of corn grown that was genetically engineered decreased and the use of insecticides decreased.
- D** The percentage of corn grown that was genetically engineered decreased and the use of insecticides increased.

40 What could be undesirable effects of deforestation?

- 1 extinction of species
- 2 increased flooding
- 3 reduction of atmospheric carbon dioxide

A 1 and 2 only **B** 1 and 3 only **C** 1, 2 and 3 **D** 2 and 3 only

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Cambridge IGCSE™ (9–1)

BIOLOGY

0970/12

Paper 1 Multiple Choice (Core)

October/November 2022

MARK SCHEME

Maximum Mark: 40

Published

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Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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Question	Answer	Marks
1	C	1
2	C	1
3	C	1
4	B	1
5	B	1
6	A	1
7	C	1
8	B	1
9	B	1
10	D	1
11	C	1
12	D	1
13	C	1
14	D	1
15	B	1
16	C	1
17	C	1
18	D	1
19	D	1
20	A	1
21	C	1
22	C	1
23	C	1
24	B	1
25	A	1
26	A	1
27	B	1
28	D	1

Question	Answer	Marks
29	C	1
30	A	1
31	D	1
32	C	1
33	A	1
34	C	1
35	D	1
36	B	1
37	C	1
38	C	1
39	A	1
40	A	1



Cambridge IGCSE™ (9–1)

BIOLOGY

0970/11

Paper 1 Multiple Choice (Core)

May/June 2023

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
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- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.



1 What are characteristics of all organisms?

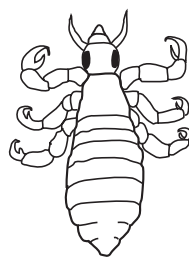
- A egestion and excretion
- B egestion and nutrition
- C excretion and nutrition
- D excretion and photosynthesis

2 The scientific name for the golden eagle is *Aquila chrysaetos*.

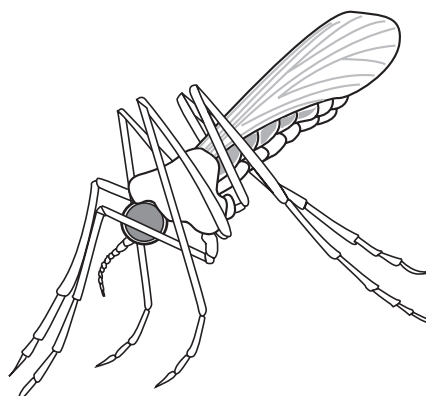
What is the genus of the golden eagle?

- A *Aquila*
- B *chrysaetos*
- C eagle
- D golden

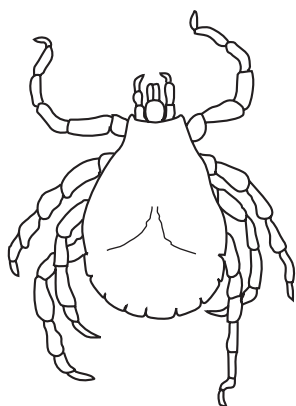
3 The diagram shows four arthropods.



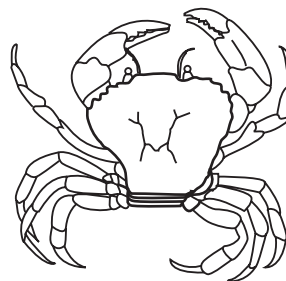
Pediculus ×20



Anopheles ×10



Dermacentor ×7



Carcinus ×0.5

How many of these arthropods are insects?

- A 1
- B 2
- C 3
- D 4

- 4 Which row shows structures that are present in both root hair cells and palisade mesophyll cells?

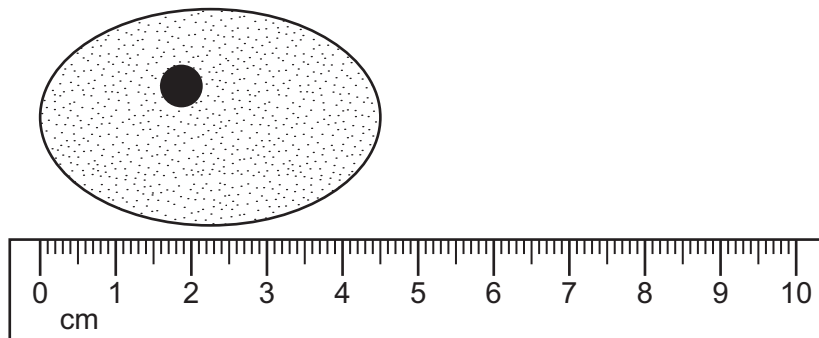
	cell wall	chloroplasts	cytoplasm	vacuole
A	✓	✓	✓	✗
B	✗	✓	✓	✓
C	✓	✗	✓	✓
D	✓	✓	✗	✓

key

✓ = present

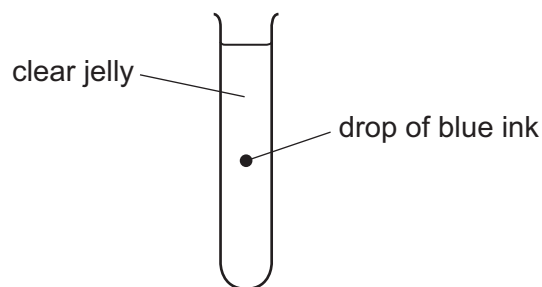
✗ = absent

- 5 The diagram shows a cell at $\times 1500$ magnification.



What is the actual size of the cell?

- A** 0.003 mm **B** 0.03 mm **C** 33 mm **D** 333 mm
- 6 The diagram shows a test-tube containing clear jelly. A drop of blue ink is injected into the middle of the jelly.

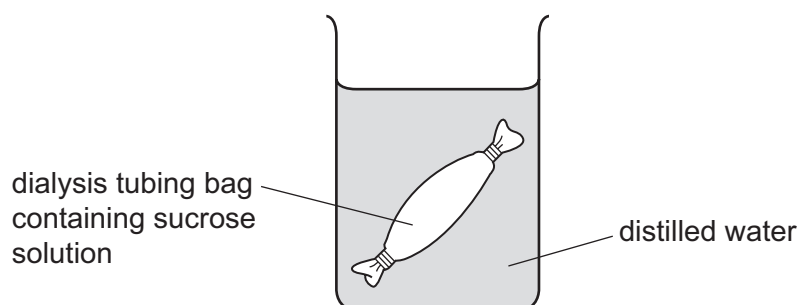


The blue colour of the ink spreads throughout the jelly.

By which process does the blue ink spread through the jelly?

- A** active transport
B catalysis
C diffusion
D osmosis

- 7 The diagram shows some of the apparatus used in an osmosis investigation.



In this investigation a dialysis tubing bag was filled with sucrose solution, sealed and weighed.

The dialysis tubing bag was then immersed in distilled water for one hour.

After one hour the dialysis tubing bag was removed from the beaker, the surface was dried and the bag was reweighed.

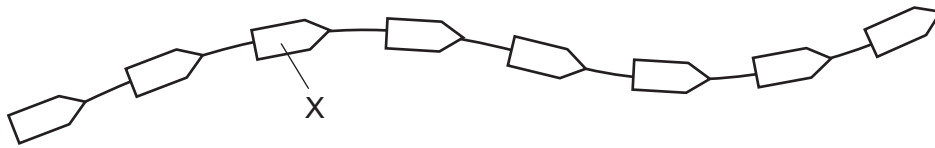
Which row explains what will happen during the investigation?

	mass of the dialysis tubing bag at the end of the investigation	net movement of sucrose molecules	net movement of water molecules
A	decreased	none	out of the bag
B	decreased	into the bag	out of the bag
C	increased	none	into the bag
D	increased	out of the bag	into the bag

- 8 Which row describes active transport of ions?

	direction of movement of ions	requires energy from respiration
A	from high concentration to low concentration	yes
B	from low concentration to high concentration	no
C	from high concentration to low concentration	no
D	from low concentration to high concentration	yes

- 9 The diagram shows part of a protein molecule.

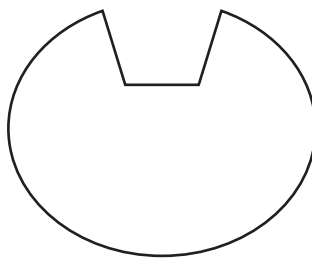


What does X represent?

- A** amino acid
B fatty acid
C glycerol
D sugar
- 10 What is true of **all** enzymes?

	they are sugars	they are most effective at pH 7	key ✓ = yes x = no
A	✓	✓	
B	✓	x	
C	x	✓	
D	x	x	

- 11 What is the substrate for enzyme X?

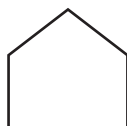


enzyme X

A



B



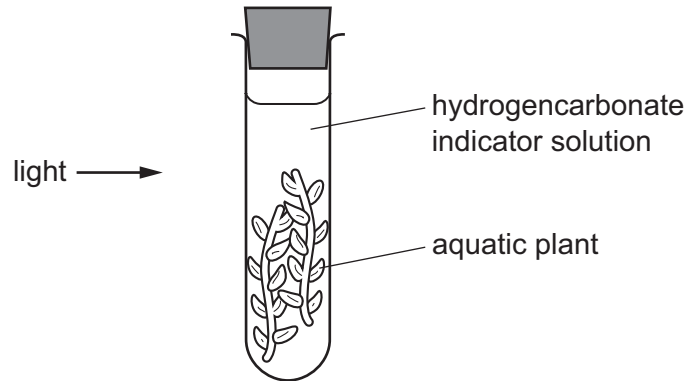
C



D



- 12 An experiment is set up to investigate gas exchange in aquatic plants.



The hydrogencarbonate indicator solution is orange at the start.

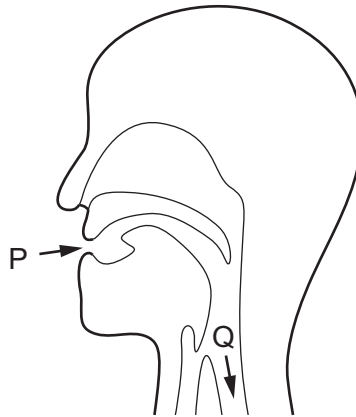
Which colour is it after three hours?

- A blue-black
 - B orange
 - C purple
 - D yellow
- 13 How are most leaves adapted for photosynthesis?
- A They have a large surface area and are thick.
 - B They have a large surface area and are thin.
 - C They have a small surface area and are thick.
 - D They have a small surface area and are thin.
- 14 The haemoglobin concentration in the blood of a person is 80g/dm^3 . The accepted normal concentration is 120g/dm^3 or above.
- Which substance may be lacking in their diet?
- A calcium
 - B fats
 - C fibre
 - D iron

15 Which statement about physical digestion is correct?

- A It increases the surface area of food.
- B It involves enzymes.
- C It takes place in the mouth only.
- D It produces smaller molecules.

16 Solid food enters the mouth at P and enters the oesophagus at Q.



How does the food at Q differ from the food at P?

- A It contains less fibre.
- B It contains less water.
- C It contains less protein.
- D It contains less starch.

17 Which row shows the functions of xylem and phloem?

	transports amino acids	transports mineral ions	transports sucrose
A	phloem	xylem	phloem
B	xylem	xylem	phloem
C	xylem	phloem	xylem
D	phloem	phloem	xylem

18 What is the effect on a plant of **not** having any root hairs?

- A The plant absorbs less water from the soil.
- B The plant absorbs more water from the soil.
- C The plant loses water to the soil.
- D The plant loses mineral ions to the soil.

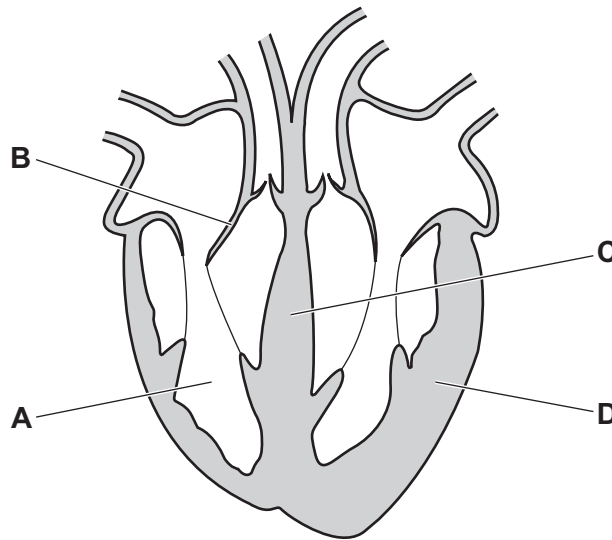
19 Which statements describe how the structures in the circulatory system function?

- 1 a muscular pump to push blood into vessels
- 2 valves to ensure one-way blood flow
- 3 veins to take blood away from the heart
- 4 vessels to return blood to the heart

A 1, 2 and 3 **B** 1, 2 and 4 **C** 1, 3 and 4 **D** 2, 3 and 4

20 The diagram shows a section through the heart.

Which part is the septum?



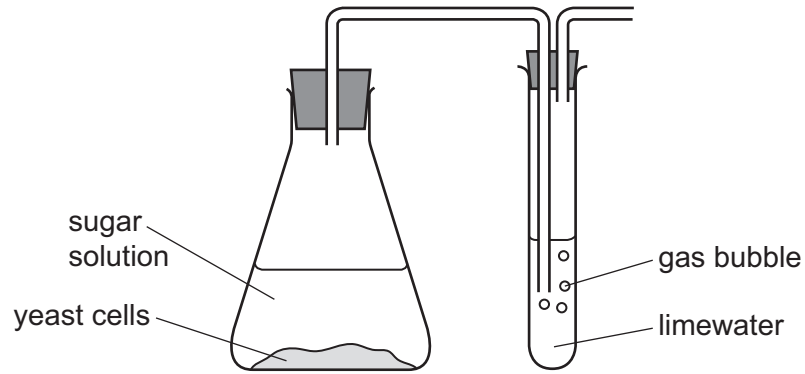
21 What is an example of good personal hygiene in the kitchen?

- A** cooking food at a high temperature
- B** disposing of waste food in sealed containers
- C** storing uncooked meat in a fridge
- D** washing hands before eating food

22 What is the pathway of expired air as it travels out of the body?

- A** alveoli → trachea → bronchiole → bronchi
- B** alveoli → bronchiole → bronchi → trachea
- C** trachea → alveoli → bronchiole → bronchi
- D** trachea → bronchi → bronchiole → alveoli

23 The diagram shows the activity of some yeast cells in a sugar solution.



Which statement explains what happens to the limewater?

- A The limewater changes from colourless to cloudy due to the presence of carbon dioxide.
- B The limewater changes from colourless to cloudy due to the presence of oxygen.
- C The limewater changes from cloudy to colourless due to the presence of carbon dioxide.
- D The limewater changes from cloudy to colourless due to the presence of oxygen.

24 What is the word equation for aerobic respiration?

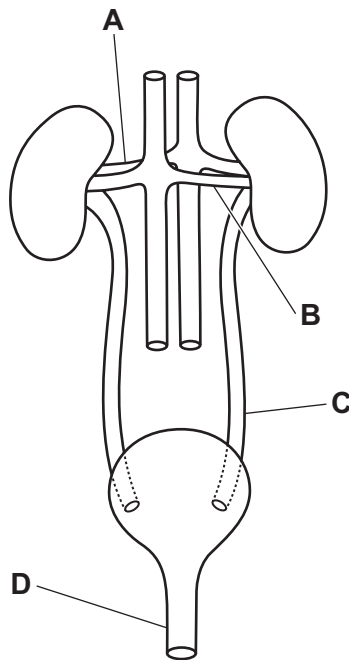
- A carbon dioxide + water $\rightarrow$ glucose + oxygen
- B glucose + oxygen $\rightarrow$ carbon dioxide + water
- C glycogen + oxygen $\rightarrow$ carbon dioxide + water
- D water + oxygen $\rightarrow$ glucose + carbon dioxide

25 Which part of the body excretes urea, excess water and excess ions?

- A gall bladder
- B heart
- C kidney
- D lungs

26 The diagram shows the kidneys and associated organs.

Which label is the ureter?



27 What is the name of the junction between two neurones?

- A** effector
- B** receptor
- C** gland
- D** synapse

28 Which statement about adrenaline is correct?

- A** It decreases the heart rate.
- B** It decreases the diameter of the pupil.
- C** It increases the breathing rate.
- D** It is made in the pancreas.

29 Which words complete the statements?

Plant growth towards light is called1..... .

When a plant shoot grows towards a light source, it is showing a2..... to light.

Light acts as the3..... .

	1	2	3
A	movement	response	stimulus
B	movement	stimulus	response
C	phototropism	response	stimulus
D	phototropism	stimulus	response

30 Which statements about antibiotics are correct?

- 1 Antibiotics can be used to treat bacterial infections.
- 2 Antibiotics can be used to treat viral infections.
- 3 Antibiotics are **not** effective against resistant bacteria.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

31 Which statement about sexual reproduction is correct?

- A** Fertilisation is involved.
- B** Zygotes fuse.
- C** Offspring are always genetically identical.
- D** One female gamete fuses with many male gametes.

32 Which row about asexual reproduction is correct?

	number of parents	offspring identical to parents	offspring identical to each other
A	one	no	yes
B	one	yes	yes
C	two	no	no
D	two	yes	no

33 What is defined as a length of DNA that codes for a protein?

- A** amino acid
- B** antibody
- C** chromosome
- D** gene

34 A man has three sons.

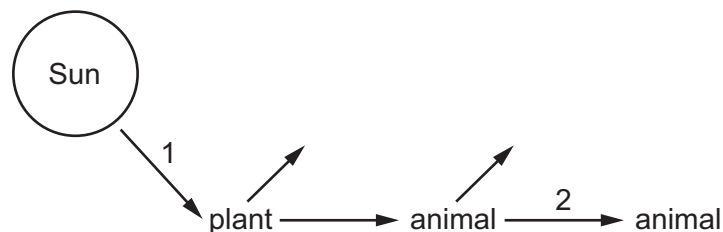
What is the chance of his next child being a daughter?

- A** 0%
- B** 25%
- C** 50%
- D** 100%

35 Which term is a genetic change?

- A** allele
- B** genotype
- C** mutation
- D** phenotype

36 The diagram shows energy flow from the Sun, through a food chain and into the environment.



What is the form of energy for each numbered arrow?

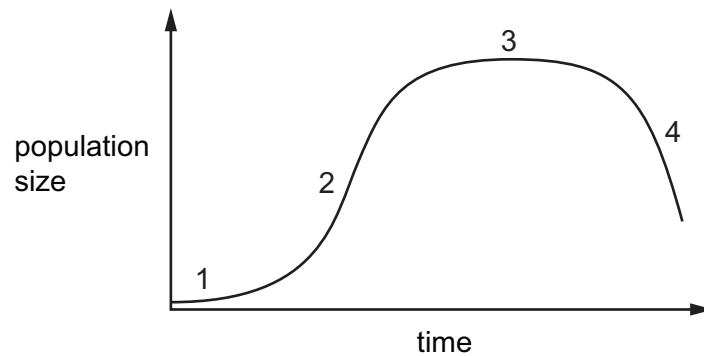
	stage 1	stage 2
A	heat	chemical
B	heat	kinetic
C	light	chemical
D	light	kinetic

37 In a pyramid of numbers, what does the top of the pyramid represent?

- A carnivores
- B decomposers
- C herbivores
- D Sun

38 A few yeast cells were placed in a container of nutrient solution.

The graph shows how their population size changed over time.



Which row shows when the reproduction rate was greater than the death rate for the numbered phases on the graph?

	reproduction rate greater than death rate			
	1	2	3	4
A	✓	✓	✓	✗
B	✓	✓	✗	✗
C	✓	✗	✗	✗
D	✗	✓	✓	✓

key

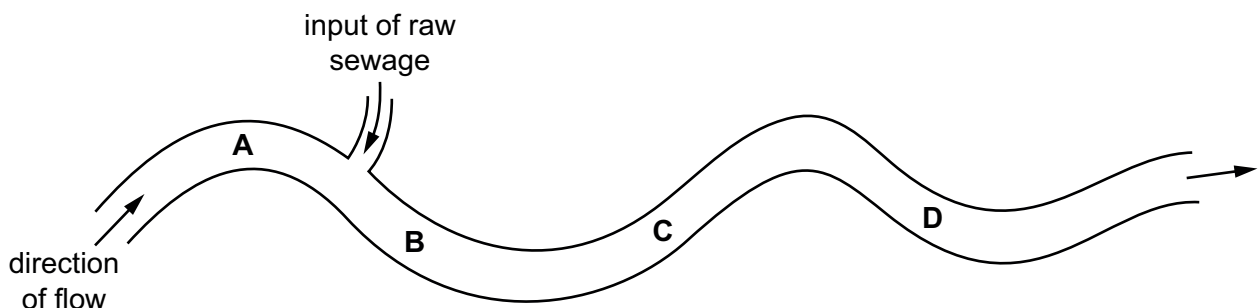
✓ = yes

✗ = no

39 The bloodworm is an organism that is found in heavily polluted water.

The diagram shows where raw sewage flows into a river.

Where would there be fewest bloodworms?



40 Which statement about genetic modification is correct?

- A** It involves choosing which individual organisms are used for breeding.
- B** It is always done using genes from the same species.
- C** It produces a new combination of genes.
- D** It produces exact copies of individual organisms.

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Cambridge IGCSE™ (9–1)

BIOLOGY (9–1)

0970/11

Paper 1 Multiple Choice (Core)

May/June 2023

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

Question	Answer	Marks
1	C	1
2	A	1
3	B	1
4	C	1
5	B	1
6	C	1
7	C	1
8	D	1
9	A	1
10	D	1
11	D	1
12	C	1
13	B	1
14	D	1
15	A	1
16	D	1
17	A	1
18	A	1
19	B	1
20	C	1
21	D	1
22	B	1
23	A	1
24	B	1
25	C	1
26	C	1
27	D	1
28	C	1

Question	Answer	Marks
29	C	1
30	C	1
31	A	1
32	B	1
33	D	1
34	C	1
35	C	1
36	C	1
37	A	1
38	B	1
39	A	1
40	C	1



Cambridge IGCSE™ (9–1)

BIOLOGY

0970/12

Paper 1 Multiple Choice (Core)

May/June 2023

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 Living animals release carbon dioxide.

This is an example of which life process?

- A excretion
- B movement
- C nutrition
- D sensitivity

- 2 Some statements about species are given.

- 1 Members of a species all look identical.
- 2 Members of a species belong to the same genus.
- 3 Members of a species can produce fertile offspring.
- 4 Species are named using an international system.

Which statements are correct?

- A 1, 2, 3 and 4
- B 1, 2 and 3 only
- C 2 and 4 only
- D 2, 3 and 4 only

- 3 The table shows some features of different animals.

Which animal is a reptile?

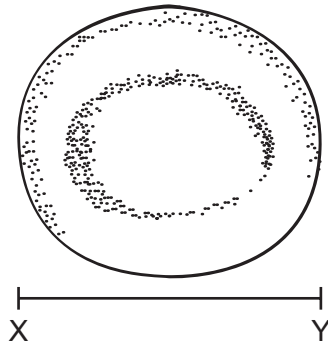
	wings	hair	scales
A	x	x	✓
B	x	x	x
C	✓	x	✓
D	x	✓	x

- 4 Which structures are found in the cells of all living organisms?

- 1 cell membrane
- 2 chloroplast
- 3 cytoplasm

- A 1, 2 and 3
- B 1 and 2 only
- C 1 and 3 only
- D 2 and 3 only

- 5 A diagram of a human red blood cell is shown. The length of line XY on the diagram is 40 mm. The actual width of the cell is 0.008 mm.



What is the magnification of the diagram?

- A $\times 500$ B $\times 5000$ C $\times 50\,000$ D $\times 500\,000$

- 6 Which statements are correct for **both** diffusion and osmosis?

	involves movement of water only	requires energy from respiration	molecules move from higher concentration to lower concentration
A	✓	✓	✓
B	✓	✓	✗
C	✗	✗	✓
D	✗	✗	✗

key

✓ = yes

✗ = no

- 7 What is a role of water in digestion?

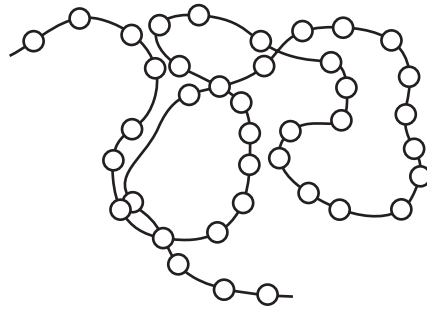
- A It is an enzyme.
B It is a solvent.
C It is soluble.
D It is a mineral.

- 8 The concentration of sodium ions in the soil is lower than the concentration inside root hair cells.

Which process will be used when sodium ions are taken into root hair cells from the soil?

- A active transport
B diffusion
C osmosis
D transpiration

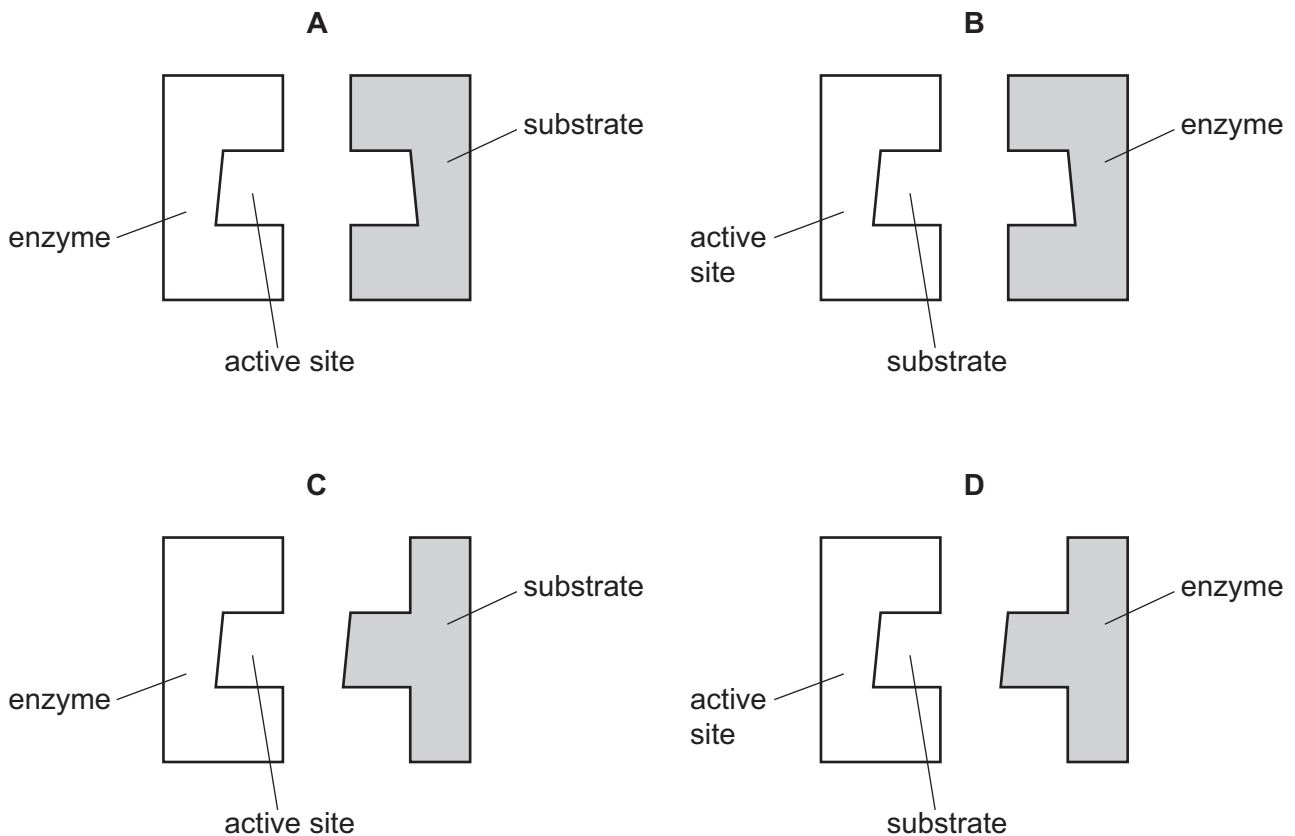
- 9 The diagram represents a protein molecule.



What do the small circles represent?

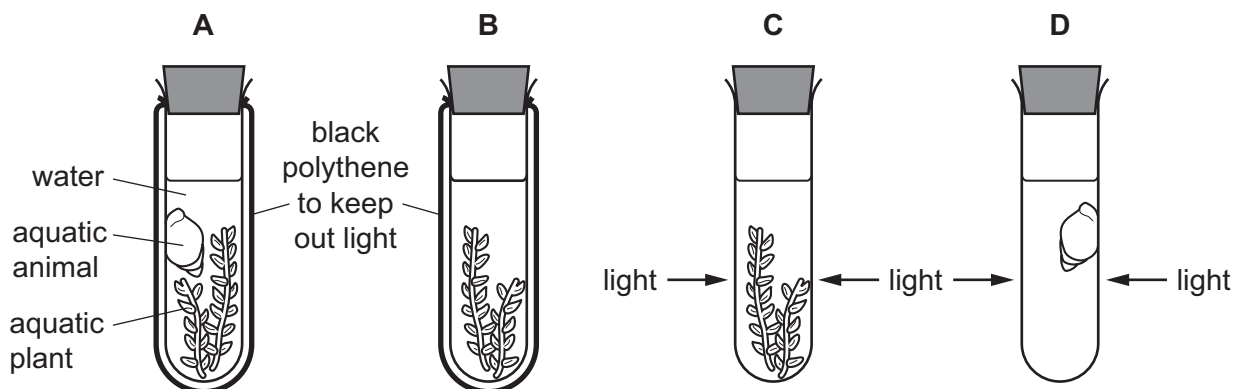
- A amino acids
 - B fatty acids
 - C glycerol
 - D simple sugars
- 10 Which processes depend on the action of enzymes?
- 1 digestion
 - 2 osmosis
 - 3 respiration
- A 1 and 2 B 1 and 3 C 1 only D 2 and 3

11 Which diagram of an enzyme, active site and substrate is correct?



12 Four test-tubes are set up as shown.

Which test-tube contains the least carbon dioxide after one hour?



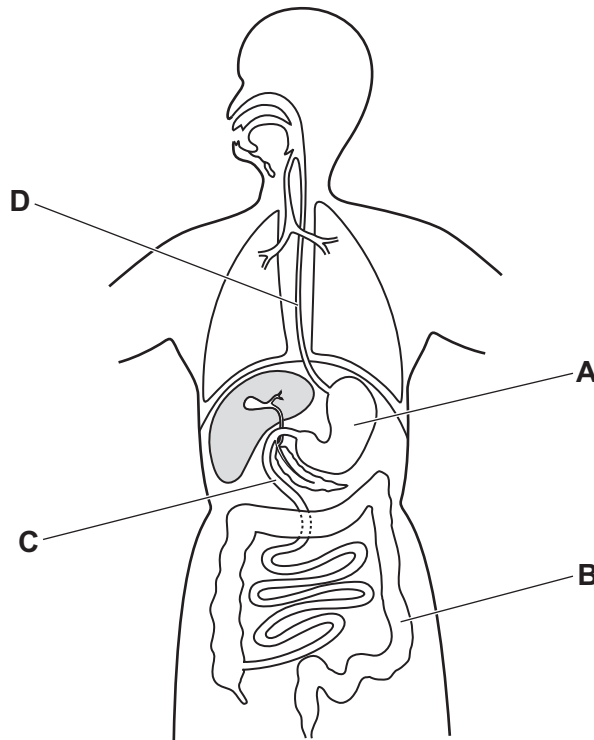
13 Which row shows the uses of some carbohydrates in plants?

	glucose	starch	sucrose
A	cell walls	transport	energy storage
B	energy storage	cell walls	respiration
C	respiration	energy storage	transport
D	transport	respiration	cell walls

14 The lack of which component of a balanced diet will lead to the development of scurvy?

- A** calcium
- B** iron
- C** vitamin C
- D** vitamin D

15 Which structure is the duodenum?



- 16 The food label is from a packet of cereal.

The label can help someone who is concerned about their diet.

Nutrition	
Typical values	100 g contains
Energy	985 kJ 235 kcal
Fat	1.5 g
of which saturates	0.3 g
Carbohydrate	45.5 g
of which sugars	3.8 g
Fibre	2.8 g
Protein	7.7 g
Salt	0.5 g

A person eats 45 g of cereal.

One of the food types listed in the label can help prevent constipation.

How many grams of this food type does the person eat?

- A** 1.3 g **B** 2.8 g **C** 3.5 g **D** 7.7 g

- 17 Which pathway is taken by water through a plant?

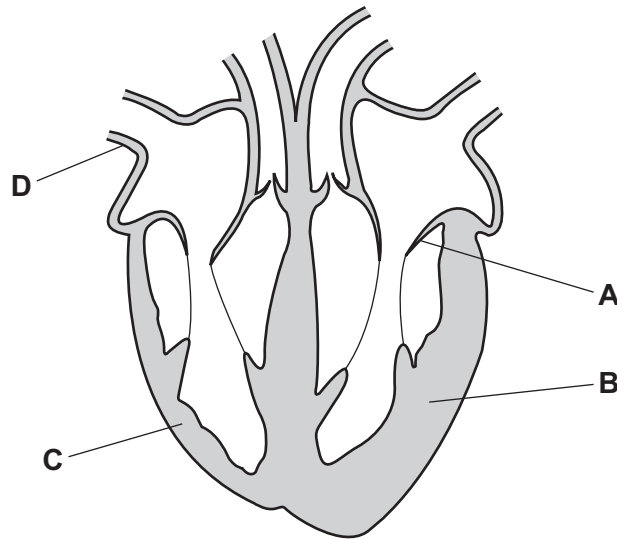
- A** root hair → root cortex → xylem → mesophyll → stomata
B root hair → xylem → mesophyll → root cortex → stomata
C stomata → root cortex → xylem → mesophyll → root hair
D stomata → root hair → root cortex → xylem → mesophyll

- 18 Which row shows the transpiration rate for the conditions described?

	wind speed	atmospheric temperature	transpiration rate
A	low	low	fast
B	high	high	slow
C	low	high	slow
D	low	low	slow

19 The diagram shows the human heart.

Which label shows the left ventricle?



20 Which row shows the features of an artery or a vein?

	blood vessel	diameter of lumen	thickness of wall
A	artery	large	thick
B	artery	small	thin
C	vein	large	thin
D	vein	small	thick

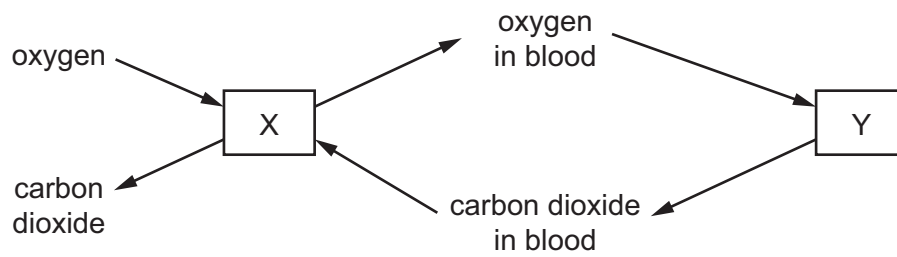
21 What are features of gas exchange surfaces in animals?

- A** thick-walled and large surface area
- B** thick-walled and small surface area
- C** thin-walled and small surface area
- D** thin-walled and large surface area

22 Which chemical can be used to show the presence of carbon dioxide gas?

- A** Benedict's solution
- B** biuret solution
- C** ethanol
- D** limewater

- 23** The diagram represents the exchange of gases during breathing and during respiration in the body.

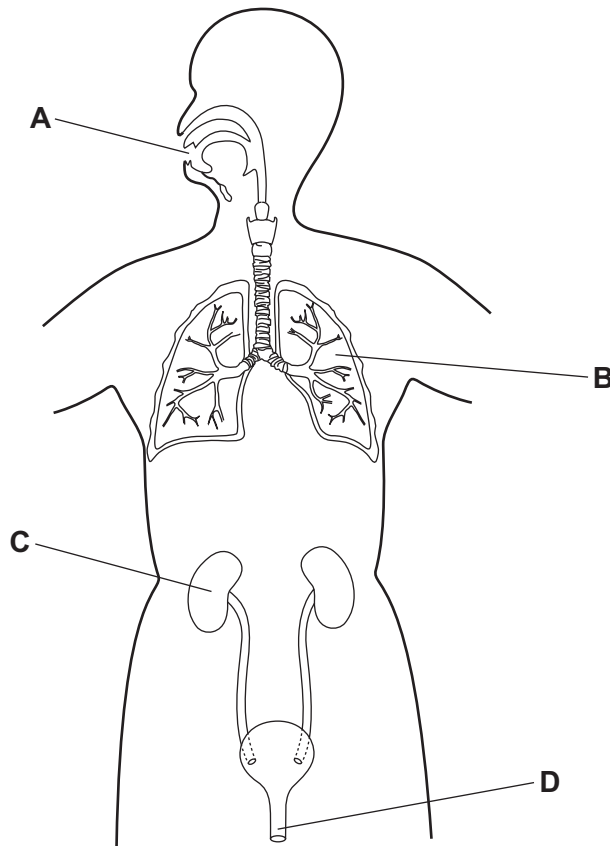


What is represented by X?

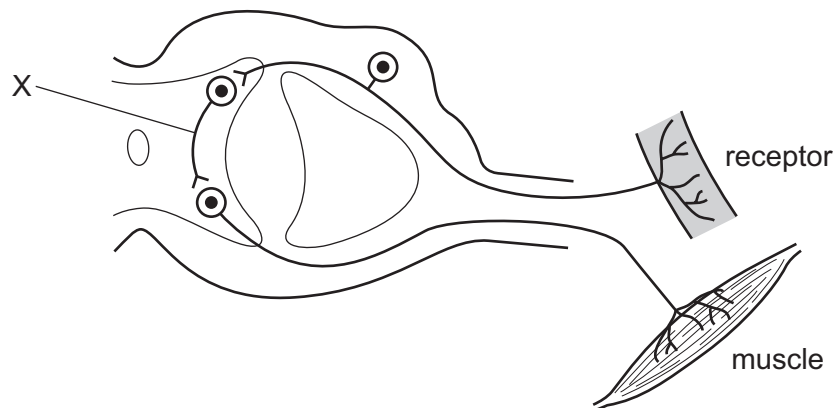
- A** heart
 - B** kidneys
 - C** liver
 - D** lungs
- 24** Which molecule is produced by anaerobic respiration in yeast?
- A** carbon dioxide
 - B** lactic acid
 - C** oxygen
 - D** water

25 The diagram shows the human body and some of its organs.

Which structure removes urea from the blood?



26 The diagram shows the structures in a reflex arc.



What is X?

- A effector
- B relay neurone
- C sensory neurone
- D synapse

- 27 Changes in light intensity cause changes in the diameter of the pupil in the eye.

Which rows show the pupil reflex?

	light intensity	pupil diameter
1	high	increases
2	high	decreases
3	low	increases
4	low	decreases

- A 1 and 3 B 1 and 4 C 2 and 3 D 2 and 4

- 28 A hormone is injected into a person to reduce their blood glucose concentration.

Which endocrine gland is **not** functioning properly in a person who needs to have this injection?

- A adrenal gland
B ovary
C pancreas
D testis

- 29 The table shows which antibiotics are effective against different types of bacteria.

The shaded areas show the antibiotic is effective against that type of bacteria.

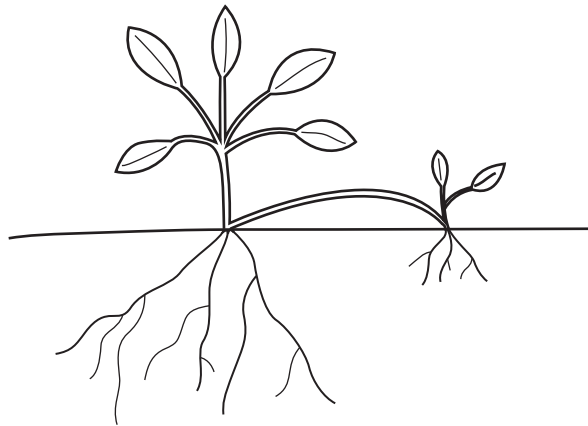
antibiotic	type of bacteria			
	MRSA	<i>Streptococcus</i>	<i>Pseudomonas</i>	<i>Anaerobes</i>
1				
2				
3				
4				
5				
6				

A person has a disease caused by a type of *Streptococcus* bacteria and a second infection caused by a type of *Pseudomonas* bacteria.

Which antibiotic should be taken?

- A 1 or 6 B 2 or 4 C 3 or 4 D 5 or 6

30 The diagram shows one way that a plant can reproduce.



What is a feature of this type of reproduction?

- A The offspring show genetic variation.
- B Only one parent is required.
- C Offspring are easily dispersed over a wide area.
- D Pollination by insects is required.

31 Which human characteristics are inherited?

	blood group	eye colour	language	sex
A	✓	✓	x	✓
B	✓	✓	✓	x
C	x	✓	x	✓
D	✓	x	✓	✓

key

✓ = inherited

x = not inherited

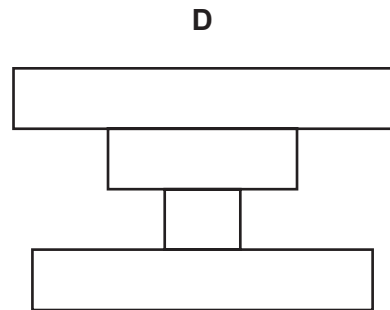
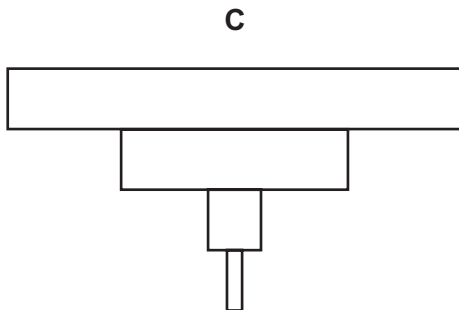
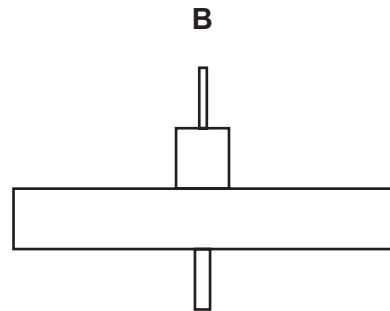
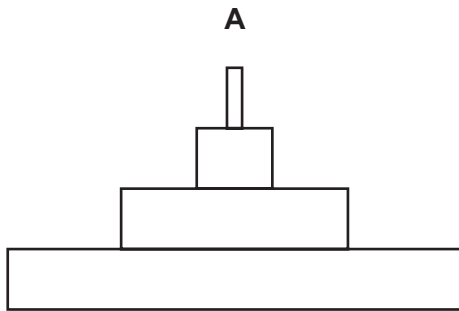
32 In a sperm cell, which structure contains enzymes that can digest the jelly coat of an egg cell?

- A acrosome
- B flagellum
- C mitochondria
- D nucleus

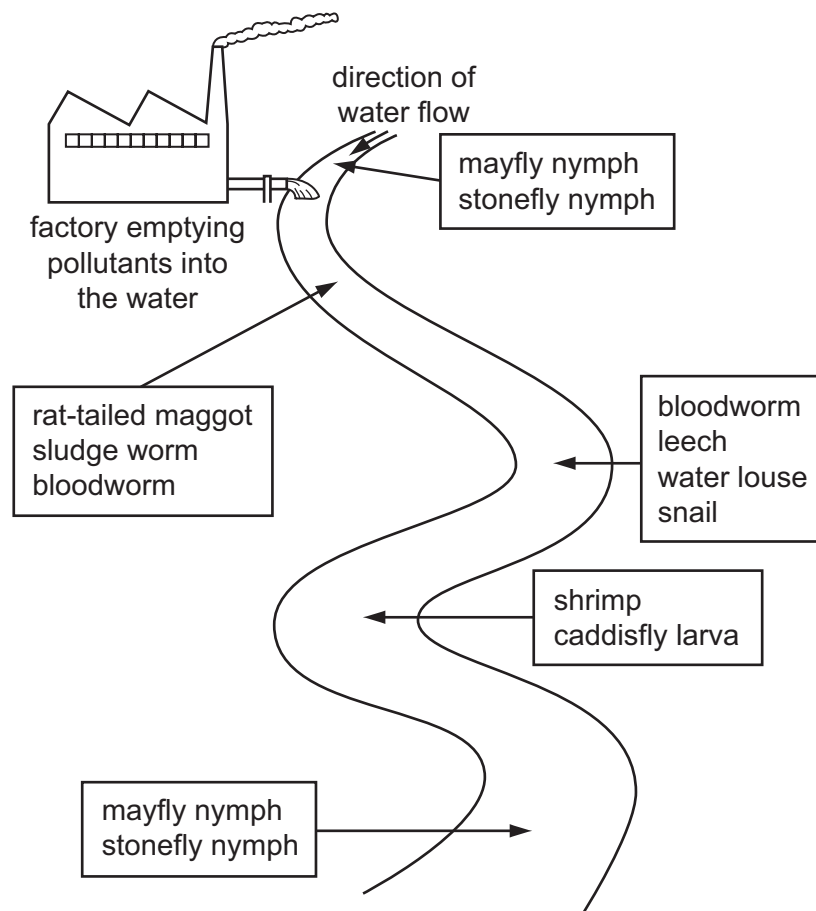
- 33** When breeding, which pair of parents will be pure-breeding for a particular characteristic?
- A** a heterozygous and a homozygous individual
 - B** two heterozygous individuals
 - C** two homozygous individuals
 - D** two individuals with the same phenotype
- 34** In a pea plant, the allele for round seeds, R, is dominant to the allele for wrinkled seeds, r.
- If plants with the genotype Rr are crossed, what are the likely proportions of offspring?
- A** all with round seeds
 - B** all with wrinkled seeds
 - C** 1 with round seeds : 1 with wrinkled seeds
 - D** 3 with round seeds : 1 with wrinkled seeds
- 35** Which definition of continuous variation is correct?
- A** variation that results in a limited number of phenotypes between two extremes
 - B** variation that results in a limited number of phenotypes with no intermediates
 - C** variation that results in a range of phenotypes between two extremes
 - D** variation that results in a range of phenotypes with no intermediates
- 36** Which term is used to describe an organism that makes its own organic nutrients?
- A** carnivore
 - B** decomposer
 - C** herbivore
 - D** producer
- 37** What is the principal source of energy for most food chains?
- A** carbon dioxide gas
 - B** glucose
 - C** oxygen
 - D** sunlight

- 38** A single tree is food for a large population of caterpillars. Several small birds eat the caterpillars. The small birds are eaten by a bird of prey.

Which is the pyramid of biomass?



- 39 The diagram shows the results of a survey on the types of animals found along a stretch of river near to a factory.



Which animal lives in the least polluted water?

- A bloodworm
 - B caddisfly larva
 - C leech
 - D stonefly nymph
- 40 Bacteria are useful for manufacturing products for human use.

Which statement explains why they are useful?

- A Bacteria can be grown without nutrients.
- B Bacteria do not have a nucleus.
- C Bacteria infect human cells.
- D Bacteria reproduce very rapidly.

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Cambridge IGCSE™ (9–1)

BIOLOGY (9–1)

0970/12

Paper 1 Multiple Choice (Core)

May/June 2023

MARK SCHEME

Maximum Mark: 40

Published

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Question	Answer	Marks
1	A	1
2	D	1
3	A	1
4	C	1
5	B	1
6	C	1
7	B	1
8	A	1
9	A	1
10	B	1
11	C	1
12	C	1
13	C	1
14	C	1
15	C	1
16	A	1
17	A	1
18	D	1
19	B	1
20	C	1
21	D	1
22	D	1
23	D	1
24	A	1
25	C	1
26	B	1
27	C	1
28	C	1

Question	Answer	Marks
29	B	1
30	B	1
31	A	1
32	A	1
33	C	1
34	D	1
35	C	1
36	D	1
37	D	1
38	A	1
39	D	1
40	D	1



Cambridge IGCSE™ (9–1)

BIOLOGY

0970/12

Paper 1 Multiple Choice (Core)

October/November 2023

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
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- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.



1 Which processes are characteristics of **all** living organisms?

- 1 digestion
- 2 respiration
- 3 excretion

A 1, 2 and 3 **B** 1 only **C** 2 and 3 only **D** 2 only

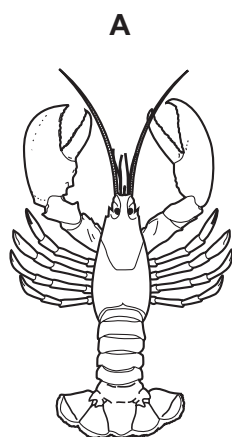
2 Using the binomial system, the Arctic fox is called *Vulpes lagopus*.

Which row is correct?

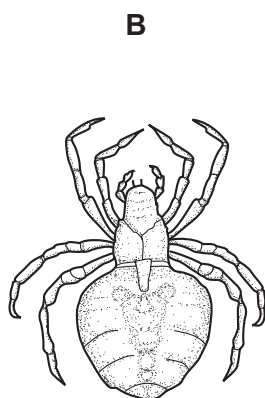
	<i>Vulpes</i>	<i>lagopus</i>
A	genus	kingdom
B	genus	species
C	species	genus
D	species	kingdom

3 The diagrams show four different types of arthropod.

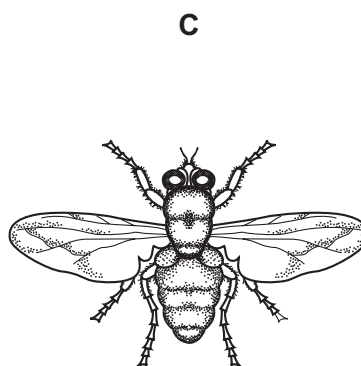
Which organism is labelled with its correct arthropod group?



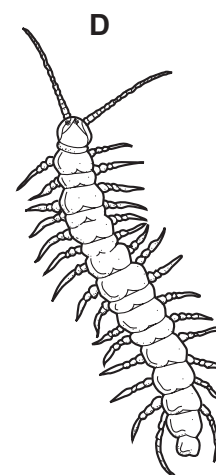
crustacean



insect



myriapod



arachnid

4 Which features are possessed by **all** plant cells?

	a cell wall	chloroplasts
A	✓	✓
B	✓	x
C	x	✓
D	x	x

key

✓ = yes

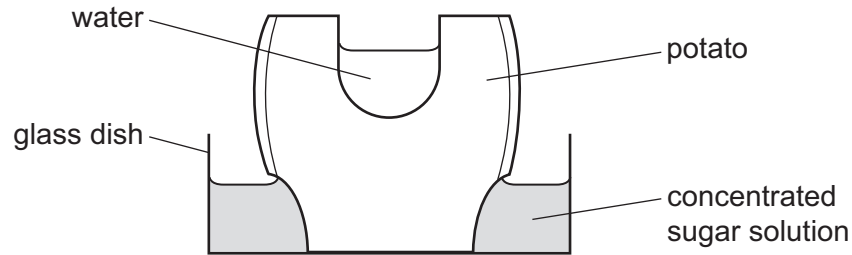
x = no

- 5 The diameter of an egg is 6.5 cm. A student draws the egg. The diameter of the egg in the drawing is 19.5 cm.

What is the magnification of the drawing?

- A** $\times 0.3$ **B** $\times 3.0$ **C** $\times 6.5$ **D** $\times 300$

- 6 The diagram shows an experiment to investigate osmosis in living cells.



What happens to the volumes of water and sugar solution after 12 hours?

	volume of water	volume of sugar solution
A	decreases	increases
B	increases	increases
C	increases	remains the same
D	remains the same	decreases

- 7 Which process causes oxygen to pass from an alveolus in the lung into a blood capillary?

- A** assimilation
B diffusion
C excretion
D osmosis

- 8 Which row identifies the chemical elements in proteins?

	carbon	hydrogen	oxygen	nitrogen
A	✓	✓	✓	✓
B	✓	x	✓	x
C	x	✓	✓	✓
D	✓	✓	x	x

key

✓ = present

x = absent

- 9 The table shows some large biological molecules and some small biological molecules.

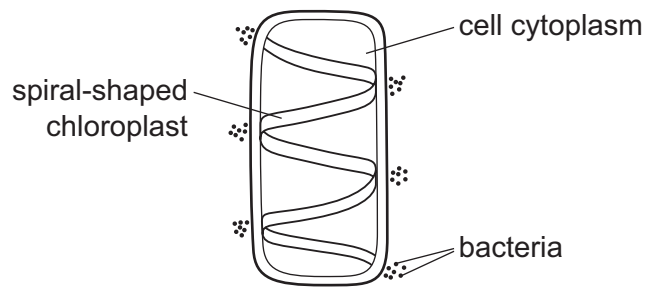
	large biological molecules	small biological molecules
1	cellulose	glucose
2	oil	amino acids
3	glycogen	glucose
4	protein	ethanol

Which rows correctly pair large molecules with the smaller molecules used to make them?

- A** 1 and 2 **B** 1 and 3 **C** 2 and 3 **D** 3 and 4
- 10 Which changes could cause an enzyme to denature?
- 1 change in pH
 - 2 change in temperature
 - 3 change in substrate concentration
- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only
- 11 Which features of a leaf allow more diffusion of carbon dioxide into the palisade cells for photosynthesis?

	number of stomata	thickness of the leaf
A	few	thick
B	few	thin
C	many	thick
D	many	thin

- 12 The diagram shows a cell with groups of bacteria around its edge.

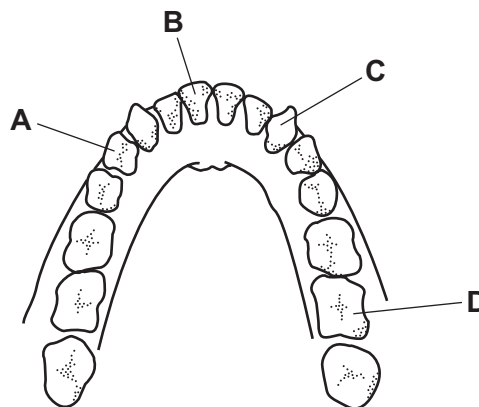


The bacteria have moved to areas of high oxygen concentration.

Which process in the cell causes the bacteria to form these groups?

- A digestion
 - B photosynthesis
 - C reproduction
 - D respiration
- 13 The diagram shows the teeth in the lower jaw of a human.

Which tooth is a premolar?



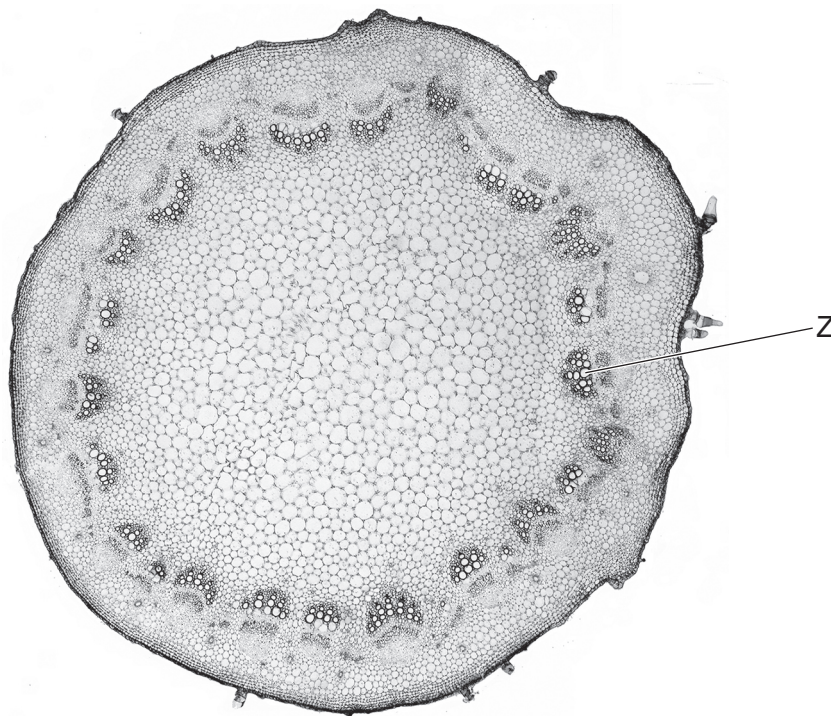
- 14 Some medicines are made into tablets which are coated in a starch-like substance.

The coating protects the medicine from the effects of gastric juice.

Which enzyme digests the coating and what is produced by this action?

	enzyme	products
A	amylase	fatty acids and glycerol
B	amylase	simple reducing sugars
C	lipase	fatty acids and glycerol
D	lipase	simple reducing sugars

- 15 In which part of the alimentary canal are **most** nutrients absorbed into the blood?
- A colon
 - B oesophagus
 - C small intestine
 - D stomach
- 16 From which part of a leaf does **most** water evaporate during transpiration?
- A the cuticle
 - B the guard cells
 - C the spongy mesophyll cells
 - D the xylem vessels
- 17 The photograph shows a cross-section through a sunflower stem.



What is the name and function of the tissue labelled Z?

	name of tissue	function of tissue
A	phloem	transport of mineral ions
B	phloem	transport of sucrose
C	xylem	transport of mineral ions
D	xylem	transport of sucrose

18 What ensures the one-way flow of blood in the human circulatory system?

- A** capillaries
- B** muscles
- C** valves
- D** ventricles

19 In which row is the component of blood correctly linked to its function?

	blood component	function
A	plasma	transport of carbon dioxide
B	platelets	phagocytosis
C	red blood cells	antibody production
D	white blood cells	blood clotting

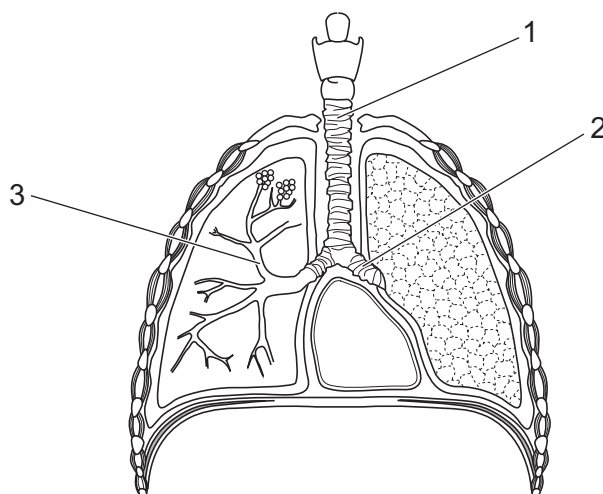
20 Which statement describes a transmissible disease?

- A** a disease caused when a pathogen passes from one host to another
- B** a disease caused when a pathogen passes through a host's body
- C** a disease caused when a pathogen passes to the host's body only by direct contact
- D** a disease caused when a pathogen passes to the host's body only by indirect contact

21 What is the composition of expired air compared with inspired air?

	carbon dioxide	oxygen	water vapour
A	decreased	increased	increased
B	decreased	decreased	increased
C	increased	decreased	decreased
D	increased	decreased	increased

22 The diagram shows the human breathing system.



What are the labelled structures?

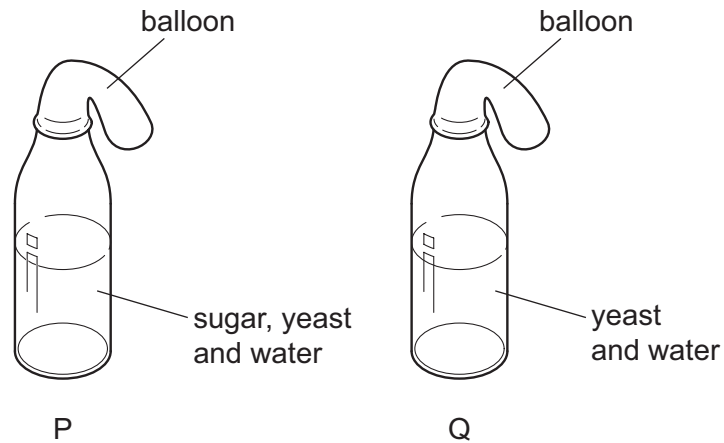
	1	2	3
A	bronchus	bronchiole	trachea
B	bronchiole	bronchus	trachea
C	trachea	bronchus	bronchiole
D	trachea	diaphragm	bronchus

23 Yeast cells can convert glucose into alcohol and carbon dioxide.

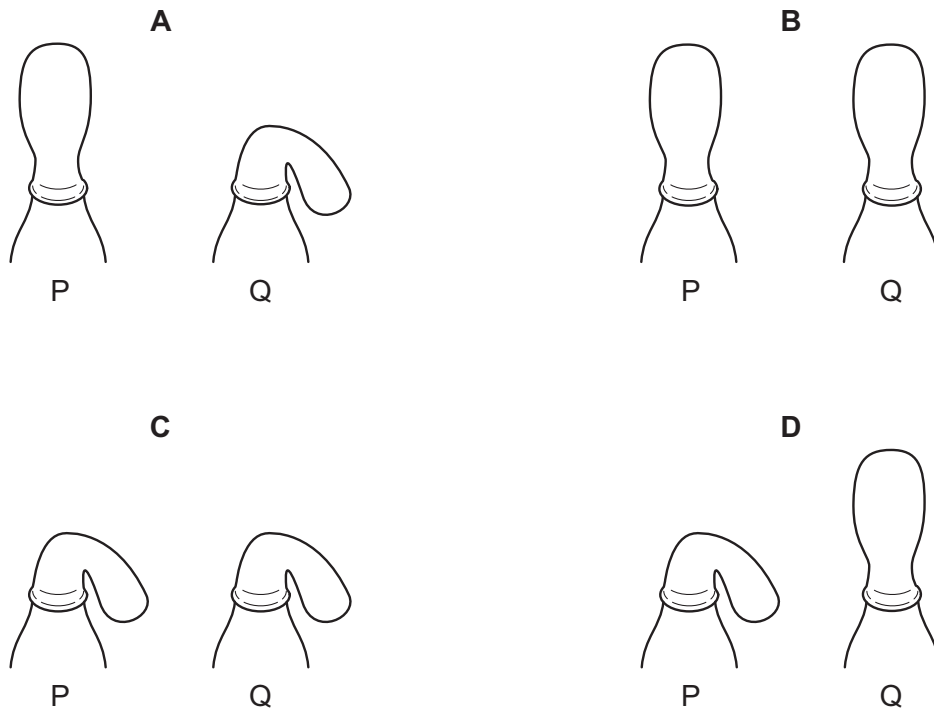
Which statement about this process is correct?

- A** The alcohol produced can be used to make bread rise.
- B** The carbon dioxide produced can be burnt as a biofuel.
- C** The yeast cells are using oxygen for this process.
- D** The yeast cells are carrying out anaerobic respiration.

- 24** In an experiment to investigate anaerobic respiration, two bottles are set up in a warm room, as shown.



What would happen to each balloon after one day?

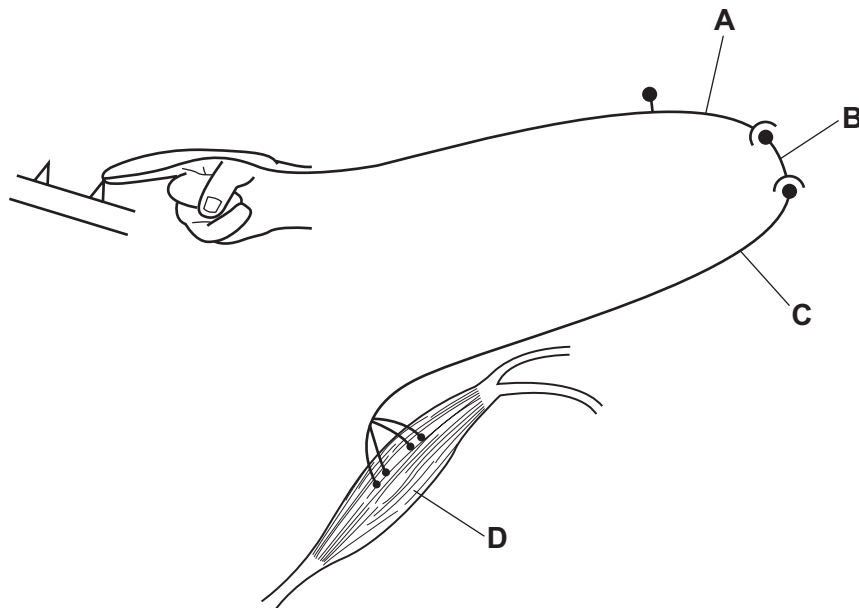


- 25** Which row shows an organ and the substance it excretes?

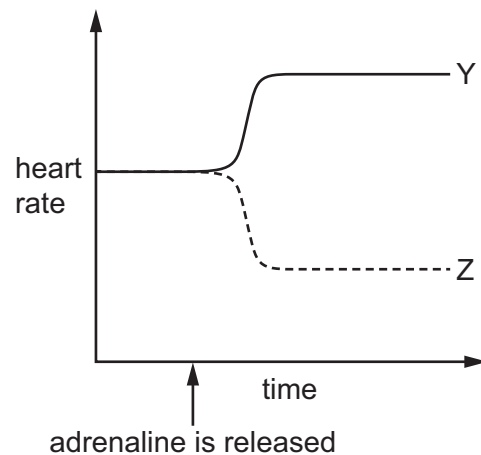
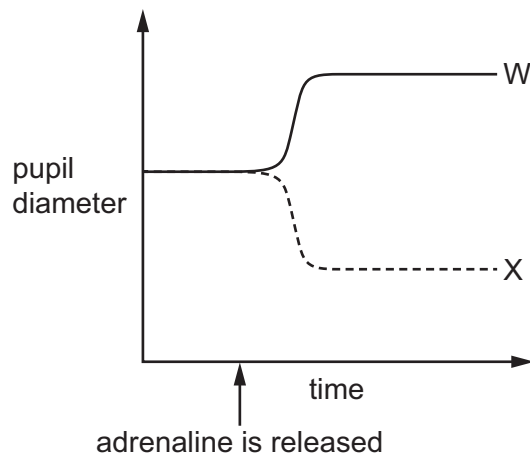
	organ	substance excreted
A	bladder	water
B	kidney	ions
C	lung	oxygen
D	urethra	urea

- 26** The diagram shows a reflex arc in a human nervous system. The person's finger has just made contact with a sharp object.

Which part is the motor neurone?



- 27** Which lines on the graphs show the effects of adrenaline on pupil diameter and heart rate?



- A** W and Y **B** W and Z **C** X and Y **D** X and Z

- 28** Which statements about tropisms in plants are correct?

- 1 Shoots grow towards light and away from gravity.
- 2 Shoots grow away from light and towards gravity.
- 3 Roots grow towards light and away from gravity.
- 4 Roots grow away from light and towards gravity.

- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

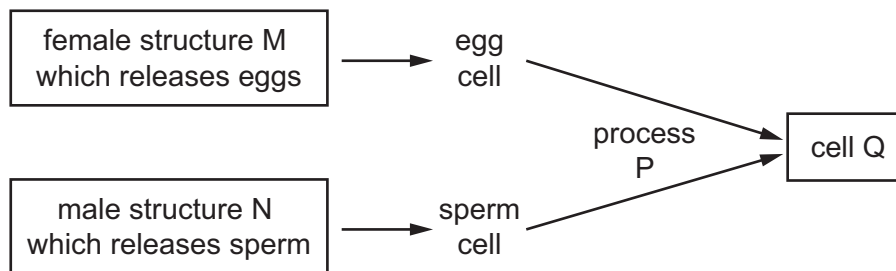
29 What is defined as a substance that is taken into the body and modifies or affects chemical reactions in the body?

- A** antibody
- B** drug
- C** hormone
- D** pathogen

30 Which row is correct for asexual reproduction?

	number of parents	offspring compared with parent or parents
A	one	genetically different
B	one	genetically identical
C	two	genetically different
D	two	genetically identical

31 The diagram shows some stages in reproduction.

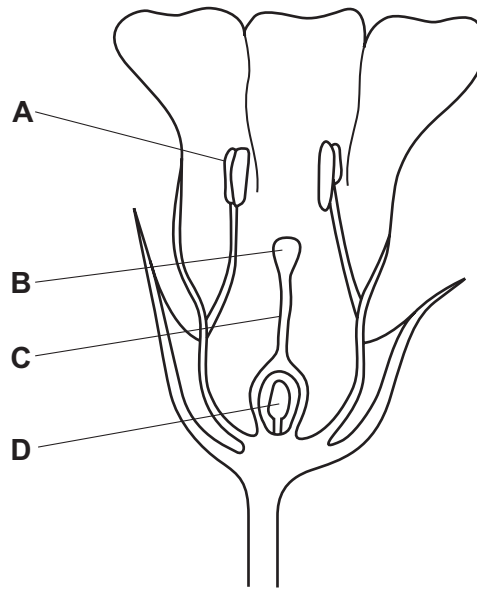


Which letter correctly identifies the structure or process?

- A** M is the oviduct.
- B** N is the sperm duct.
- C** P is fertilisation.
- D** Q is the gamete.

32 The diagram shows half a flower.

Which structure produces pollen?



33 Two parents have the genotypes Bb and bb for a particular feature.

What is the probability of these parents having offspring that are homozygous recessive for this feature?

- A** 100% **B** 75% **C** 50% **D** 25%

34 A genetic cross between two organisms may be shown as Tt × Tt.

What does T represent?

- A** a dominant allele
B a dominant chromosome
C a recessive allele
D a recessive chromosome

35 Some phenotypes are listed.

- 1 blood group
- 2 body mass
- 3 height
- 4 seed colour in peas

Which phenotypes are examples of continuous variation?

- A** 1, 2 and 3 **B** 2 and 3 only **C** 2 and 4 **D** 3 and 4

36 A food chain is shown.

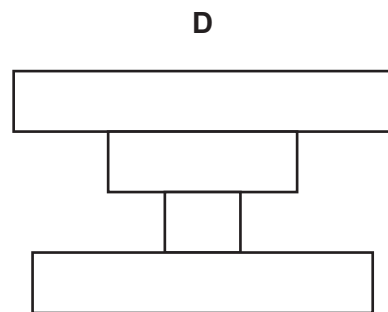
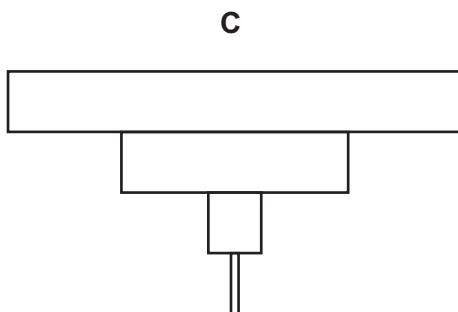
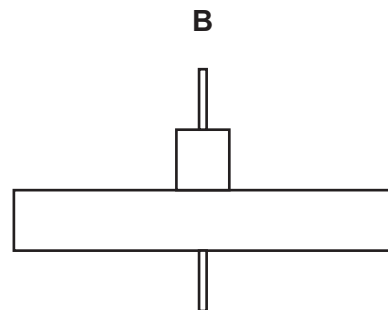
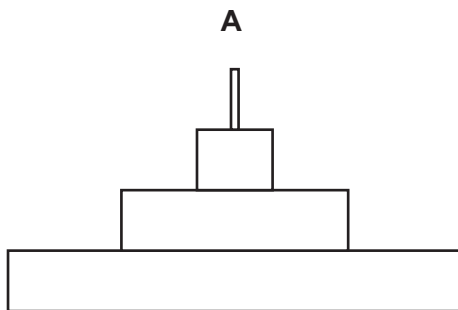
lettuce plant → slug → hedgehog

By which process is energy transferred from the lettuce plant to the slug?

- A egestion
- B excretion
- C ingestion
- D respiration

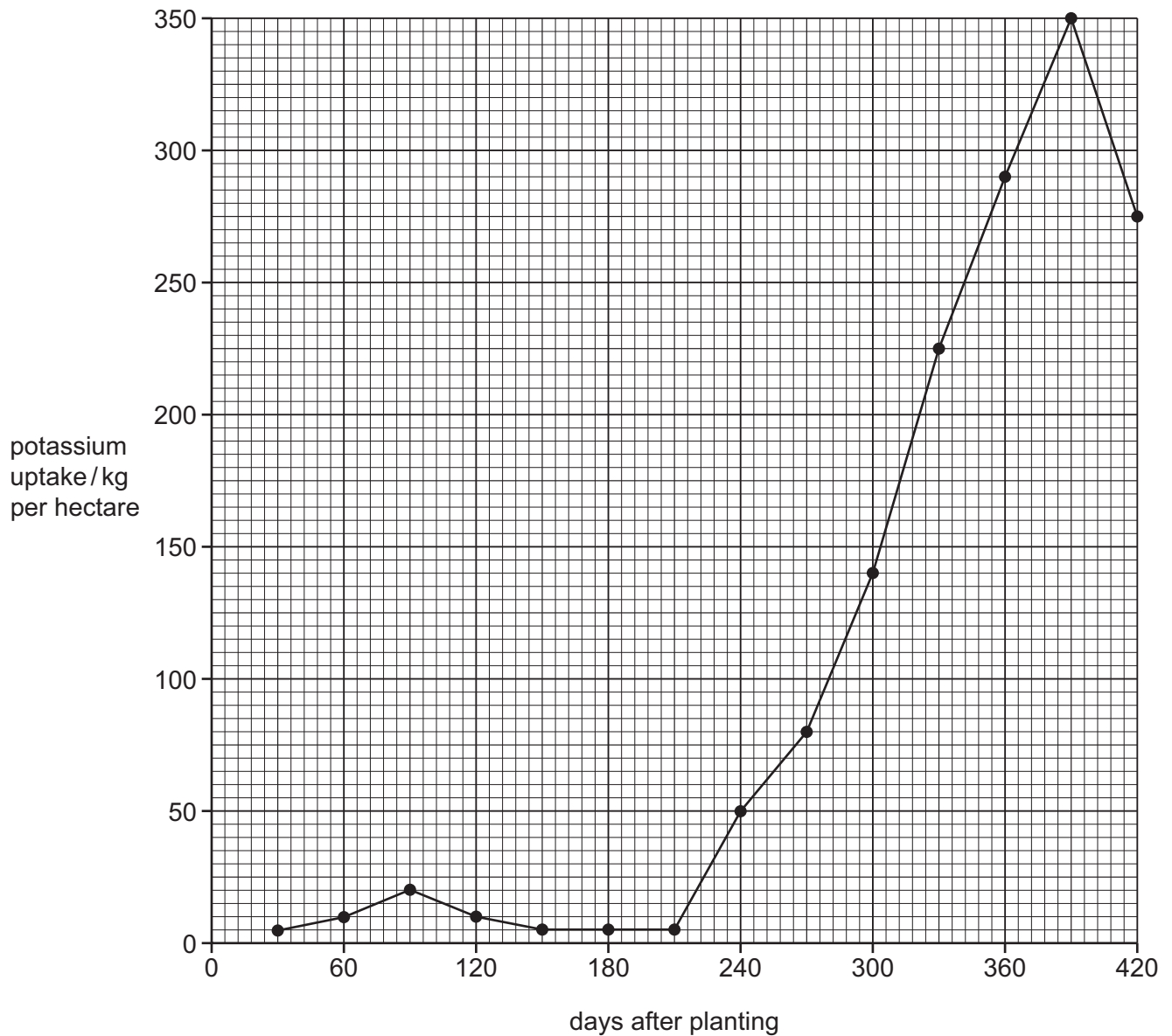
37 A single bush is food for a large population of greenfly. Several ladybirds eat the greenfly. The ladybirds are eaten by a bird.

Which diagram shows the pyramid of numbers for this food chain?



38 Scientists studied the uptake of potassium ions from chemical fertilisers by wheat plants.

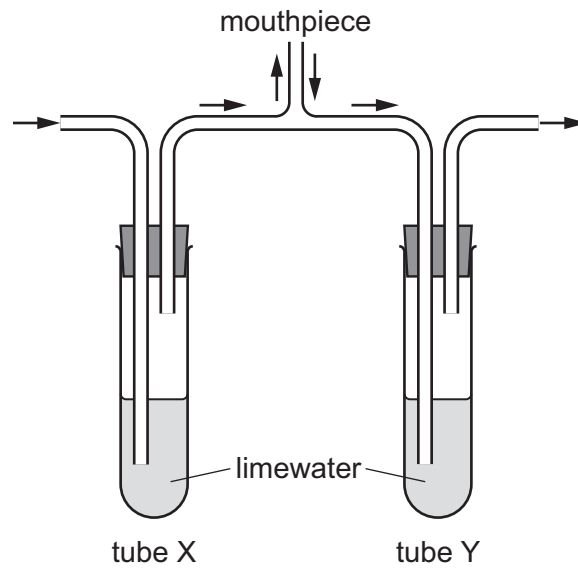
The results are shown.



What is the difference in the uptake of potassium ions at 360 days after planting compared to 30 days after planting?

- A 295 kg per hectare
- B 290 kg per hectare
- C 285 kg per hectare
- D 280 kg per hectare

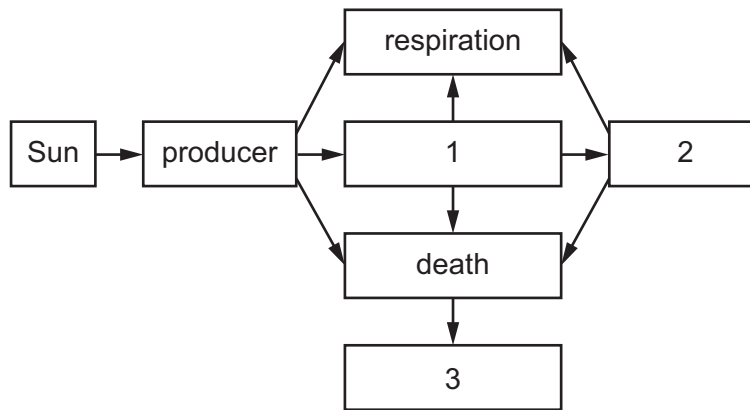
39 A student investigated gas exchange using the apparatus shown.



What would be the appearance of the limewater in tube X and tube Y after breathing in and out of the mouthpiece for one minute?

	tube X	tube Y
A	clear	clear
B	clear	cloudy
C	cloudy	clear
D	cloudy	cloudy

40 The diagram shows the flow of energy in an ecosystem.



Which row correctly completes boxes 1, 2 and 3?

	1	2	3
A	herbivore	carnivore	decomposer
B	carnivore	herbivore	decomposer
C	decomposer	carnivore	herbivore
D	carnivore	decomposer	herbivore

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Cambridge IGCSE™ (9–1)

BIOLOGY

0970/12

Paper 1 Multiple Choice (Core)

October/November 2023

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

Question	Answer	Marks
1	C	1
2	B	1
3	A	1
4	B	1
5	B	1
6	A	1
7	B	1
8	A	1
9	B	1
10	B	1
11	D	1
12	B	1
13	A	1
14	B	1
15	C	1
16	C	1
17	C	1
18	C	1
19	A	1
20	A	1
21	D	1
22	C	1
23	D	1
24	A	1
25	B	1
26	C	1
27	A	1
28	B	1

Question	Answer	Marks
29	B	1
30	B	1
31	C	1
32	A	1
33	C	1
34	A	1
35	B	1
36	C	1
37	B	1
38	C	1
39	B	1
40	A	1



Cambridge IGCSE™ (9–1)

BIOLOGY

0970/11

Paper 1 Multiple Choice (Core)

May/June 2024

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.



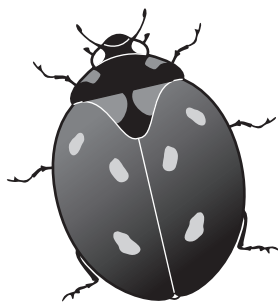
- 1 Which process do living organisms use to remove excess carbon dioxide from their bodies?
- A excretion
 - B movement
 - C nutrition
 - D respiration
- 2 The photograph shows an elephant hawk-moth.



Using the features visible in the photograph, to which group of arthropods does the elephant hawk-moth belong?

- A arachnids
 - B crustaceans
 - C insects
 - D myriapods
- 3 What does the term species mean?
- A a group of animals that reproduce asexually
 - B a group of organisms that can reproduce to produce fertile offspring
 - C all the plants that live in the same area at the same time
 - D a group of vertebrates that reproduce sexually

- 4 The image shows an animal.



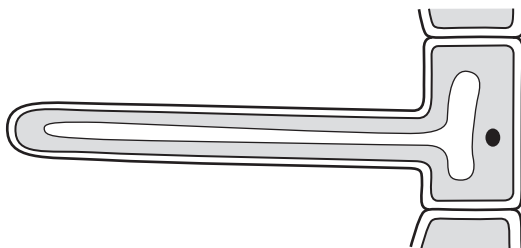
The length of the animal in the image is 36 mm.

The actual length of the animal is 12 mm.

What is the magnification of the image?

- A** $\times 0.3$ **B** $\times 3$ **C** $\times 43.2$ **D** $\times 432$

- 5 The diagram shows a plant cell.



Which features show that it is a plant cell?

- A** It has a cell wall and a vacuole.
B It has a nucleus and cytoplasm.
C It has a nucleus but no chloroplasts.
D It has chloroplasts but no vacuole.

- 6** A scientist investigates the effect of two types of antibiotics on bacteria.

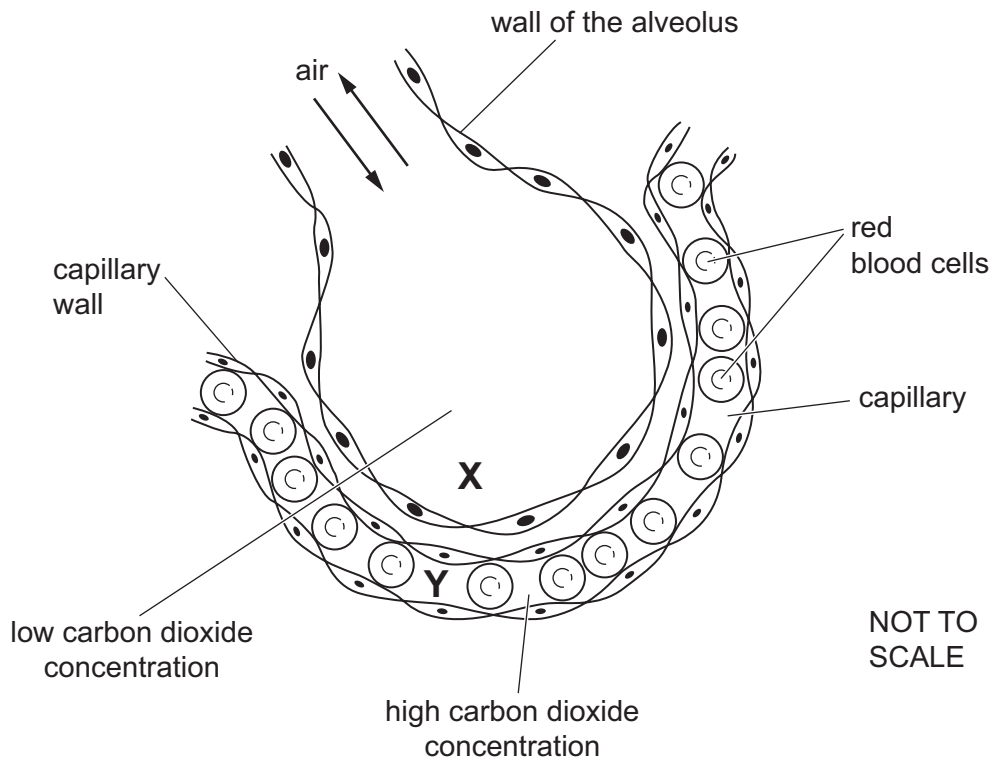
The table shows the results.

type of antibiotic	effect on bacteria
penicillin	cells burst
tetracycline	stops protein synthesis

Which cell structure does each antibiotic affect?

- A** Penicillin affects the cell wall and tetracycline affects the ribosomes.
- B** Penicillin affects the nucleus and tetracycline affects the cytoplasm.
- C** Penicillin affects the ribosomes and tetracycline affects the cell wall.
- D** Penicillin affects the vacuole and tetracycline affects the nucleus.

- 7 The diagram shows a section through an alveolus and through a capillary.



How does carbon dioxide move from **Y** to **X**?

- A** by active transport
 - B** by diffusion
 - C** by osmosis
 - D** by transpiration
- 8 In which row will the diffusion of ions into a cell be the quickest?

	ion concentration in the cell	ion concentration in the external solution	surface area of the cell	width of the cell membrane
A	high	low	large	thin
B	low	high	small	thick
C	high	low	small	thick
D	low	high	large	thin

- 9 Large biological molecules are made of smaller molecules.

Which row correctly matches the large molecule with the smaller molecules?

	large molecule	smaller molecules
A	cellulose	fatty acids
B	fat	amino acids
C	protein	amino acids
D	starch	fatty acids

- 10 A student tests a solution for protein.

Which row shows the correct food test and a positive result?

	name of test	colour seen with a positive result
A	Benedict's test	blue-black
B	Benedict's test	purple
C	biuret test	blue-black
D	biuret test	purple

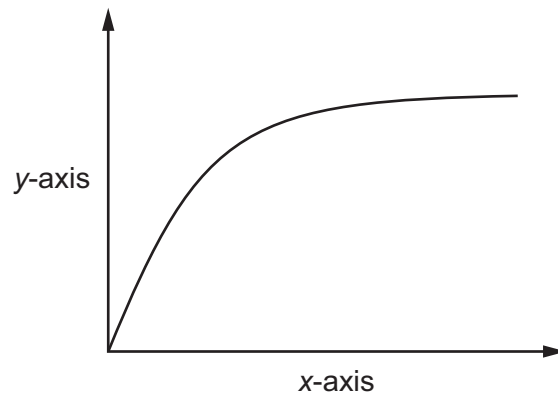
- 11 What is a correct statement about enzymes?

- A** The pH has no effect on the rate of reaction.
- B** They are denatured at low temperatures.
- C** They are made of carbohydrates which have specific shapes.
- D** They speed up chemical reactions.

- 12 Which statement about enzyme action is correct?

- A** The shape of the enzyme's active site is complementary to all proteins.
- B** The shape of the enzyme's active site is complementary to the substrate.
- C** The shape of the substrate's active site is complementary to the product.
- D** The shape of the substrate's active site is complementary to the enzyme.

- 13 The graph shows the results of an experiment that investigated the effect of light intensity on the rate of photosynthesis.

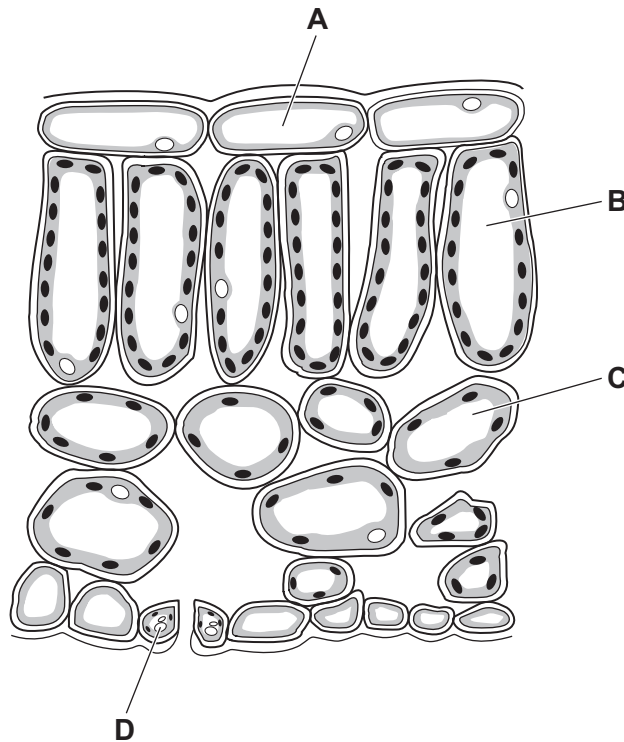


What are the labels for the x-axis and the y-axis?

	x-axis	y-axis
A	light intensity	rate of oxygen production
B	light intensity	rate of carbon dioxide production
C	rate of carbon dioxide production	light intensity
D	rate of oxygen production	light intensity

14 The diagram shows a cross-section of a leaf.

Which type of cell controls the entry of carbon dioxide into the leaf?



15 The table shows vitamins that should be included in a balanced diet.

Which row shows the correct use in the body and food source for one of these vitamins?

	vitamin	use in body	food source
A	C	strengthens bones and teeth	sugar
B	C	resistance to disease	fish oil
C	D	strengthens bones and teeth	fish oil
D	D	resistance to disease	sugar

16 Which substance is an ion that is used to make chlorophyll?

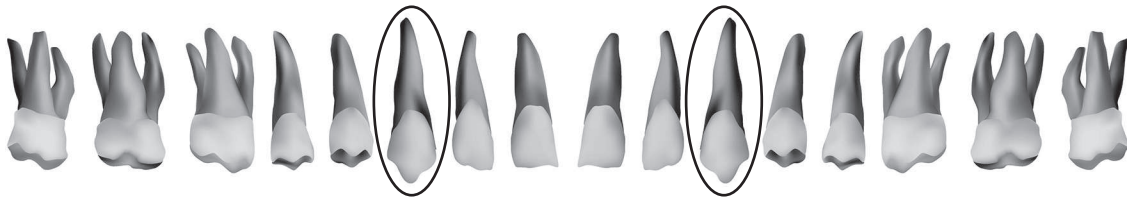
- A** calcium
- B** iron
- C** magnesium
- D** starch

17 What is the function of the gall bladder?

- A production of bile
- B production of gastric juice
- C storage of bile
- D storage of gastric juice

18 The diagram shows the different types of teeth found in the upper jaw of a human.

They are in the same order as they are found in the jaw.



Which type of tooth is circled?

- A canine
- B incisor
- C molar
- D premolar

19 A person ate food X. Chemical digestion of food X occurred in the mouth and small intestine but **not** in the stomach.

What was in food X?

- A fat
- B ions
- C protein
- D starch

20 What is present in a vascular bundle?

- A anthers
- B guard cells
- C palisade cells
- D xylem vessels

21 What is the pathway taken by water from the soil to the leaf?

- A** root hair cells → root cortex → mesophyll → xylem
- B** root hair cells → root cortex → xylem → mesophyll
- C** root cortex → root hair cells → mesophyll → xylem
- D** root cortex → root hair cells → xylem → mesophyll

22 What are methods for monitoring the activity of the heart?

	ECG	measuring pulse rate	listening to sounds of valves closing
A	✓	✗	✗
B	✗	✓	✓
C	✓	✗	✓
D	✓	✓	✓

key

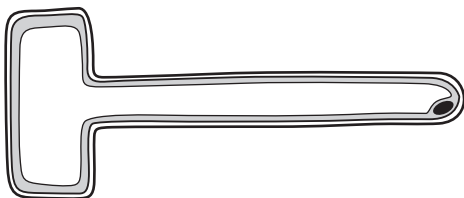
✓ = yes

✗ = no

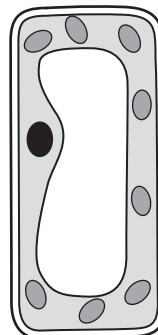
23 The diagrams show four different types of cells.

Which cell contains haemoglobin?

A

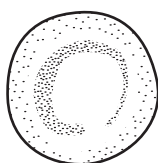


B



NOT TO SCALE

C



D

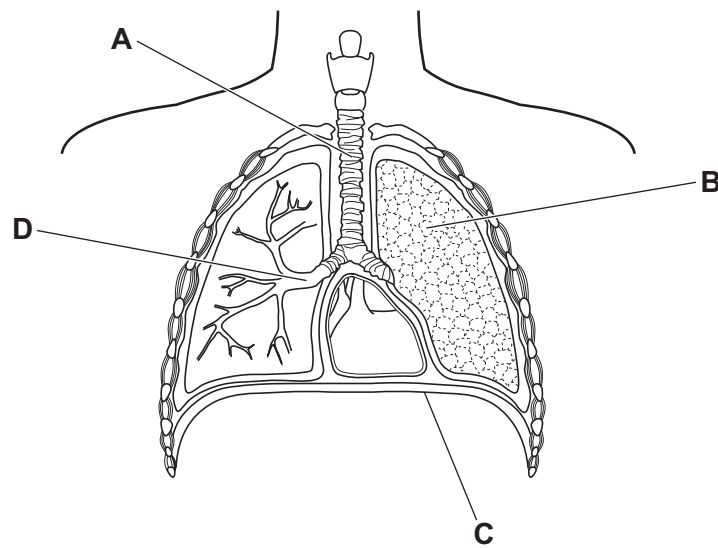


24 What is a direct way of transmitting a pathogen?

- A by contact with contaminated air
- B by contact with contaminated blood
- C by contact with contaminated food
- D by contact with contaminated surfaces

25 The diagram shows the breathing system.

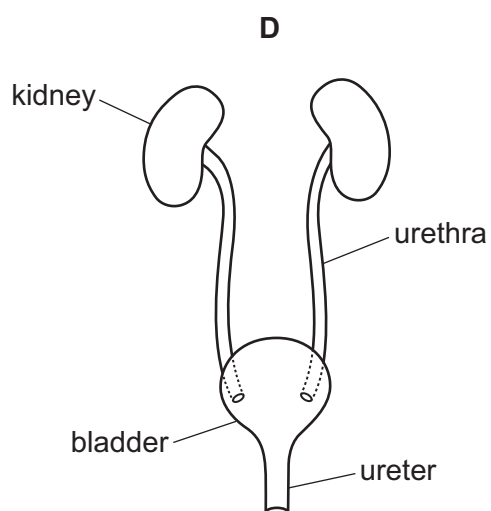
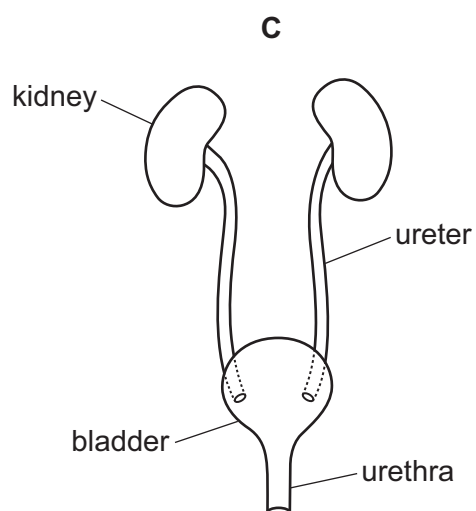
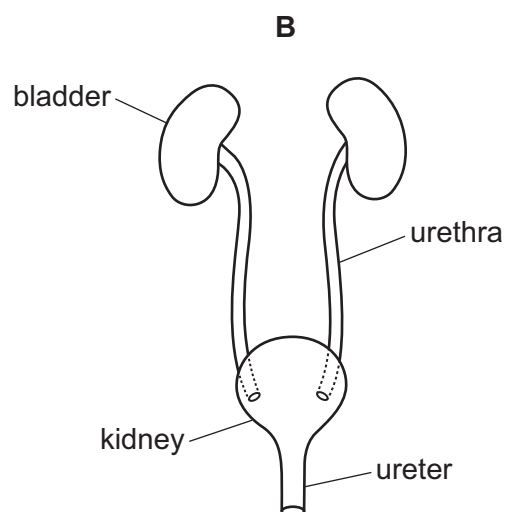
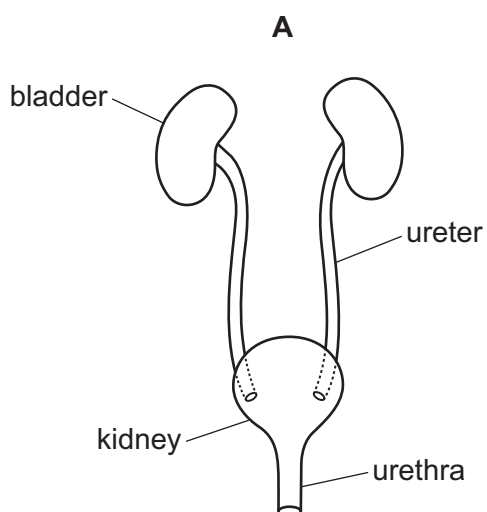
Which structure is the diaphragm?



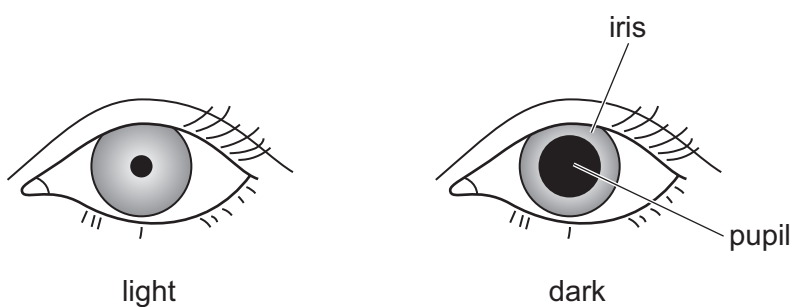
26 What is the word equation for aerobic respiration?

- A carbon dioxide + glucose $\rightarrow$ oxygen + water
- B carbon dioxide + water $\rightarrow$ oxygen + glucose
- C oxygen + glucose $\rightarrow$ carbon dioxide + water
- D oxygen + water $\rightarrow$ carbon dioxide + glucose

27 Which diagram of the excretory system is labelled correctly?



28 The diagram shows an iris and a pupil in different light conditions.



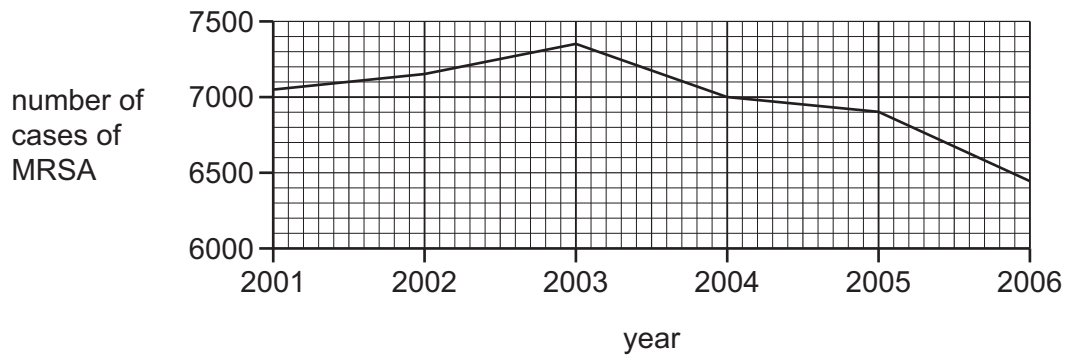
Which characteristic of living organisms is this an example of?

- A growth
- B movement
- C excretion
- D sensitivity

29 Where is insulin secreted, and what is its effect on blood glucose concentration?

	secreted by	effect on blood glucose concentration
A	liver	increases
B	liver	decreases
C	pancreas	increases
D	pancreas	decreases

- 30 The graph shows the number of cases of MRSA in one country between 2001 and 2006.



Between which years was the **greatest** change in the number of cases of MRSA?

- A 2002 and 2003
 - B 2003 and 2004
 - C 2004 and 2005
 - D 2005 and 2006
- 31 Which row describes an insect-pollinated flower?

	anther	pollen grain	stigma
A	inside flower	large and sticky	inside flower
B	outside flower	large and sticky	inside flower
C	inside flower	small and light	outside flower
D	outside flower	small and light	outside flower

- 32 Which part of the female reproductive system releases egg cells?

- A cervix
- B ovary
- C oviduct
- D uterus

33 Which adaptive feature of sperm provides the energy that enables it to swim?

- A** the acrosome
- B** the cell membrane
- C** the flagellum
- D** the mitochondria

34 What is a use of yeast in biotechnology?

- A** aerobic respiration to make biofuels
- B** anaerobic respiration to make bread
- C** aerobic respiration to make alcohol
- D** anaerobic respiration to make human proteins

35 Many mammals have coloured fur and many birds have coloured feathers.

Which uses of colour could help males survive long enough to pass on their features to their offspring?

	hiding from animals which might eat them	hiding from animals which they could eat	warning animals that they are poisonous
A	✓	✓	✓
B	✓	✓	✗
C	✗	✓	✗
D	✓	✗	✓

key

✓ = yes

✗ = no

36 The list shows statements about selection.

- 1 Individuals reproduce and pass on their alleles to the next generation.
- 2 Genetic variation exists between individuals and many offspring are produced.
- 3 Individuals struggle for survival and compete for resources.
- 4 Individuals that are better adapted to the environment survive.
- 5 Individuals with desirable features are selected by humans.

Which statements are correct descriptions of natural selection?

- A** 1, 2, 3 and 4
B 1, 2 and 5 only
C 2, 3 and 4 only
D 3, 4 and 5 only

37 Some of the processes involved in the carbon cycle are listed.

- 1 combustion
- 2 decomposition
- 3 photosynthesis
- 4 respiration

During the carbon cycle, which processes release carbon dioxide into the atmosphere?

- A** 1, 2 and 3 **B** 1, 2 and 4 **C** 1 and 4 only **D** 2 and 3 only

38 What are the effects of sewage pollution on the biodiversity and the number of bacteria in a river?

	biodiversity	number of bacteria
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

39 The increase of which gases contributes to an enhanced greenhouse effect?

- A** carbon dioxide and methane
- B** methane and nitrogen
- C** nitrogen and oxygen
- D** oxygen and carbon dioxide

40 Why are bacteria used in biotechnology?

- A** Bacteria reproduce rapidly.
- B** Bacterial cells contain a nucleus.
- C** Bacterial cells contain cytoplasm.
- D** Bacterial cells do **not** produce complex molecules.

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Cambridge IGCSE™ (9–1)

BIOLOGY (9–1)**0970/11**

Paper 1 Multiple Choice (Core)

May/June 2024**MARK SCHEME**Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	A	1
2	C	1
3	B	1
4	B	1
5	A	1
6	A	1
7	B	1
8	D	1
9	C	1
10	D	1
11	D	1
12	B	1
13	A	1
14	D	1
15	C	1
16	C	1
17	C	1
18	A	1
19	D	1
20	D	1
21	B	1
22	D	1
23	C	1
24	B	1
25	C	1
26	C	1
27	C	1
28	D	1

Question	Answer	Marks
29	D	1
30	D	1
31	A	1
32	B	1
33	D	1
34	B	1
35	A	1
36	A	1
37	B	1
38	B	1
39	A	1
40	A	1



Cambridge IGCSE™ (9–1)

BIOLOGY

0970/12

Paper 1 Multiple Choice (Core)

May/June 2024

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

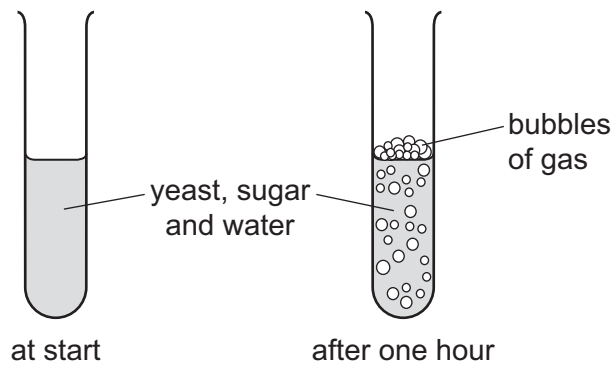
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.



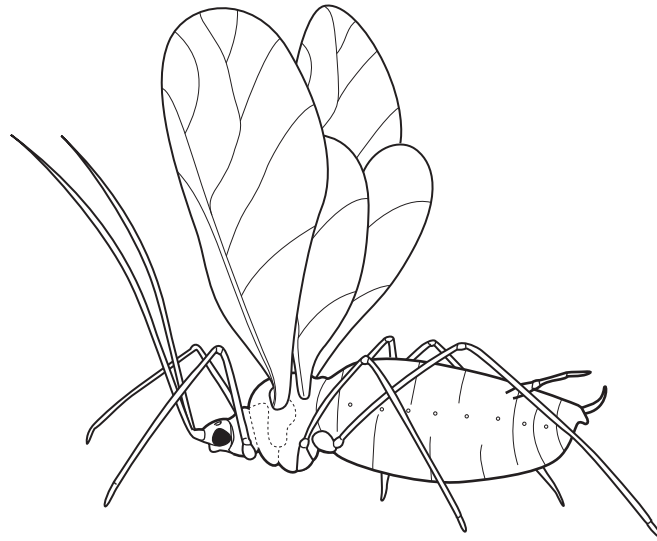
- 1 Some yeast, sugar and water are mixed in a test-tube. The diagrams show the test-tube at the start and after one hour.



Which process causes this change?

- A growth
- B reproduction
- C respiration
- D sensitivity

2 The diagram shows an animal.



Use the key to identify the animal.

1	has two wings	go to 2
	has four wings	go to 3
2	has antennae much shorter than legs	A
	has antennae about the same length as legs	B
3	has feathery antennae	C
	has smooth antennae	D

3 Donkeys and zebras are different species. They can breed to produce an animal called a zedonk.

Zedonks are not fertile.

Which statement is correct?

- A** Zedonks and donkeys are the same species.
- B** Zedonks and zebras are the same species.
- C** Zedonks are a species.
- D** Zedonks are **not** a species.

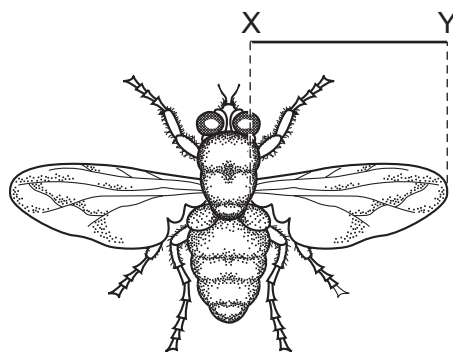
- 4 Crabs are arthropods with 10 jointed legs and two pairs of antennae.

To which arthropod class do they belong?

- A arachnids
 - B crustaceans
 - C insects
 - D myriapods
- 5 Heart muscle cells have a high rate of metabolism.

Which structure do they require to be present in high numbers?

- A chloroplasts
 - B mitochondria
 - C cell walls
 - D vacuoles
- 6 The diagram shows a fly.



The line XY represents the length of the wing.

The length of line XY is 26 mm.

The actual size of the wing between X and Y is 4 mm.

What is the magnification of the diagram?

- A $\times 0.15$
- B $\times 6.5$
- C $\times 22$
- D $\times 104$

7 What is an example of diffusion?

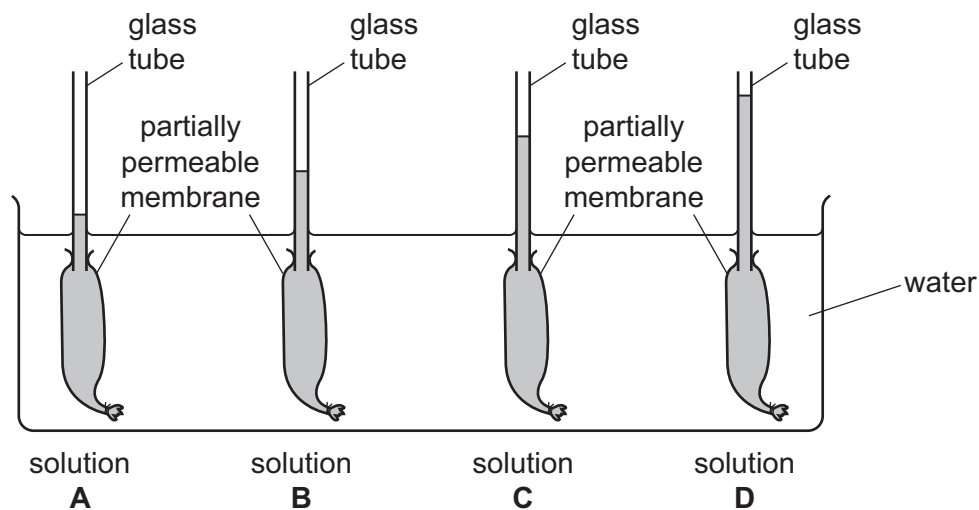
- A** dust particles being moved by ciliated cells in the trachea
- B** oxygen molecules moving into a red blood cell in the lungs
- C** pollen grains moving from anthers to stigmas in the wind
- D** red blood cells moving in a blood capillary in a muscle

8 The diagram shows apparatus used to investigate osmosis.

The volumes of solutions **A**, **B**, **C** and **D** were the same at the start of the investigation.

After one hour, the solutions had moved up the glass tubes.

Which solution was the most concentrated at the start of the investigation?



9 Which row describes active transport?

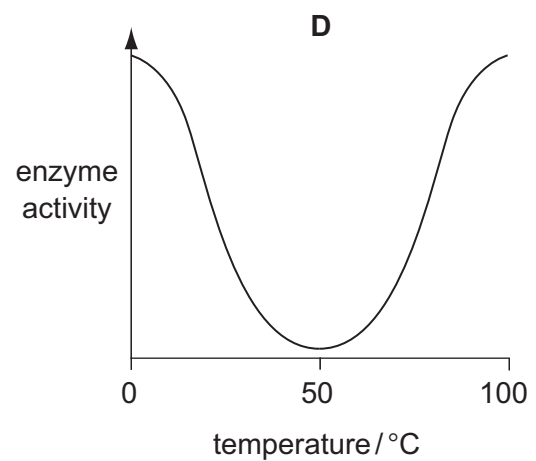
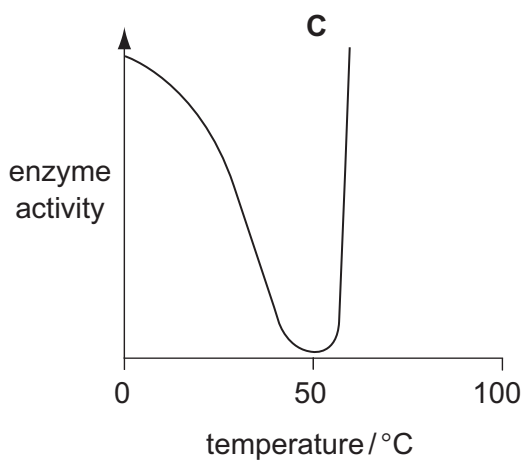
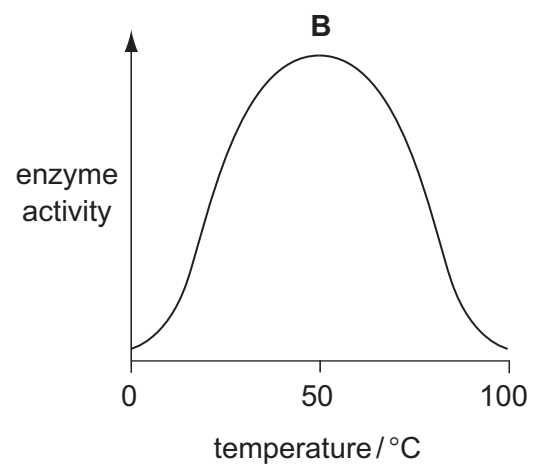
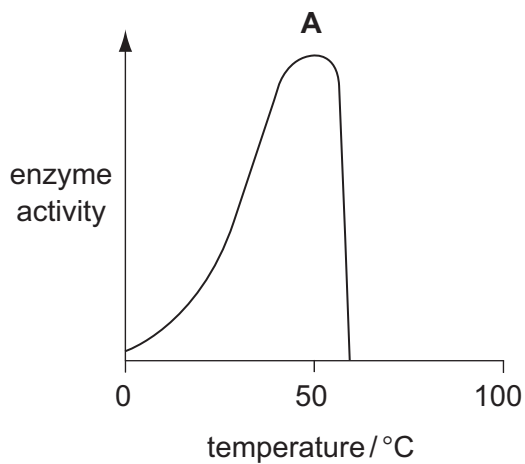
	movement of water	uses energy from respiration	movement through a cell membrane
A	yes	no	no
B	yes	no	yes
C	no	yes	no
D	no	yes	yes

10 Iodine solution is added to a food sample that contains starch.

What is the colour of the sample after the iodine solution has been added?

- A** blue-black
- B** purple
- C** red
- D** yellow

11 Which graph shows the effect of temperature on the activity of an enzyme?



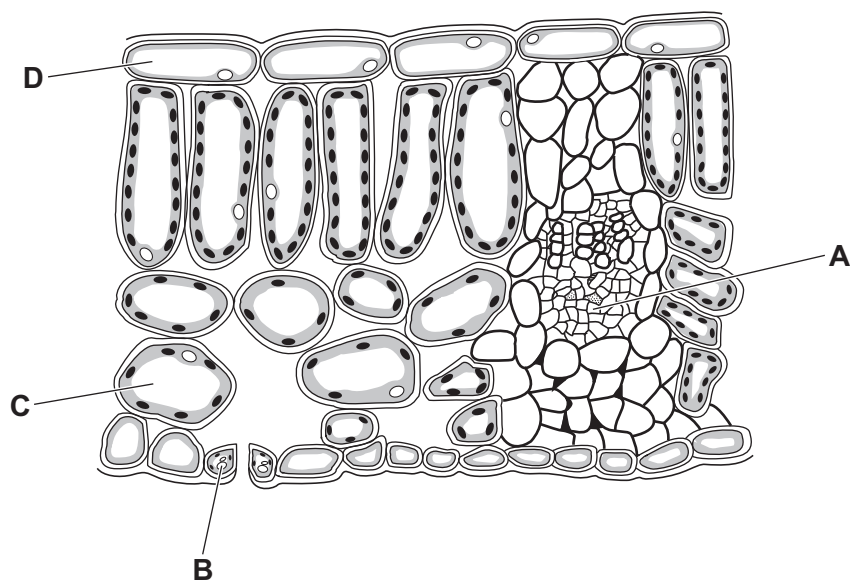
12 Which of these statements about enzymes are correct?

- 1 enzymes are proteins
- 2 enzymes are used up during chemical reactions
- 3 enzymes speed up metabolic reactions

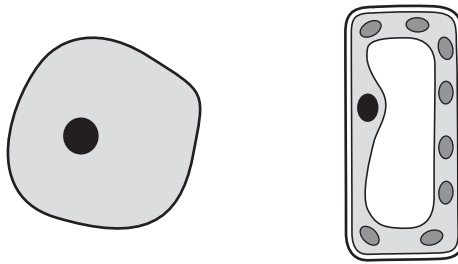
A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

13 The diagram shows a cross-section of part of a leaf.

Which cell controls the rate of gas exchange?



14 The diagram shows two cells.



Which process happens in only one of these cells?

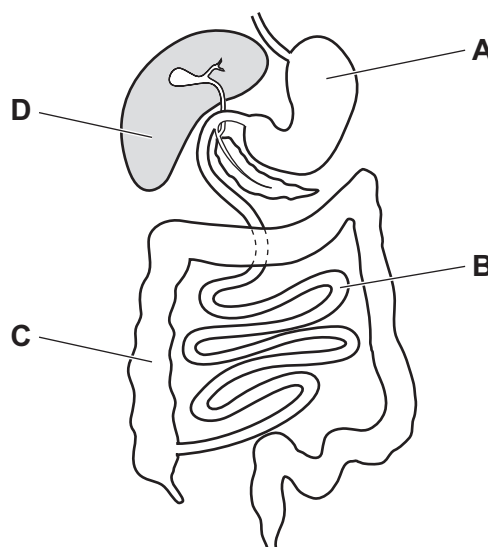
- A** controlling the chemical reactions in the cell
- B** controlling the movement of substances into the cell
- C** making starch inside the cell
- D** using glucose inside the cell

15 Which organ is part of the digestive system?

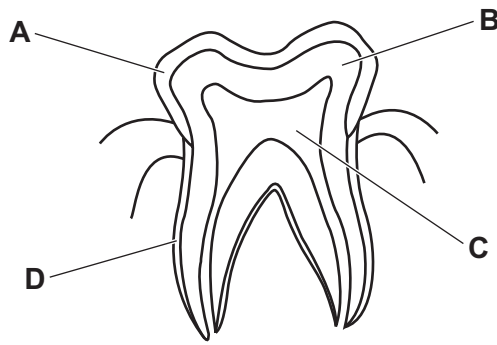
- A** colon
- B** larynx
- C** vena cava
- D** ureter

16 The diagram shows the human alimentary canal.

Where is most digested food absorbed into the blood?

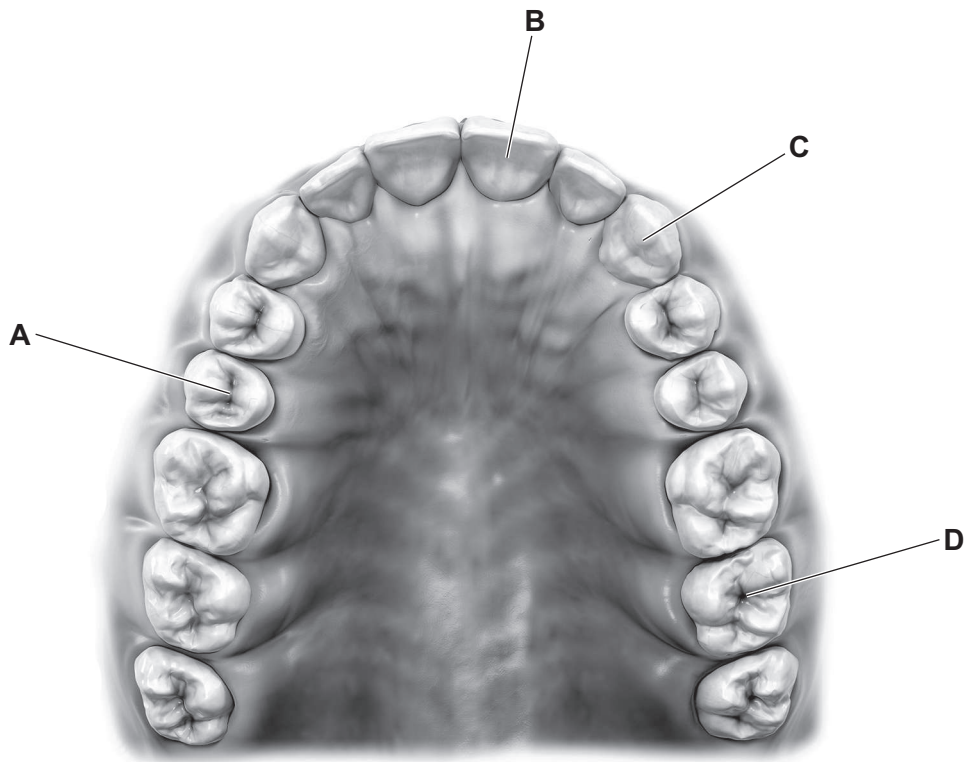


17 Which layer is dentine in the diagram of a human tooth?

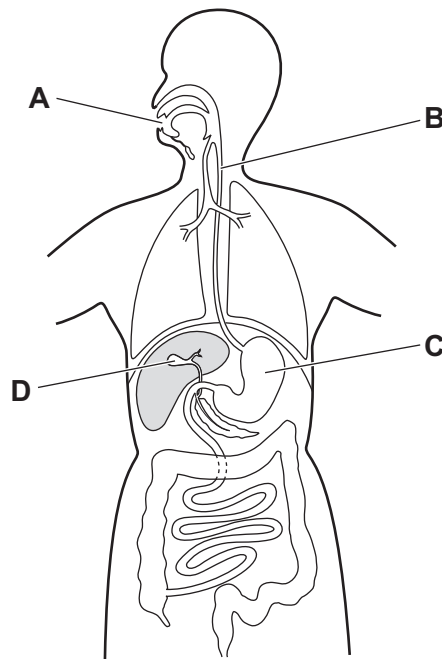


18 The diagram shows a human jaw.

Which labelled tooth is an incisor?



19 Where does protease act?



20 Which statement describes transpiration?

- A** Water is pulled upwards by active transport.
- B** Water is pulled up from the roots by phloem.
- C** Water moves by osmosis inside the xylem vessels.
- D** Water vapour diffuses out of the leaves.

21 Which statement describes the circulation of blood in the human body?

- A** Blood is pumped away from the heart by the atria.
- B** Blood is pumped away from the heart through arteries.
- C** Blood is pumped away from the heart through veins.
- D** Blood is pumped to the heart through capillaries.

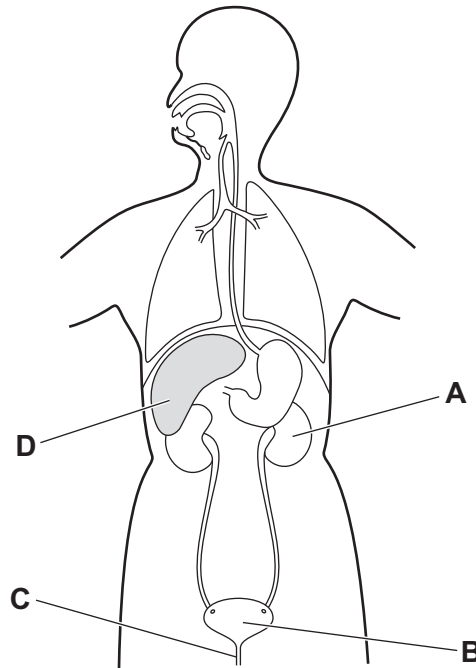
22 Which method of pathogen transmission is direct?

- A** breathing in pathogens in air
- B** eating contaminated food
- C** transfusing contaminated blood
- D** touching a door handle that has pathogens on it

23 Which statement describes anaerobic respiration?

- A** Anaerobic respiration in humans produces lactic acid and carbon dioxide.
- B** Anaerobic respiration in yeast produces water and carbon dioxide.
- C** Anaerobic respiration releases energy from glucose without using oxygen.
- D** Anaerobic respiration releases more energy per glucose molecule than aerobic respiration.

24 Which organ is responsible for the excretion of urea in the human body?



25 Which part of the eye contains light receptors?

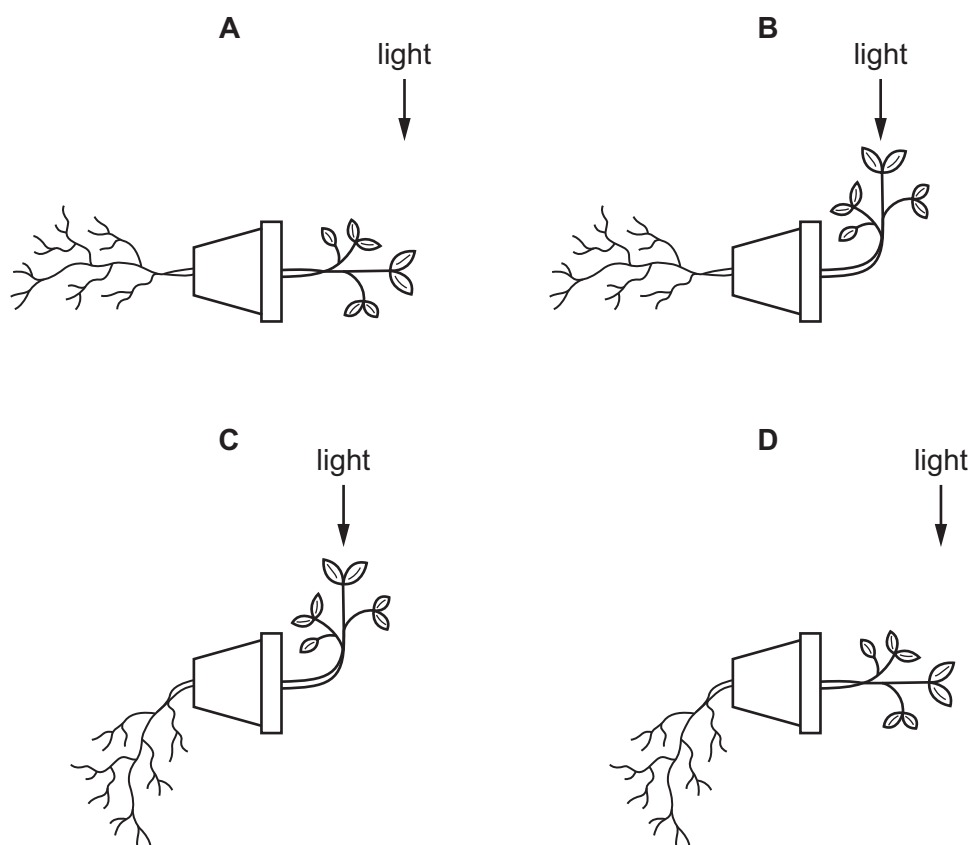
- A** cornea
- B** iris
- C** lens
- D** retina

26 Which row shows the effects of adrenaline secretion?

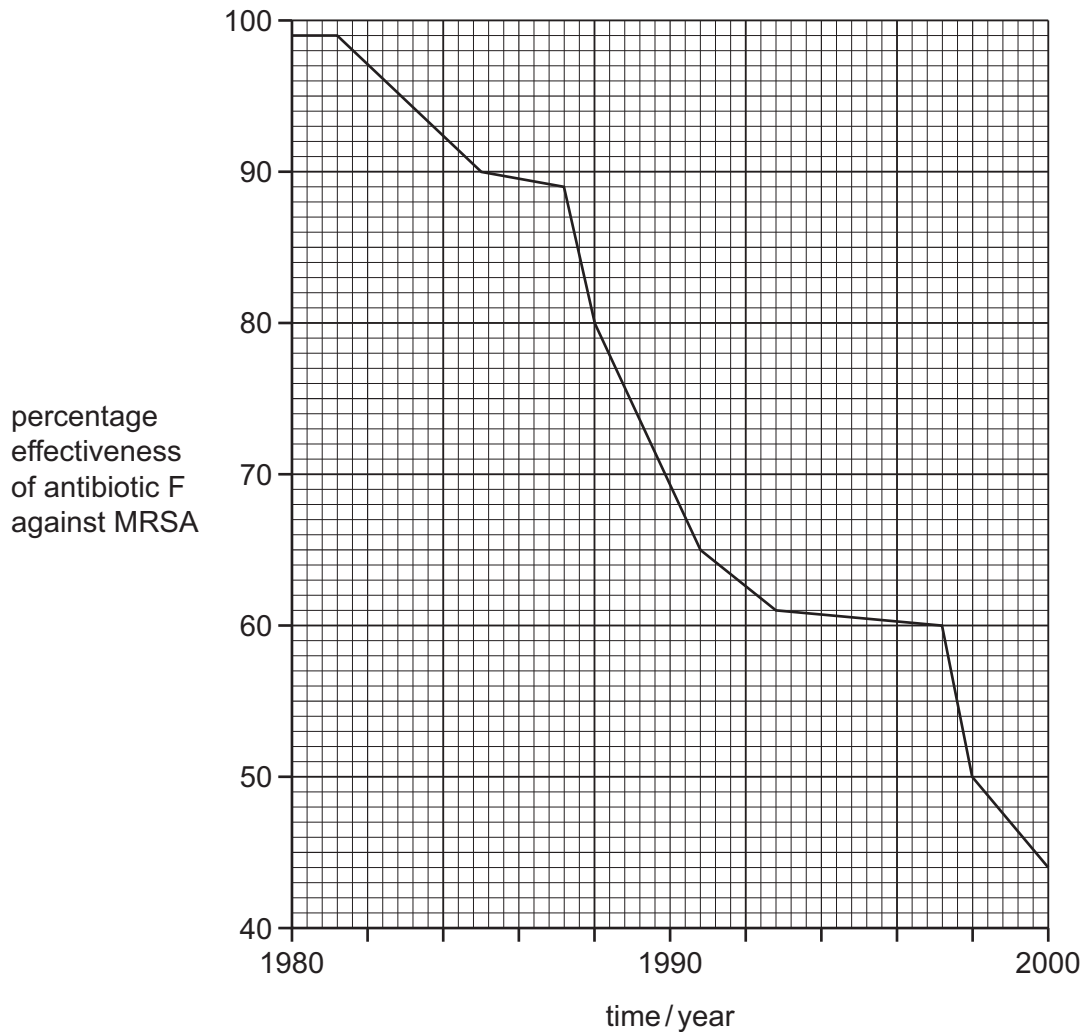
	pupil diameter	heart rate
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

27 A plant pot is placed on its side.

Which diagram shows a plant with roots showing gravitropism and shoots showing phototropism?



- 28 The graph shows how the percentage effectiveness of antibiotic F against MRSA bacteria varied between 1980 and 2000.



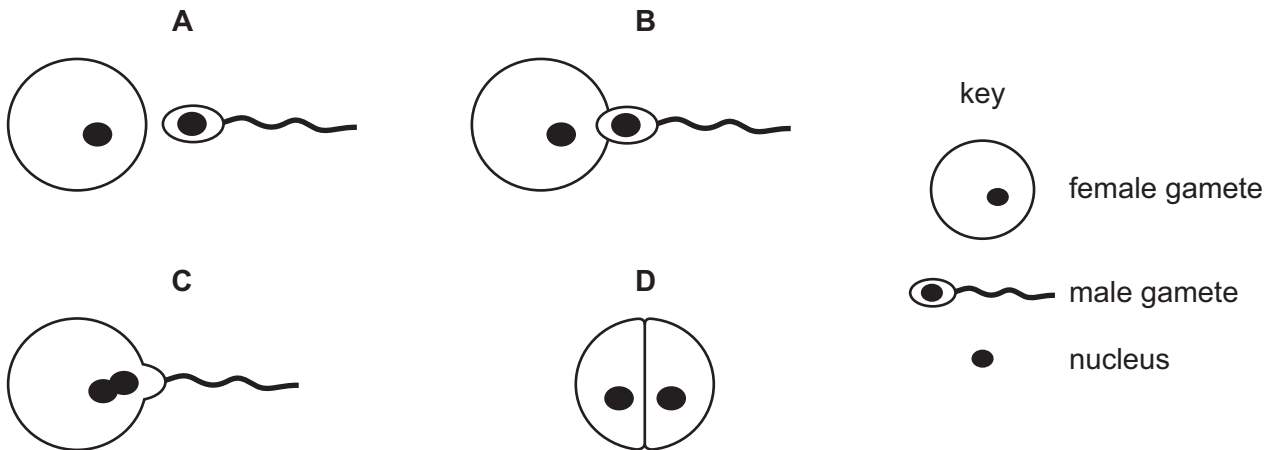
What is a correct conclusion from the graph?

- A Antibiotic F was completely ineffective against MRSA in 2000.
- B Antibiotic F was most effective in 1982.
- C MRSA developed resistance to antibiotic F between 1981 and 2000.
- D The greatest decrease in effectiveness of antibiotic F against MRSA was between 1988 and 1990.
- 29 Which statement describes asexual reproduction?
- A Gametes are involved in this type of reproduction.
- B Offspring are genetically different from each other.
- C Offspring are genetically identical to the parent.
- D Two parents are required for this type of reproduction.

30 What is the function of the acrosome in a sperm cell?

- A The acrosome contains enzymes that digest the jelly coat of an egg cell.
- B The acrosome is needed for the sperm cell to move.
- C The acrosome releases energy.
- D The acrosome stores genetic information.

31 Which diagram shows fertilisation?



32 What is HIV?

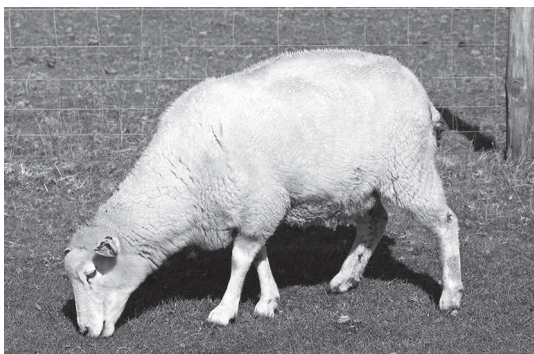
- A a bacterium
- B an enzyme
- C a pathogen
- D an antibody

33 What describes an allele?

- A an alternative form of a chromosome made of DNA
- B an alternative form of a chromosome made of protein
- C an alternative form of a gene made of DNA
- D an alternative form of a gene made of protein

34 The images show four animals.

1



sheep

2



eagle

3



tortoise

4



tiger

Which two animals show adaptive features for being secondary consumers?

A 1 and 3

B 1 and 4

C 2 and 3

D 2 and 4

35 A gene is added to the genetic material of an avocado plant.

This gene makes the avocado fruits stay ripe for longer before they start to decompose.

What is this process an example of?

- 1 adaptation
- 2 genetic modification
- 3 selective breeding

A 1, 2 and 3 **B** 1 and 3 only **C** 2 only **D** 3 only

36 The diagram shows a food chain.

plant leaf → slug → frog → fox

Which terms describe the position of the fox in this food chain?

	consumer	trophic level
A	quaternary	fourth
B	tertiary	third
C	quaternary	third
D	tertiary	fourth

37 These are processes found in the carbon cycle.

- combustion
- decomposition
- feeding
- photosynthesis
- respiration

How many of these processes release carbon dioxide into the atmosphere?

A 1 **B** 2 **C** 3 **D** 4

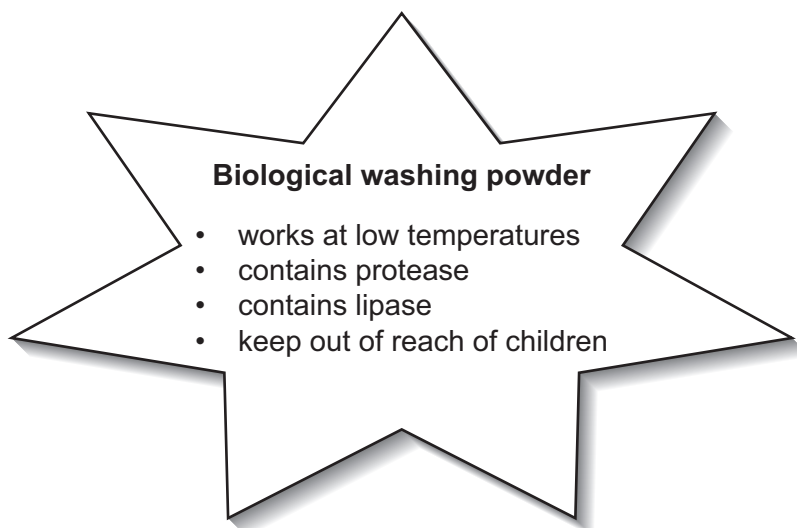
38 Which change occurs when areas of forest are cleared for farming?

- A decreased release of carbon dioxide into the atmosphere
- B increased biodiversity
- C increased loss of soil
- D increased release of oxygen into the atmosphere

39 What is a use of ethanol?

- A biofuel production
- B biological washing powder
- C bread-making
- D fruit juice production

40 The diagram shows a label found on a container of biological washing powder.



Which stains does the washing powder remove?

	food stain containing carbohydrates	food stain containing protein	food stain containing fat	food stain containing oil
A	✓	✓	x	x
B	✓	x	x	✓
C	x	✓	✓	✓
D	x	x	✓	x

key

✓ = yes

x = no

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Cambridge IGCSE™ (9–1)

BIOLOGY (9–1)**0970/12**

Paper 1 Multiple Choice (Core)

May/June 2024**MARK SCHEME**Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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This document consists of **3** printed pages.

Question	Answer	Marks
1	C	1
2	D	1
3	D	1
4	B	1
5	B	1
6	B	1
7	B	1
8	D	1
9	D	1
10	A	1
11	A	1
12	C	1
13	B	1
14	C	1
15	A	1
16	B	1
17	B	1
18	B	1
19	C	1
20	D	1
21	B	1
22	C	1
23	C	1
24	A	1
25	D	1
26	D	1
27	C	1
28	C	1

Question	Answer	Marks
29	C	1
30	A	1
31	C	1
32	C	1
33	C	1
34	D	1
35	C	1
36	D	1
37	C	1
38	C	1
39	A	1
40	C	1



Cambridge IGCSE™ (9–1)

BIOLOGY

0970/12

Paper 1 Multiple Choice (Core)

October/November 2024

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.



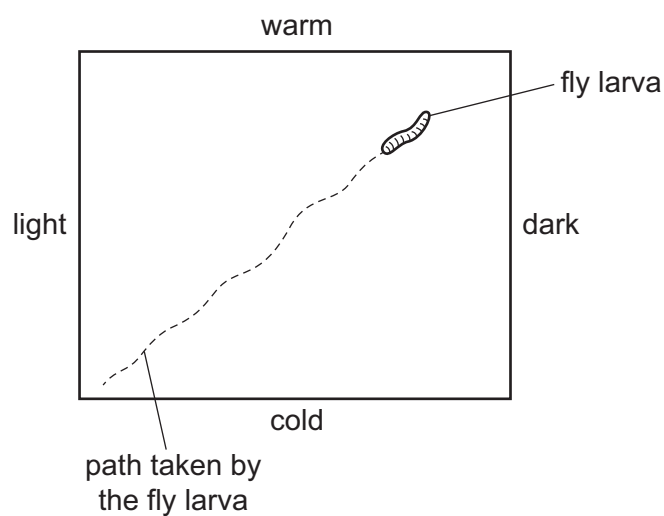
1 Which type of cell moves mucus?

- A ciliated cell
- B egg cell
- C guard cell
- D root hair cell

2 A student exposed a fly larva to light and temperature gradients.

They lit the area from one side and made one side of the area warmer than the other.

They traced the path of the fly larva's movement.



Which statement describes the response of the fly larva?

- A The fly larva does **not** respond to temperature or light intensity.
- B The fly larva moves away from a higher temperature and a higher light intensity.
- C The fly larva moves away from a higher temperature and towards a higher light intensity.
- D The fly larva moves towards a higher temperature and away from a higher light intensity.

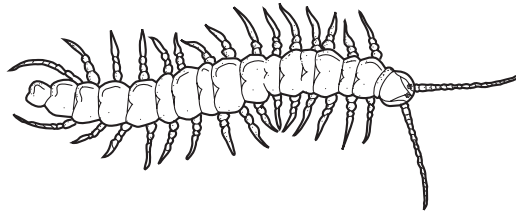
- 3 The table shows the scientific names and the common names of four plants.

scientific name	common name
<i>Althaea officinalis</i>	marsh mallow
<i>Hottonia palustris</i>	water violet
<i>Viola palustris</i>	marsh violet
<i>Viola tricolor</i>	wild pansy

Which word is a genus name?

- A marsh
- B *palustris*
- C *Viola*
- D violet

- 4 The diagram shows an arthropod.

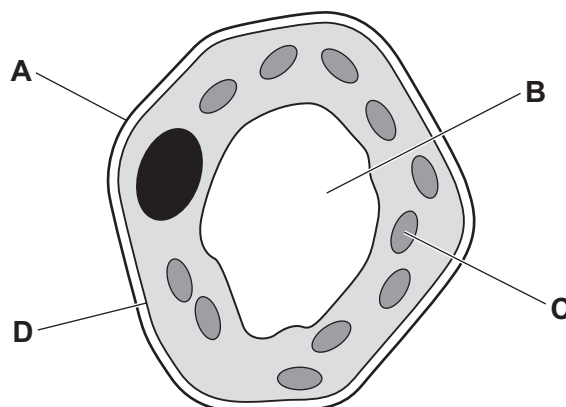


Which group does this arthropod belong to?

- A arachnid
- B crustacean
- C insect
- D myriapod

- 5 The diagram shows a plant cell.

Which letter indicates the location of the cell membrane?



6 Which magnification formula is correct?

- A magnification = actual size $\div$ image size
- B magnification = actual size + image size
- C magnification = image size $\div$ actual size
- D magnification = image size $\times$ actual size

7 A frog is an animal. The skin of a frog is permeable to oxygen and carbon dioxide.

Which directions will have a net diffusion of oxygen and carbon dioxide when a frog swims in pond water?

	from the frog into the water	from the water into the frog
A	carbon dioxide	oxygen
B	carbon dioxide and oxygen	no movement
C	oxygen	carbon dioxide
D	no movement	carbon dioxide and oxygen

8 Which statement about plant support is correct?

- A The pressure of water inside the cells presses inwards on the cell walls.
- B The pressure of water inside the cells presses outwards on the cell walls.
- C The pressure of water outside the cells presses inwards on the cell walls.
- D The pressure of water outside the cells presses outwards on the cell walls.

9 What is a correct statement about active transport?

- A Particles move into cells, using energy released by photosynthesis.
- B Particles move across a cell wall without using energy.
- C Particles move from a region of higher concentration to a region of lower concentration.
- D Particles move across a cell membrane, using energy released by respiration.

10 A sample of food was tested with biuret reagent.

Which colour would be seen if the food sample contained protein?

- A blue-black
- B brick red
- C yellow-brown
- D purple

11 Which elements are found in fats?

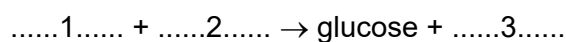
- A carbon, hydrogen and iron
- B carbon, hydrogen and oxygen
- C carbon, hydrogen and nitrogen
- D carbon, oxygen and nitrogen

12 Which statements about enzymes are correct?

- 1 Enzymes are biological catalysts.
- 2 Enzymes are carbohydrates.
- 3 Enzymes have an active site.

- A 1 and 2 B 1 and 3 C 2 and 3 D 3 only

13 The equation for photosynthesis is shown.



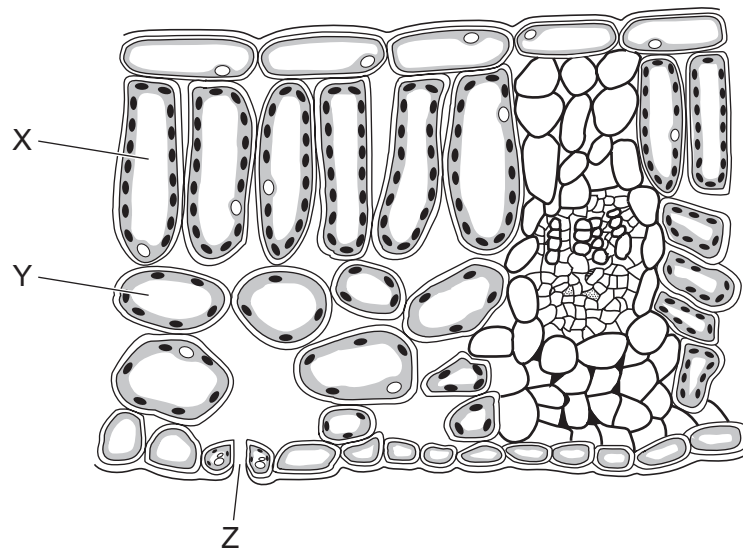
Which words correctly complete gaps 1, 2 and 3?

	1	2	3
A	carbon dioxide	light	oxygen
B	carbon dioxide	water	oxygen
C	oxygen	light	carbon dioxide
D	oxygen	water	carbon dioxide

14 Which carbohydrate is used to make cell walls?

- A cellulose
- B glycogen
- C starch
- D sucrose

15 The diagram shows a cross-section of a leaf.



Which row shows the correct labels?

	palisade mesophyll cell	spongy mesophyll cell	stoma
A	X	Y	Z
B	X	Z	Y
C	Y	X	Z
D	Y	Z	X

16 The table shows some of the nutrients found in yoghurt.

nutrient	content per 100 g of yoghurt / g
carbohydrate	6.90
protein	4.80
fat	1.50
calcium	0.16

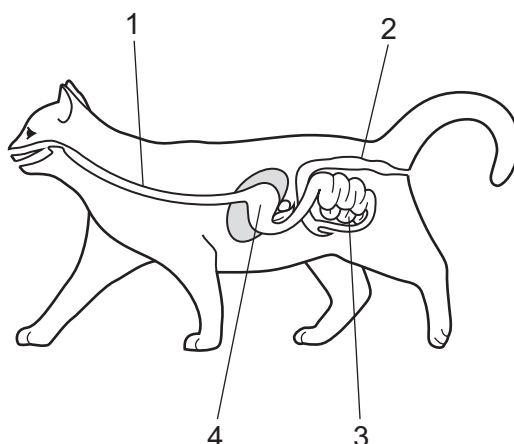
One serving size of yoghurt is 150 g.

One of the nutrients in the yoghurt is required for healthy bones and teeth.

What is the mass of this nutrient in one serving of yoghurt?

- A** 0.24 g **B** 2.25 g **C** 7.20 g **D** 10.35 g

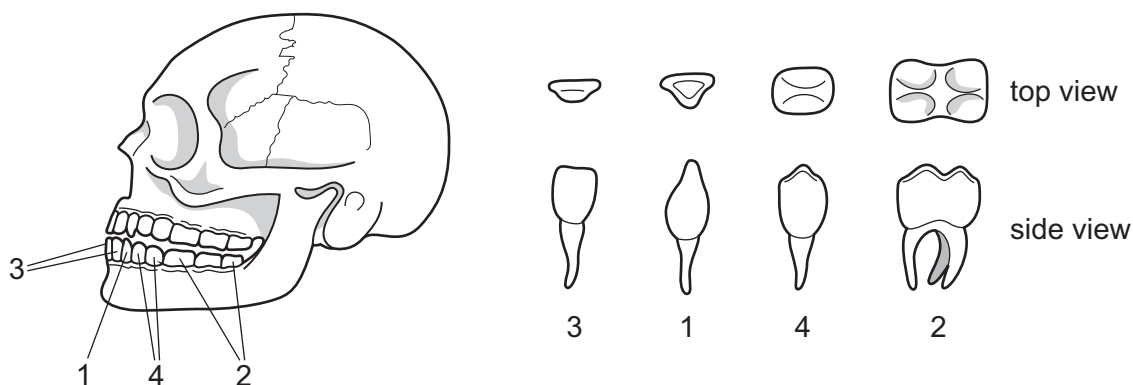
- 17 The diagram shows the alimentary canal and some of the associated organs in a cat. The alimentary canal of a cat is similar to that of a human.



Which row identifies the structures labelled 1–4?

	1	2	3	4
A	oesophagus	anus	large intestine	liver
B	oesophagus	large intestine	small intestine	stomach
C	trachea	anus	large intestine	stomach
D	trachea	large intestine	small intestine	liver

- 18 The diagram shows the four types of teeth in a human jaw.



Which row correctly identifies all four types of teeth?

	1	2	3	4
A	canine	incisor	molar	premolar
B	canine	molar	incisor	premolar
C	premolar	incisor	molar	canine
D	premolar	molar	incisor	canine

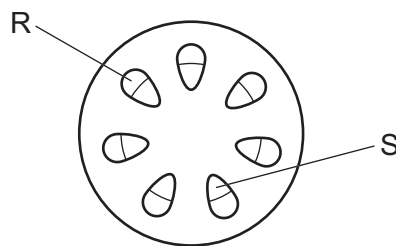
19 What are the products of protein digestion in the stomach?

- A** amino acids
- B** fatty acids
- C** glycerol
- D** proteases

20 Where is water absorbed?

- A** colon and small intestine
- B** large intestine only
- C** gall bladder and colon
- D** small intestine only

21 The diagram shows a cross-section of a stem.



Which statement correctly describes the location and function of phloem?

- A** Structure R is phloem and transports amino acids and starch.
- B** Structure S is phloem and transports amino acids and starch.
- C** Structure R is phloem and transports amino acids and sucrose.
- D** Structure S is phloem and transports amino acids and sucrose.

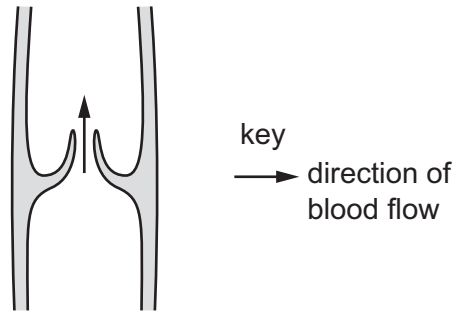
22 Some environmental factors are listed.

- 1 carbon dioxide concentration
- 2 temperature
- 3 oxygen concentration
- 4 wind speed

Which factors directly affect the rate of transpiration?

- A** 1 and 3
- B** 1 and 4
- C** 2 and 3
- D** 2 and 4

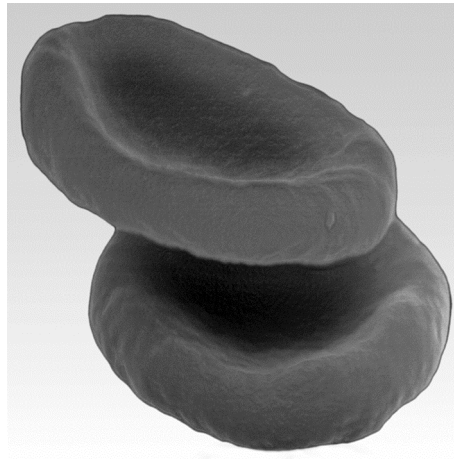
23 The diagram shows part of a blood vessel.



Which evidence in the diagram shows that it is a vein and **not** an artery?

- A There is a valve inside the blood vessel.
- B Blood is flowing in an upwards direction.
- C The blood vessel has a relatively small lumen.
- D The blood vessel has a relatively thick wall.

24 The photomicrograph shows two blood cells.



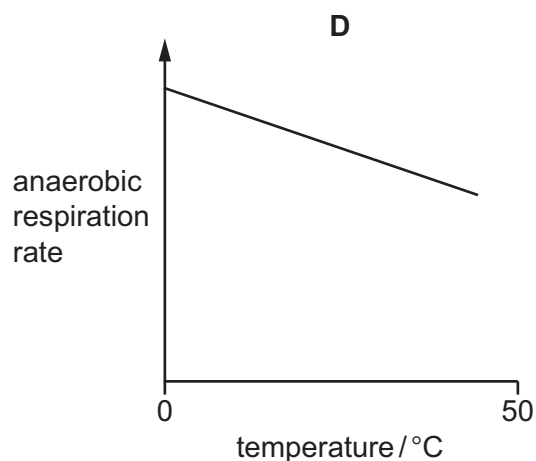
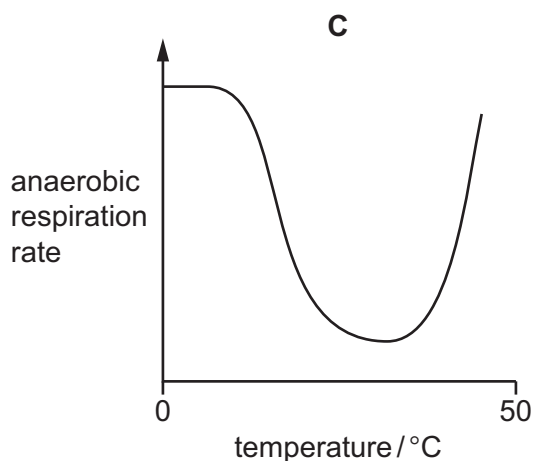
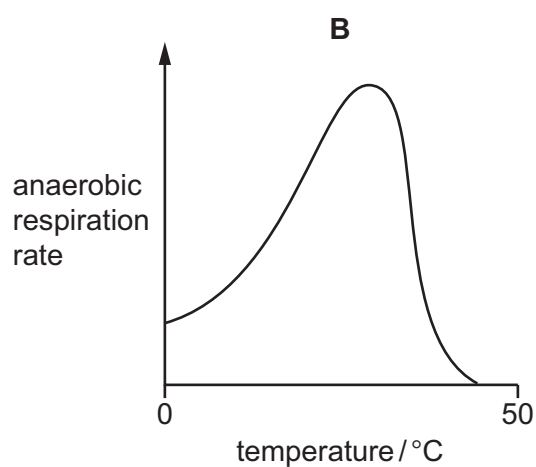
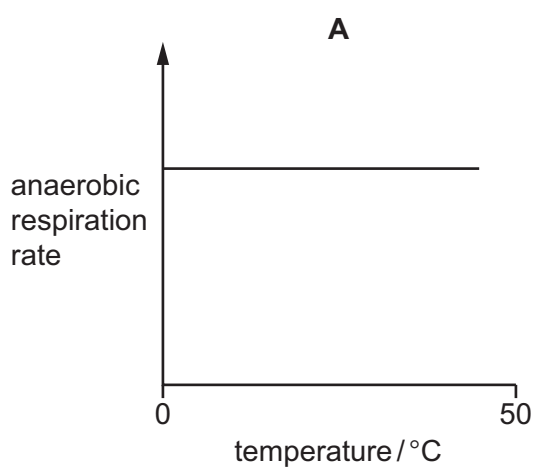
What is the function of these cells?

- A to carry out phagocytosis
- B to produce antibodies
- C to transport oxygen
- D to transport vitamins

- 25 Which row matches a gas in expired air with a substance that can be used to test for the presence of that gas?

	gas present in expired air	test substance
A	carbon dioxide	DCPIP
B	carbon dioxide	limewater
C	oxygen	DCPIP
D	oxygen	limewater

- 26 Which graph shows how temperature affects the rate of anaerobic respiration in yeast?



- 27 What is the pathway of a reflex arc from the receptor to the effector?

- A** motor neurone → relay neurone → sensory neurone
- B** motor neurone → sensory neurone → relay neurone
- C** sensory neurone → motor neurone → relay neurone
- D** sensory neurone → relay neurone → motor neurone

28 What is an effect of adrenaline secretion?

- A decreased breathing rate
- B decreased pupil diameter
- C development of secondary sexual characteristics
- D increased heart rate

29 What is phototropism?

- A absorbing mineral ions
- B absorbing water
- C directional growth in response to gravity
- D directional growth in response to light

30 A gardener wants to grow some new plants that are identical to plants that she already has.

Which method of reproduction is used, and how many parents are needed for her to grow some new identical plants?

	method of reproduction	number of parents needed
A	asexual	one
B	asexual	two
C	sexual	one
D	sexual	two

31 What must **always** be present for seeds to germinate?

- A chlorophyll
- B light
- C soil
- D water

32 What is a flagellum?

- A a ball of cells that implants into the lining of the uterus
- B a cell structure that releases energy
- C a cell that forms after the sperm cell fuses with the egg cell
- D an adaptive feature used for movement of sperm cells

33 Which word describes a fertilised cell?

- A** acrosome
- B** embryo
- C** gamete
- D** zygote

34 Which secondary sexual characteristic is caused by both testosterone and oestrogen?

- A** breast development
- B** changes in body shape
- C** growth of facial hair
- D** sudden deepening of voice

35 Some ways that diseases can be transmitted are listed.

- 1 airborne droplets
- 2 blood and needles
- 3 food and drink
- 4 unprotected sex

Which ways can transmit HIV?

- A** 1 and 3 **B** 1 only **C** 2 and 4 **D** 4 only

36 What do genes code for?

- A** fats
- B** proteins
- C** starch
- D** sugars

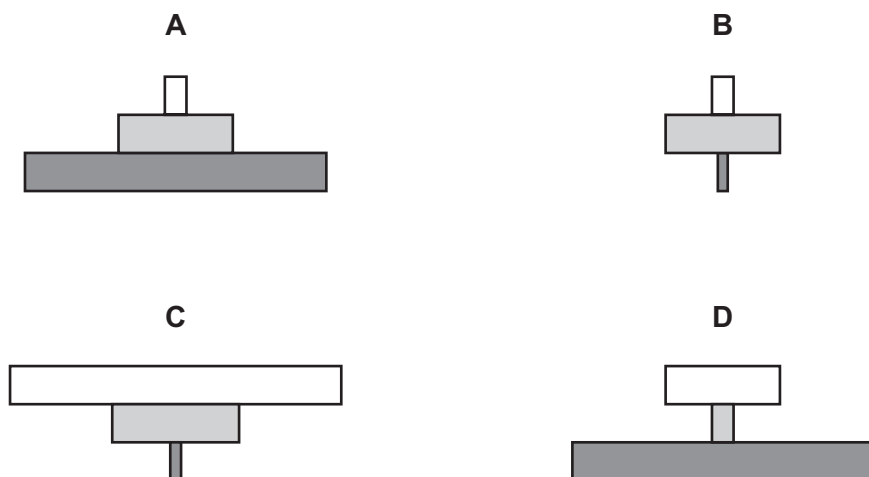
- 37** When white-flowered pea plants are crossed with red-flowered pea plants, the offspring plants all have red flowers.

If these offspring plants are crossed, the next generation of plants contains both red-flowered and white-flowered plants.

Which statement explains this?

- A** The allele for red flowers is dominant, and the offspring plants are heterozygous.
 - B** The allele for red flowers is dominant, and the offspring plants are homozygous.
 - C** The allele for red flowers is recessive, and the offspring plants are heterozygous.
 - D** The allele for red flowers is recessive, and the offspring plants are homozygous.
- 38** What is a description of an adaptive feature?
- A** an inherited characteristic that helps an organism to survive and reproduce
 - B** the differences between individuals of the same species
 - C** the genetic make-up of an organism
 - D** the transmission of information from generation to generation

- 39** Which diagram shows a pyramid of biomass for the food chain shown?



- 40** Which statement about conservation is correct?
- A** Captive breeding programmes can be used to conserve endangered species.
 - B** Climate change is reducing the number of species that are endangered.
 - C** Habitat protection increases the number of species that are endangered.
 - D** Sustainable resources are those that are used more rapidly than they are replaced.

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Cambridge IGCSE™ (9–1)

BIOLOGY**0970/12**

Paper 1 Multiple Choice (Core)

October/November 2024**MARK SCHEME**Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	A	1
2	D	1
3	C	1
4	D	1
5	D	1
6	C	1
7	A	1
8	B	1
9	D	1
10	D	1
11	B	1
12	B	1
13	B	1
14	A	1
15	A	1
16	A	1
17	B	1
18	B	1
19	A	1
20	A	1
21	C	1
22	D	1
23	A	1
24	C	1
25	B	1
26	B	1
27	D	1
28	D	1

Question	Answer	Marks
29	D	1
30	A	1
31	D	1
32	D	1
33	D	1
34	B	1
35	C	1
36	B	1
37	A	1
38	A	1
39	A	1
40	A	1



Cambridge IGCSE™ (9–1)

BIOLOGY

0970/11

Paper 1 Multiple Choice (Core)

May/June 2025

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
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- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

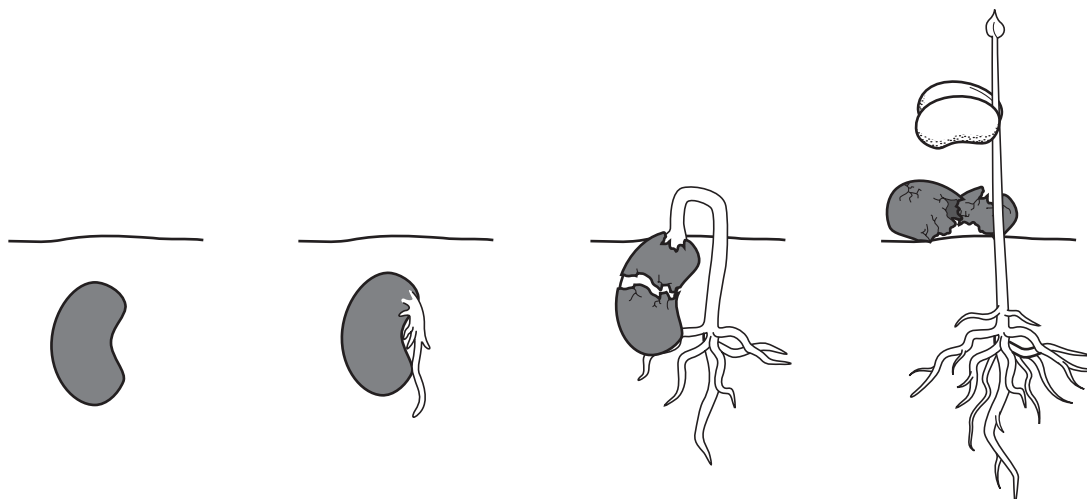
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.



- 1 The diagram shows how a seed changes after it is planted in soil and watered.



Which characteristics of living things are demonstrated by the sequence shown in the diagram?

- A** growth only
B growth and sensitivity
C nutrition and reproduction
D reproduction and sensitivity
- 2 Which row describes each part of the binominal system name for humans?

	<i>Homo</i>	<i>sapiens</i>
A	genus	kingdom
B	genus	species
C	species	genus
D	species	kingdom

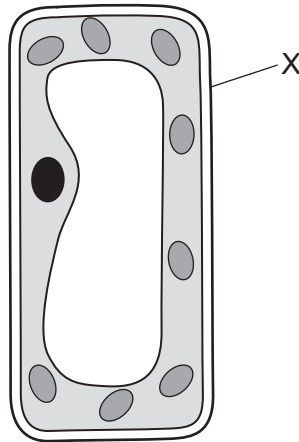
- 3 What are the characteristics of fish?

	maintain a constant body temperature	eggs covered in jelly	external ears present	scales present
A	yes	no	yes	no
B	yes	yes	no	yes
C	no	no	yes	no
D	no	yes	no	yes

4 Which type of cell transports oxygen?

- A ciliated cell
- B neurone
- C palisade mesophyll cell
- D red blood cell

5 The diagram shows a plant cell.



What is the function of the part labelled X?

- A controls what enters and leaves the cell
 - B strength and support
 - C photosynthesis
 - D protein synthesis
- 6 Four students used a microscope to view some onion plant cells. Each student drew one of the cells.

Each student calculated the actual length of the cell and the magnification of their drawing.

Which student produced the longest drawing of a cell?

	actual length of onion plant cell / mm	magnification of drawing
A	0.120	$\times 150$
B	0.125	$\times 300$
C	0.150	$\times 225$
D	0.200	$\times 175$

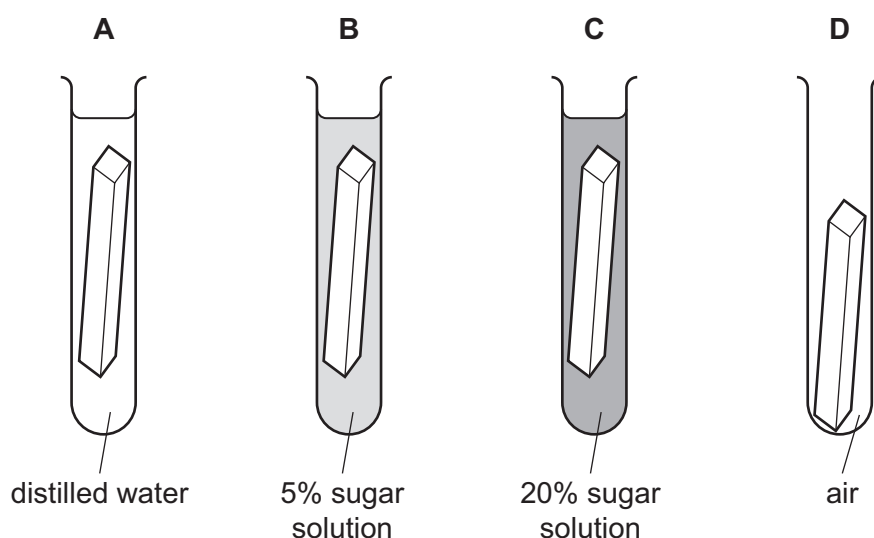
- 7 The table shows the concentration of gases in a blood vessel and in an alveolus.

Which row shows the conditions that cause a substance **produced** in respiration in humans to diffuse from the blood vessel into the alveolus?

	substance produced in respiration	substance concentration in the blood vessel	substance concentration in the alveolus
A	carbon dioxide	low	high
B	carbon dioxide	high	low
C	oxygen	low	high
D	oxygen	high	low

- 8 The diagram shows four identical pieces of potato plant tissue in test-tubes. The pieces of potato were kept in the test-tubes for six hours.

Which piece of potato would have the greatest increase in mass after six hours?



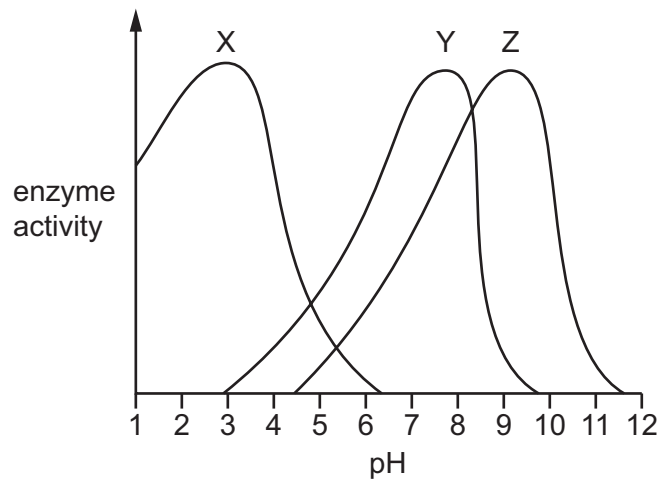
- 9 Which row correctly shows the elements and small molecules that are found in a large molecule?

	elements	small molecules	large molecule
A	carbon, hydrogen, oxygen, nitrogen	glucose	fat
B	carbon, hydrogen, oxygen, nitrogen	glycerol	fat
C	carbon, hydrogen, oxygen	glucose	starch
D	carbon, hydrogen, oxygen	glycerol	starch

10 Which substance is the DCPIP test used for?

- A fat
- B protein
- C starch
- D vitamin C

11 The graph shows the activity of three digestive enzymes at differing pH values.



Which statement is correct?

- A Enzymes X and Y are both active at pH 7.
- B Enzymes X and Z are both active at pH 4.
- C Enzymes Y and Z are both active at pH 4.
- D Enzymes Y and Z are both active at pH 8.

- 12 A student drew a diagram to show the substances used and produced in photosynthesis in a leaf.

1 + 2 are used by the leaf

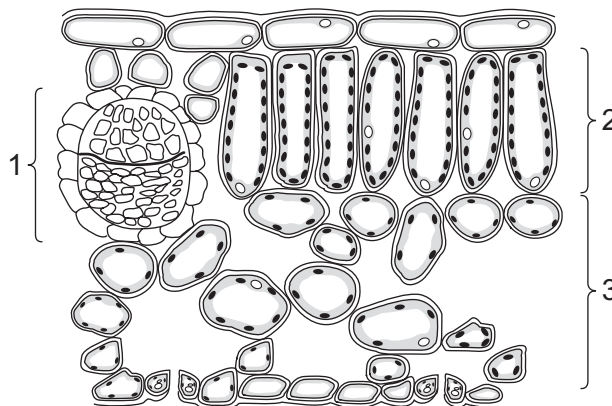


3 + 4 are produced by the leaf

Which row shows the correct labels for the diagram?

	1	2	3	4
A	carbon dioxide	glucose	oxygen	water
B	water	carbon dioxide	glucose	oxygen
C	oxygen	water	carbon dioxide	glucose
D	glucose	oxygen	water	carbon dioxide

- 13 The diagram shows a cross-section of part of a leaf.



What are the names of regions 1, 2 and 3?

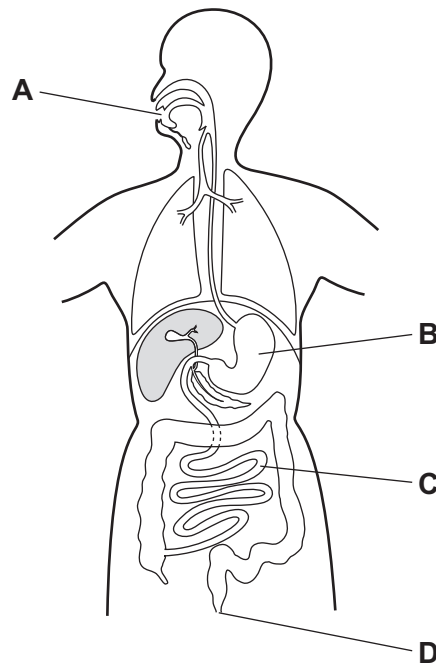
	1	2	3
A	palisade mesophyll	vascular bundle	spongy mesophyll
B	spongy mesophyll	palisade mesophyll	vascular bundle
C	vascular bundle	palisade mesophyll	spongy mesophyll
D	vascular bundle	spongy mesophyll	palisade mesophyll

14 Why is iron an important component of the human diet?

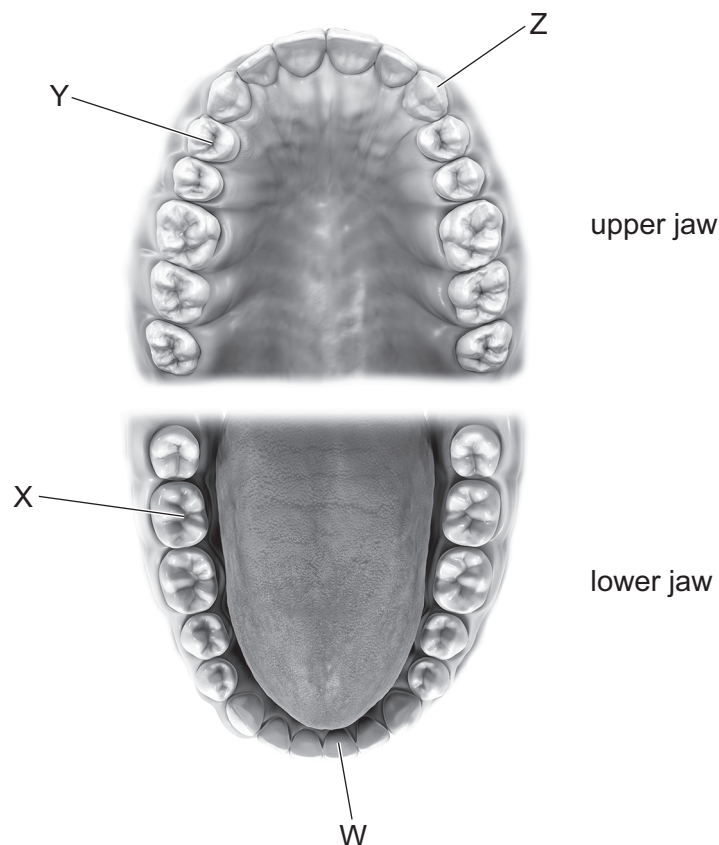
- A** Iron is needed to make haemoglobin.
- B** Iron is needed to prevent rickets.
- C** Iron is needed to prevent scurvy.
- D** Iron is needed to strengthen teeth.

15 The diagram shows the digestive system.

Where does egestion take place?



16 The diagram shows four different types of teeth in the upper and lower jaws in a human mouth.



Which letters identify the types of teeth shown in the diagram?

	W	X	Y	Z
A	canine	premolar	molar	incisor
B	canine	molar	premolar	incisor
C	incisor	premolar	molar	canine
D	incisor	molar	premolar	canine

17 Which row shows an enzyme with the correct site of production and products of digestion?

	enzyme	site of production	products of digestion
A	amylase	salivary glands	amino acids
B	amylase	stomach	simple reducing sugars
C	protease	salivary glands	simple reducing sugars
D	protease	stomach	amino acids

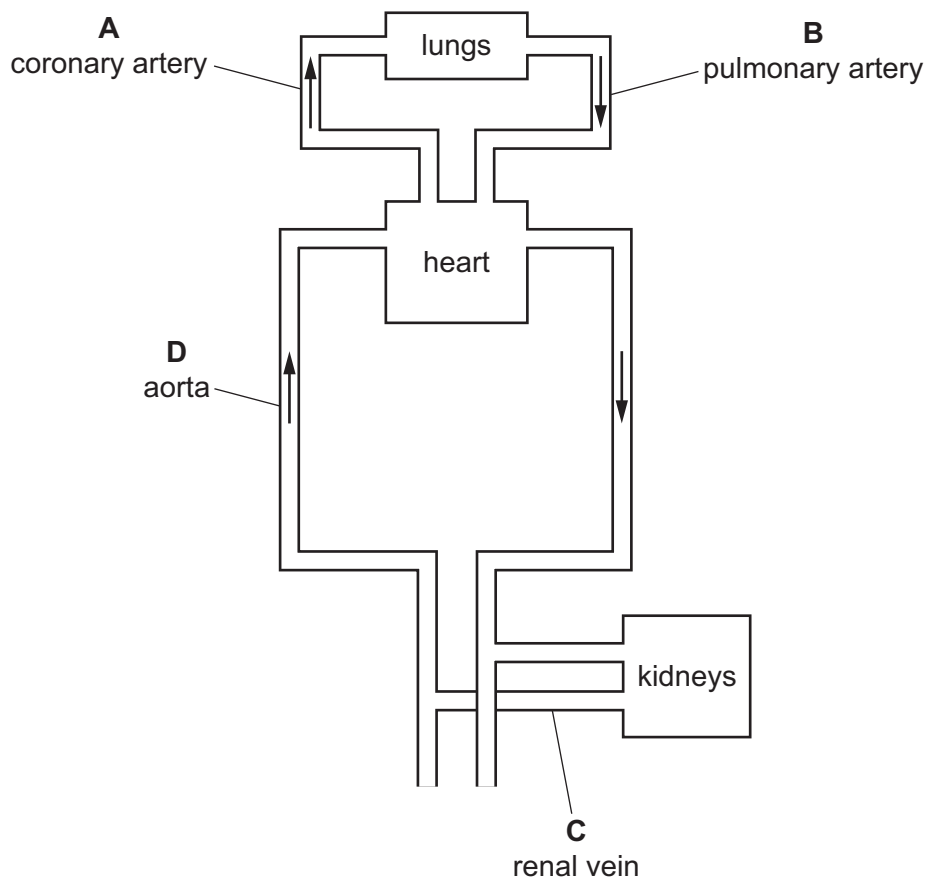
- 18** The roots of a plant are cut off and the stem is placed in a beaker containing a liquid red stain.

After a few hours, in which part of the plant stem will the red stain be visible?

- A** cuticle
- B** phloem
- C** style
- D** xylem

- 19** The diagram shows the blood vessels between the lungs, heart and kidneys in a human. The arrows indicate the direction of blood flow.

Which blood vessel is labelled correctly?

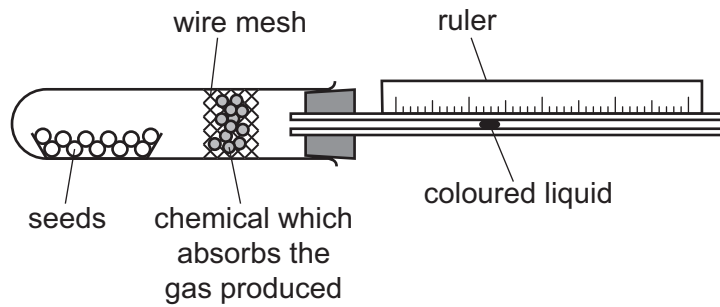


- 20** Which statements describe the difference between the composition of inspired air and expired air?

- 1 The percentage of carbon dioxide in inspired air is greater than in expired air.
- 2 The percentage of carbon dioxide in inspired air is less than in expired air.
- 3 The percentage of oxygen in inspired air is greater than in expired air.
- 4 The percentage of oxygen in inspired air is less than in expired air.

- A** 1 and 3
- B** 1 and 4
- C** 2 and 3
- D** 2 and 4

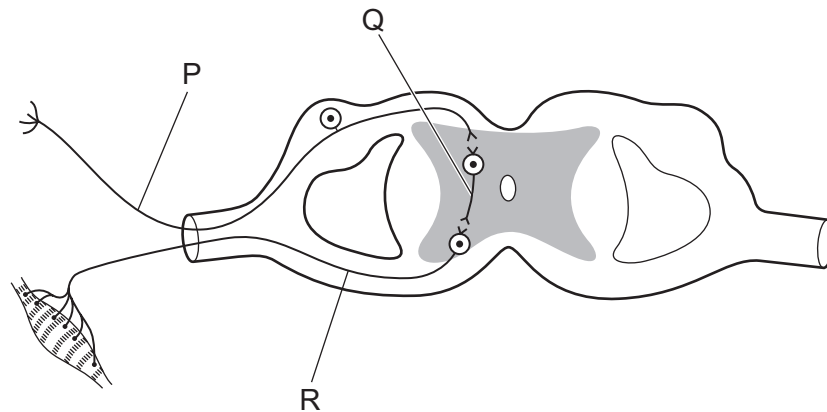
21 The diagram shows apparatus that was used to investigate germination in seeds.



Which other substances must be present in the test-tube for the seeds to germinate?

- A carbon dioxide and chlorophyll
 - B carbon dioxide and water
 - C chlorophyll and oxygen
 - D oxygen and water
- 22 What is a product of anaerobic respiration in yeast cells?
- A alcohol
 - B lactic acid
 - C oxygen
 - D water
- 23 Which organ excretes urea?
- A bladder
 - B kidney
 - C lung
 - D stomach
- 24 Which type of pathogen transmission is by direct contact?
- A contaminated air
 - B contaminated blood
 - C contaminated food
 - D contaminated surfaces

25 The diagram shows a reflex arc.



Which letters on the diagram represent the relay neurone and the sensory neurone?

	relay neurone	sensory neurone
A	P	Q
B	P	R
C	Q	R
D	Q	P

26 Which hormone is produced by the pancreas?

- A** adrenaline
- B** insulin
- C** oestrogen
- D** testosterone

27 Which statements describe correct examples of tropic responses?

- 1 Gravitropism is the growth of plant roots away from gravity.
- 2 Gravitropism is the growth of plant roots towards gravity.
- 3 Phototropism is the growth of plant shoots away from light.
- 4 Phototropism is the growth of plant shoots towards light.

- A** 1 and 3
- B** 1 and 4
- C** 2 and 3
- D** 2 and 4

28 Which statement applies to antibiotics?

- A** They kill bacteria.
- B** They kill viruses.
- C** They are produced by platelets.
- D** They are produced by white blood cells.

29 Four statements about different types of reproduction are listed.

- 1 Offspring are formed that are genetically different.
- 2 Offspring are formed that are genetically identical.
- 3 It only occurs in animals.
- 4 The nuclei of two gametes fuse in fertilisation.

Which statements describe sexual reproduction?

- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

30 What happens to the uterus lining at the start of the menstrual cycle?

- A** The uterus lining breaks down.
- B** The uterus lining grows new blood vessels.
- C** An embryo implants in the uterus lining.
- D** The uterus lining repairs and thickens.

31 What can reduce the transmission of HIV?

- A** clean water supplies
- B** hygienic food preparations
- C** use of condoms
- D** sewage treatment

32 What are alternative forms of genes called?

- A** alleles
- B** chromosomes
- C** genotypes
- D** phenotypes

- 33 Humans produce two types of ear wax, either wet or dry.

Which row describes the variation in human ear wax?

	type of variation	controlled by
A	continuous	genes and environment
B	continuous	genes only
C	discontinuous	genes and environment
D	discontinuous	genes only

- 34 Giraffes are mammals that have long necks. This enables them to eat leaves from tall trees. One theory to explain how giraffes developed long necks is natural selection.

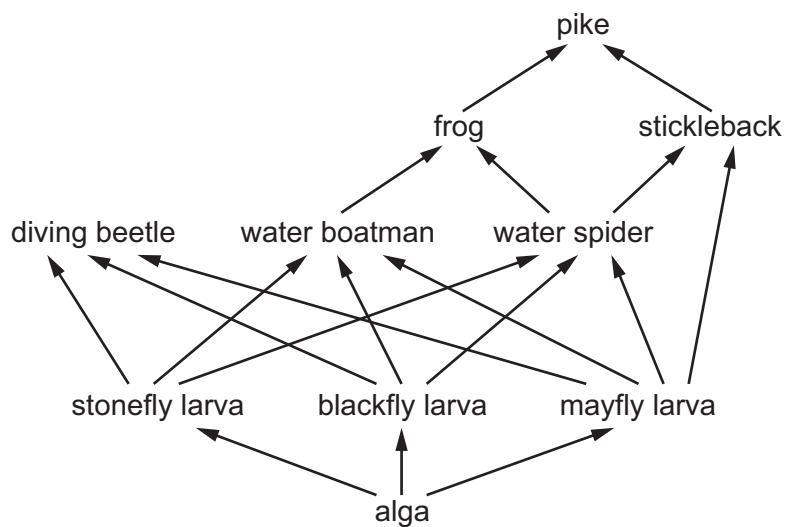
Which statement is part of the explanation for the development of long necks in giraffes by natural selection?

- A** Giraffes with longer necks were better adapted to the environment than giraffes with shorter necks.
 - B** Humans selected giraffes with the longest necks and bred them together.
 - C** There was **no** competition for food.
 - D** There were **no** trees in the environment.
- 35 Alpine plants grow on mountains at high altitude. Alpine plants can only grow for a few months because they are covered in snow for most of the year.

What are **most** likely to be adaptive features of alpine plants?

- A** They flower quickly and produce seeds quickly.
- B** They flower quickly and produce seeds slowly.
- C** They flower slowly and produce seeds quickly.
- D** They flower slowly and produce seeds slowly.

36 The diagram shows a food web.

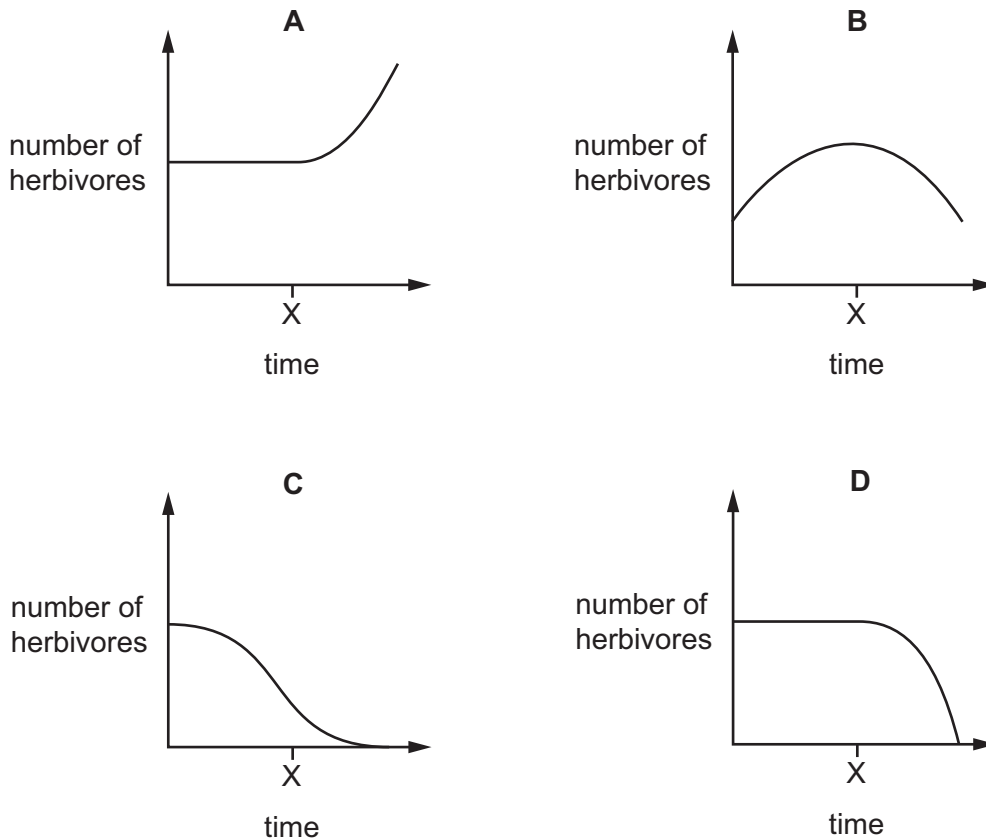


Which animal is a secondary consumer?

- A** blackfly larva
- B** diving beetle
- C** frog
- D** pike

- 37** The graphs show the number of herbivores in a habitat over a period of time. At time X, the carnivores are removed from the habitat.

Which graph shows the effect of this on the number of herbivores?



- 38** Some statements about food production are listed.

- 1 Agricultural machinery allows the efficient use of larger areas of land.
- 2 Herbicides increase the competition of crop plants with weeds.
- 3 Selective breeding improves production in livestock.

Which statements about how food production can be increased are correct?

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

- 39** What is one way endangered species can be conserved?

- A** growing monocultures
- B** overharvesting
- C** reducing biodiversity
- D** using seed banks

40 The table shows the features of four different types of bacteria.

Which type of bacteria would be the **most** useful to use for biotechnology?

	reproduction rate / number of cell divisions per hour	ability to make complex molecules
A	3.0	low
B	0.5	high
C	1.0	low
D	3.0	high

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Cambridge IGCSE™ (9–1)

BIOLOGY (9–1)**0970/11**

Paper 1 Multiple Choice (Core)

May/June 2025

MARK SCHEME

Maximum Mark: 40

Published

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Question	Answer	Marks
1	B	1
2	B	1
3	D	1
4	D	1
5	B	1
6	B	1
7	B	1
8	A	1
9	C	1
10	D	1
11	D	1
12	B	1
13	C	1
14	A	1
15	D	1
16	D	1
17	D	1
18	D	1
19	C	1
20	C	1
21	D	1
22	A	1
23	B	1
24	B	1
25	D	1
26	B	1
27	D	1
28	A	1

Question	Answer	Marks
29	B	1
30	A	1
31	C	1
32	A	1
33	D	1
34	A	1
35	A	1
36	B	1
37	A	1
38	C	1
39	D	1
40	D	1



Cambridge IGCSE™ (9–1)

BIOLOGY

0970/12

Paper 1 Multiple Choice (Core)

May/June 2025

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
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- You may use a calculator.

INFORMATION

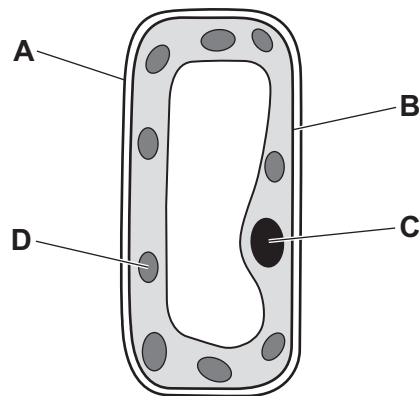
- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.



- 1 When a herbivore sees a carnivore, which characteristic of living things is involved?
- A excretion
 - B growth
 - C reproduction
 - D sensitivity
- 2 Which organism has been named using the binomial system?
- A *Brown seaweed*
 - B *Polar bear*
 - C *Red fox*
 - D *Vulpes vulpes*
- 3 Which list contains **only** arthropods?
- A amphibians, insects, myriapods
 - B arachnids, crustaceans, myriapods
 - C crustaceans, insects, fish
 - D insects, myriapods, reptiles
- 4 The diagram shows a plant cell.

Which letter identifies where photosynthesis occurs in the cell?



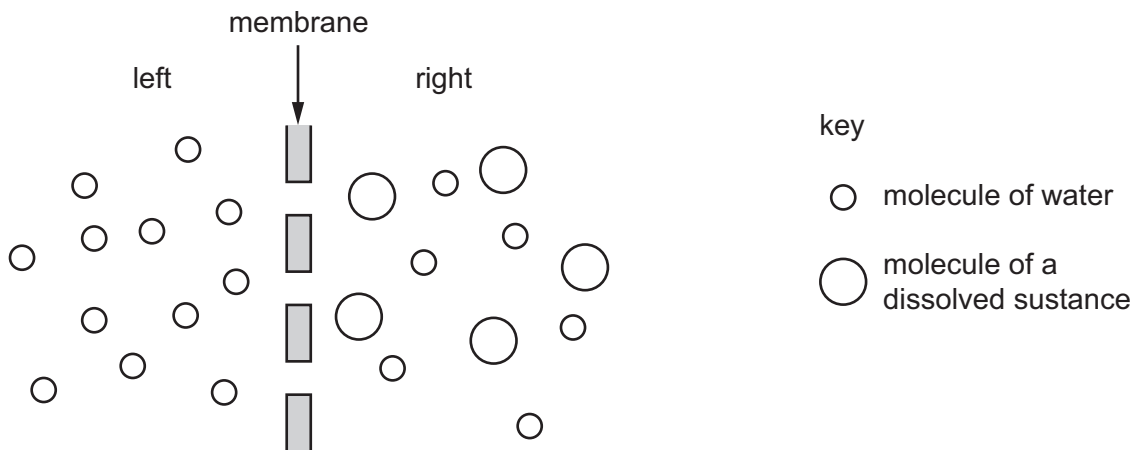
- 5 The table shows the magnifications and image lengths of three specimens when seen with a light microscope.

specimen	magnification	image length / mm
X	$\times 50$	0.20
Y	$\times 150$	0.50
Z	$\times 250$	0.40

What is the order of increasing actual lengths of the specimens, from smallest to largest?

- A** $X \rightarrow Y \rightarrow Z$ **B** $X \rightarrow Z \rightarrow Y$ **C** $Y \rightarrow X \rightarrow Z$ **D** $Z \rightarrow Y \rightarrow X$

- 6 The diagram represents two liquids separated by a partially permeable membrane.



Which statement describes how the molecules will move by osmosis?

- A** Molecules of the dissolved substance move from left to right.
B Molecules of the dissolved substance move from right to left.
C Overall, water molecules move from left to right.
D Overall, water molecules move from right to left.

- 7 Small molecules are used to synthesise large food molecules.

Which statement is correct?

- A** Amino acids are the basic units of carbohydrates.
B Fatty acids are the basic units of glycogen.
C Glycerol is a basic unit of oils.
D Simple sugar is the basic unit of protein.

- 8 A student used biuret reagent to test milk for the presence of protein.

The milk tested positive for protein.

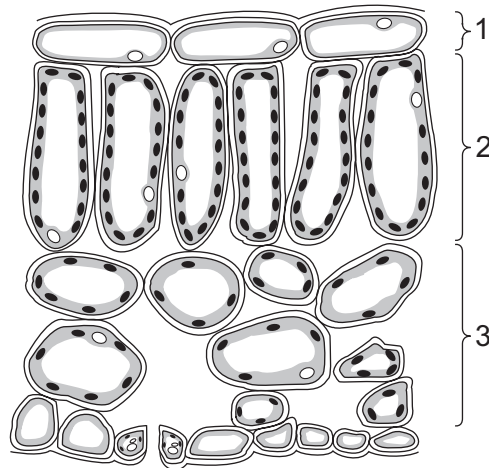
Which colour did the sample change to?

- A blue
 - B blue-black
 - C brick red
 - D purple
- 9 Which substances are used for photosynthesis?
- A carbon dioxide and glucose
 - B carbon dioxide and water
 - C glucose and oxygen
 - D glucose and water

- 10 What happens when a plant lacks magnesium ions?

	chlorophyll production	colour of leaves produced
A	decreases	green
B	decreases	yellow
C	increases	green
D	increases	yellow

11 The diagram shows part of a cross-section of a leaf.



What are tissues 1, 2 and 3?

	1	2	3
A	epidermis	palisade mesophyll	spongy mesophyll
B	epidermis	spongy mesophyll	palisade mesophyll
C	palisade mesophyll	epidermis	spongy mesophyll
D	spongy mesophyll	palisade mesophyll	epidermis

12 A diet contains a lot of plant material, but **no** meat.

Which dietary component is **unlikely** to be deficient in this diet?

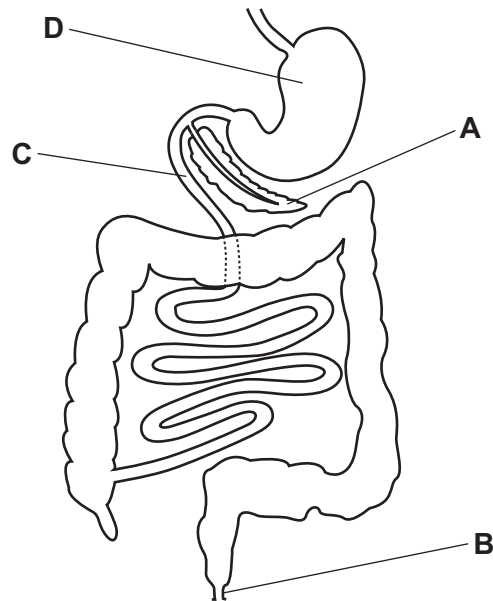
- A** fat
- B** fibre
- C** protein
- D** vitamin D

13 Which two parts of the mouth are teeth embedded in?

- A** cement and pulp
- B** enamel and dentine
- C** gums and bone
- D** pulp and bone

14 The diagram shows part of the human digestive system.

Where does amylase act on food?

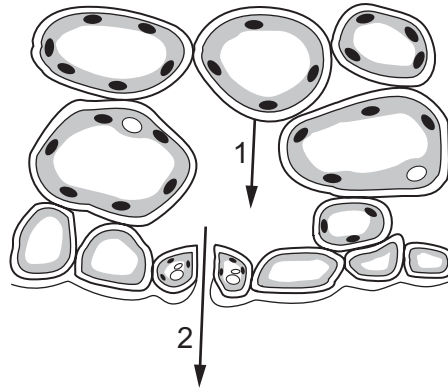


15 What is absorbed in the colon and the small intestine?

	colon	small intestine
A	digested food only	digested food and water
B	digested food only	water only
C	water	digested food only
D	water	digested food and water

16 The diagram shows the movement of water molecules from a plant leaf to its environment.

Two processes have been numbered.

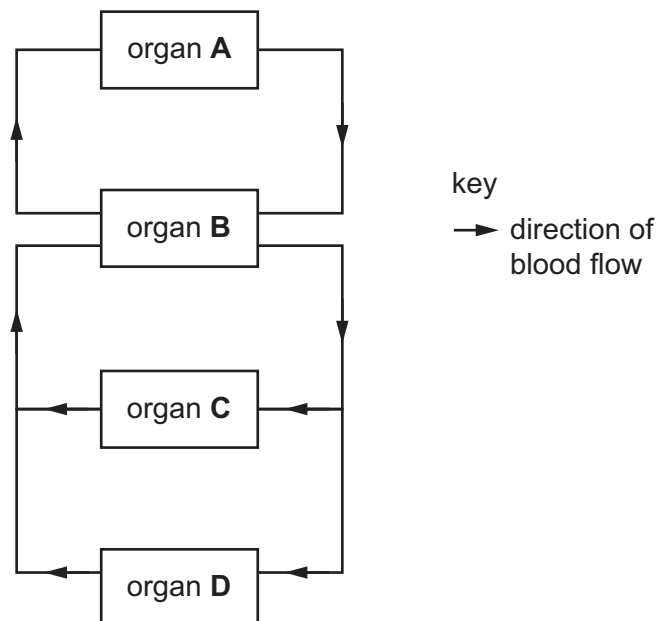


Which row identifies the numbered processes?

	process 1	process 2
A	diffusion	evaporation
B	evaporation	diffusion
C	evaporation	osmosis
D	osmosis	evaporation

17 The diagram shows a simplified circulatory system in humans.

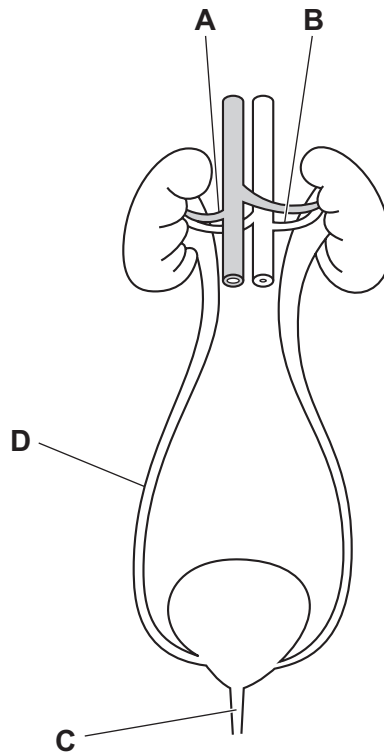
Which organ represents the pump in this circulatory system?



- 18 Which blood vessel transports blood from the lungs to the heart?
- A aorta
 - B pulmonary artery
 - C pulmonary vein
 - D vena cava
- 19 How can the spread of disease be prevented?
- A leaving food uncovered
 - B only washing hands after meals
 - C using the same equipment to prepare raw and cooked food
 - D washing hands before and after handling raw food
- 20 Which substance can be used to test for carbon dioxide?
- A DCPIP
 - B ethanol
 - C iodine solution
 - D limewater
- 21 Which statement is a correct comparison of aerobic respiration and anaerobic respiration in **yeast**?
- A Only aerobic respiration produces carbon dioxide.
 - B Only anaerobic respiration produces alcohol.
 - C They both produce water.
 - D They both use the reactants glucose and oxygen.

22 The diagram shows some of the organs involved in excretion.

Which label is a ureter?

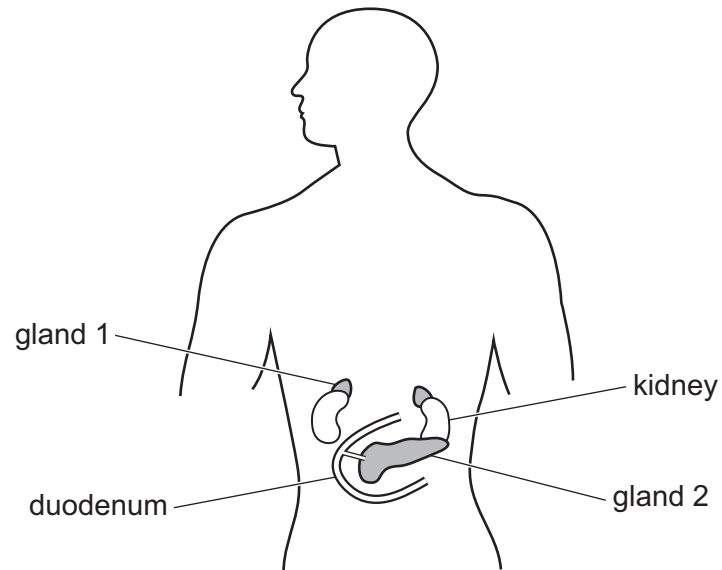


23 A person touches a hot object which stimulates structure P in the skin. This causes an electrical impulse to be sent through structure Q to the central nervous system.

Which row identifies structures P and Q?

	P	Q
A	effector	motor neurone
B	effector	sensory neurone
C	receptor	motor neurone
D	receptor	sensory neurone

24 The diagram shows the position of two glands in the body.

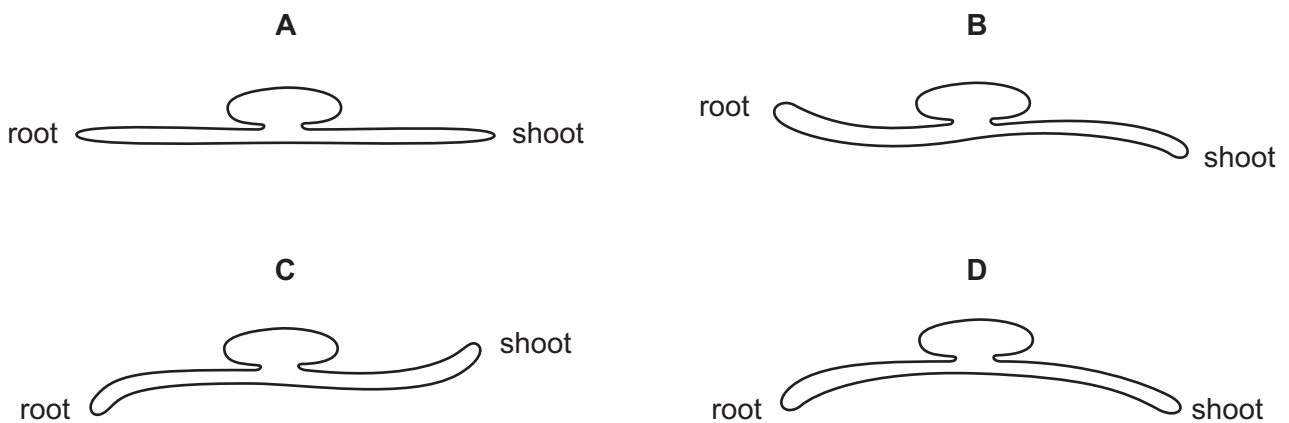


Which row states the hormones produced by these glands?

	gland 1	gland 2
A	adrenaline	insulin
B	amylase	adrenaline
C	insulin	lipase
D	lipase	amylase

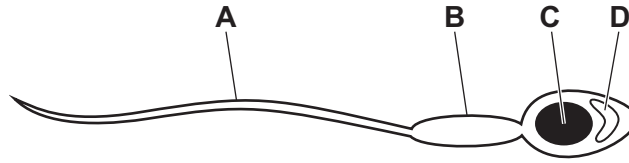
25 A seedling was placed in a horizontal position.

Which diagram shows the expected gravitropic responses in the seedling?



26 The diagram shows a sperm cell.

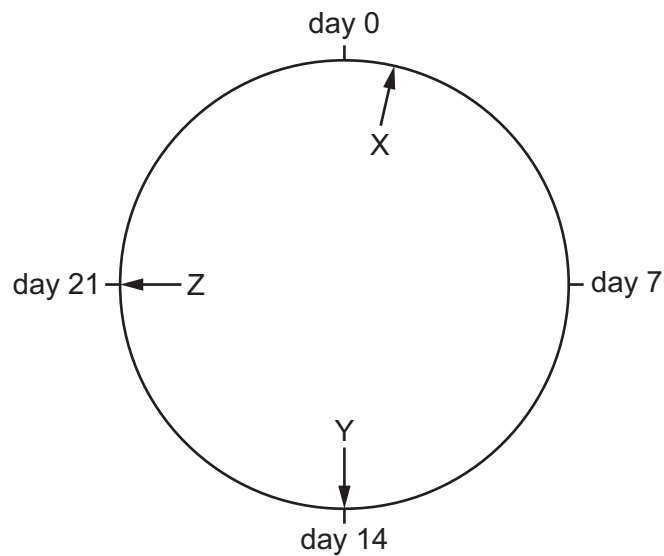
Which part contains digestive enzymes?



27 Which secondary sexual characteristic is associated with testosterone?

- A breast development
- B facial hair growth
- C start of the menstrual cycle
- D widening of hips

28 The diagram shows a human menstrual cycle.



What are represented by X, Y and Z?

	the lining of the uterus is breaking down	an ovary releases an egg	a thick uterus lining is maintained
A	X	Y	Z
B	X	Z	Y
C	Z	Y	X
D	Z	X	Y

29 What is an alternative form of a gene called?

- A** allele
- B** chromosome
- C** genotype
- D** protein

30 Cystic fibrosis is an inherited condition that occurs when an individual is homozygous for the recessive allele.

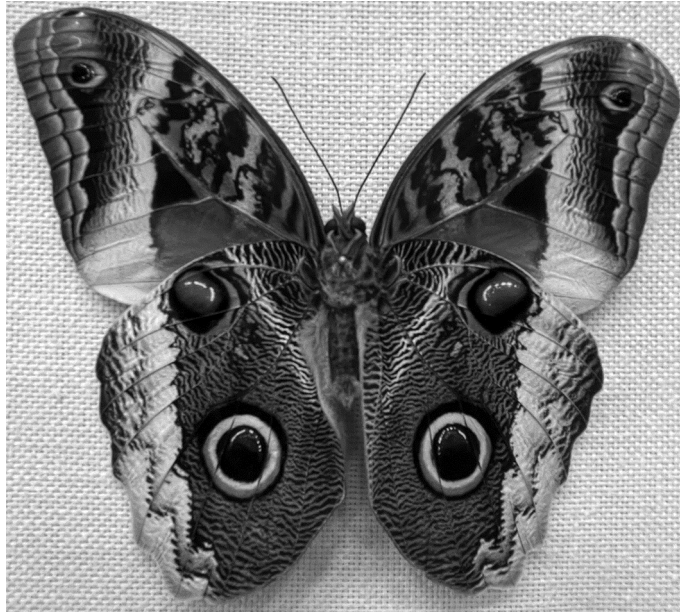
If both parents are heterozygous for this characteristic, what is the probability that their first child will have cystic fibrosis?

- A** 0% **B** 25% **C** 50% **D** 100%

31 Which characteristic is an example of continuous variation?

- A** ABO blood groups in humans
- B** body mass in cats
- C** seed colour in pea plants
- D** seed shape in pea plants

32 The photograph shows the butterfly *Caligo martia*.



Which feature is an adaptation to prevent predation?

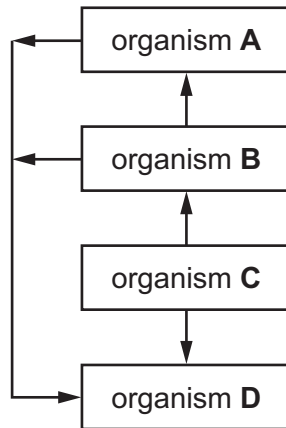
- A large wings to fly slowly
- B long antennae to detect flower nectar
- C spots on the wings that look like large eyes
- D the pattern on the wings that makes the butterfly easier to see

33 What is the principal source of energy input to biological systems?

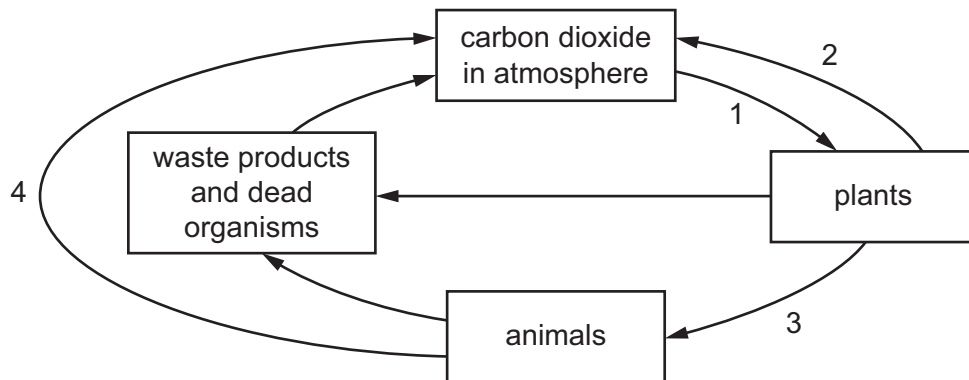
- A kinetic
- B photosynthesis
- C respiration
- D the Sun

34 The diagram shows the flow of energy in a food chain.

Which organism is the producer in the food chain?



35 The diagram shows part of the carbon cycle.

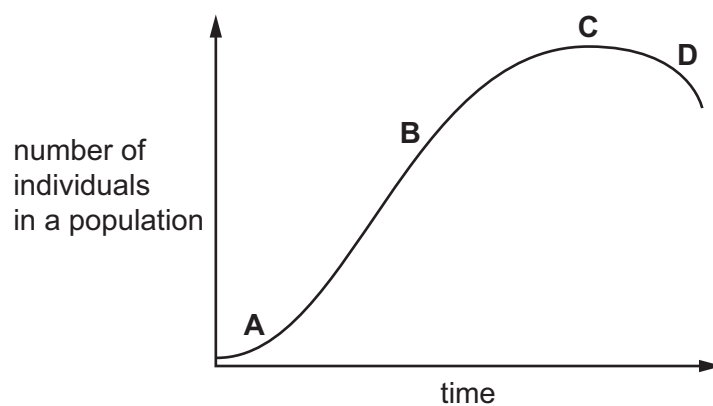


Which two numbered arrows represent respiration?

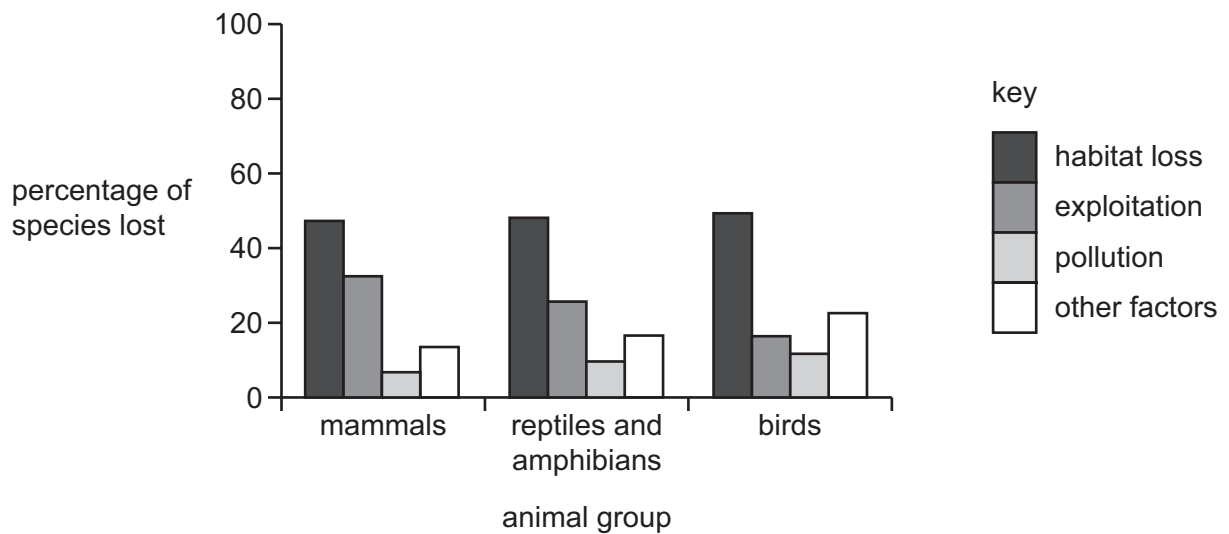
- A** 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

36 The diagram shows a population curve.

Which letter identifies the lag phase in this sigmoid curve?



- 37** The graph shows the percentage of species lost, in three different animal groups, due to different human activities during the last ten years.



Which activity has caused the highest percentage of species loss overall?

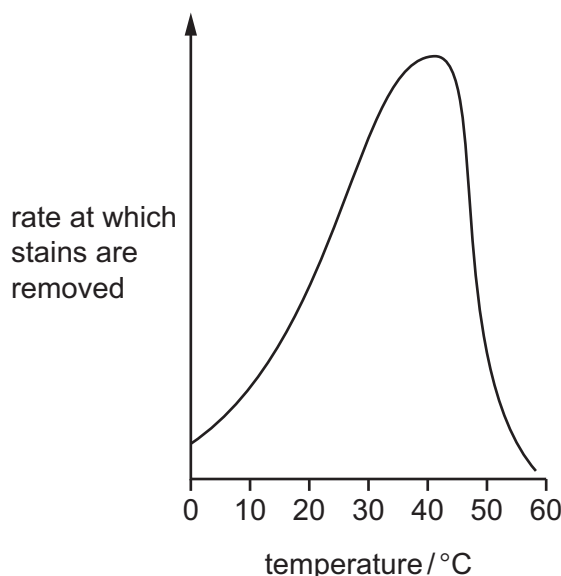
- A** habitat loss
- B** exploitation
- C** pollution
- D** other factors

- 38** Why are bacteria useful in biotechnology and genetic modification?

- 1 They can make complex molecules.
- 2 They have a rapid reproduction rate.
- 3 They have cell walls.

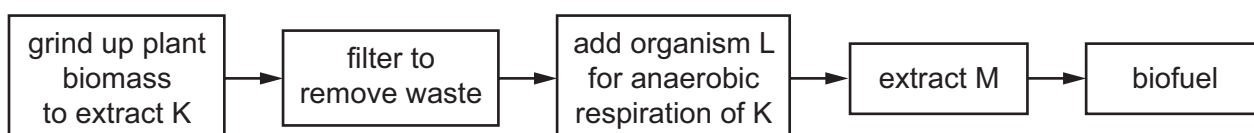
- A** 1, 2 and 3
- B** 1 and 2 only
- C** 1 and 3 only
- D** 2 and 3 only

- 39 The graph shows how changing the temperature affects the rate of stain removal from clothes, using a biological washing powder.



What does the graph suggest about biological washing powders?

- A** Biological washing powders contain enzymes.
- B** Biological washing powders work best at all temperatures above 40 °C.
- C** The optimum temperature for the reaction is between 10 °C and 30 °C.
- D** There is a directly proportional relationship between the temperature used and the rate at which stains are removed.
- 40 The flow chart shows some of the processes involved in making biofuel.



What are represented by K, L and M in the diagram?

	K	L	M
A	ethanol	bacteria	glucose
B	glucose	yeast	ethanol
C	glucose	bacteria	water
D	lactic acid	yeast	glucose

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Cambridge IGCSE™ (9–1)

BIOLOGY (9–1)**0970/12**

Paper 1 Multiple Choice (Core)

May/June 2025**MARK SCHEME**Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	D	1
2	D	1
3	B	1
4	D	1
5	D	1
6	C	1
7	C	1
8	D	1
9	B	1
10	B	1
11	A	1
12	B	1
13	C	1
14	C	1
15	D	1
16	B	1
17	B	1
18	C	1
19	D	1
20	D	1
21	B	1
22	D	1
23	D	1
24	A	1
25	C	1
26	D	1
27	B	1
28	A	1

Question	Answer	Marks
29	A	1
30	B	1
31	B	1
32	C	1
33	D	1
34	C	1
35	C	1
36	A	1
37	A	1
38	B	1
39	A	1
40	B	1



Cambridge IGCSE™ (9–1)

BIOLOGY

0970/12

Paper 1 Multiple Choice (Core)

October/November 2025

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.

1 Which process causes a permanent increase in the dry mass of an organism?

- A digestion
- B growth
- C reproduction
- D respiration

2 The scientific names of some plants are listed.

- 1 *Escallonia resinosa*
- 2 *Ilex glabra*
- 3 *Pinus glabra*
- 4 *Pinus resinosa*
- 5 *Quercus ilex*

Which plants belong to the same genus?

- A 1 and 4 B 2 and 3 C 2 and 5 D 3 and 4

3 An organism has a body with three parts, three pairs of jointed legs and one pair of antennae.

To which group does the organism belong?

- A arachnids
- B crustaceans
- C insects
- D myriapods

4 Which term describes a leaf?

- A cell
- B organ
- C organism
- D tissue

5 The length of an insect in a diagram is 150 mm.

The magnification of the diagram is $\times 20$.

What is the actual length of the insect?

- A 0.13 mm B 3.0 mm C 7.5 mm D 130 mm

6 What are features of osmosis?

	diffusion is involved	requires a cell wall	requires a partially permeable membrane
A	yes	no	yes
B	yes	no	no
C	no	yes	yes
D	no	yes	no

7 Enzymes are large molecules.

Which small molecules are they made from?

- A** amino acids
- B** carbon dioxide and water
- C** fatty acids
- D** sugars

8 Why does the active site of an enzyme fit with its substrate?

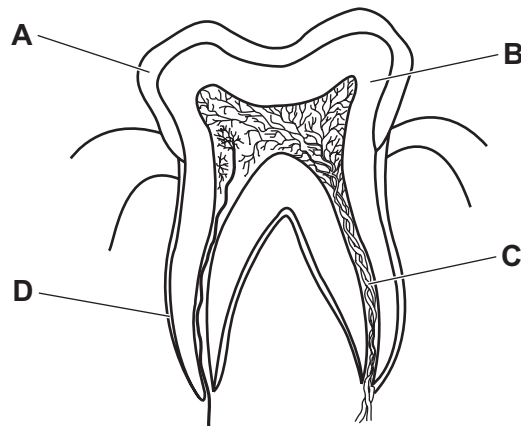
- A** They denature at the same temperature.
- B** They have complementary shapes to each other.
- C** They have the same amount of kinetic energy.
- D** They have the same shape as each other.

9 Which processes in plants transfer energy from light into energy in chemicals?

	photosynthesis	respiration
A	yes	yes
B	yes	no
C	no	yes
D	no	no

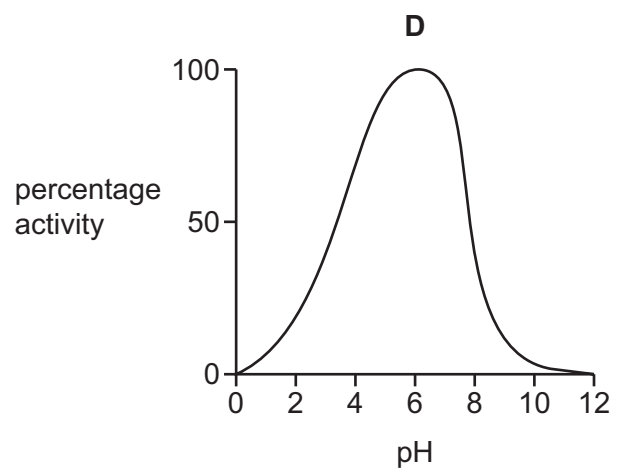
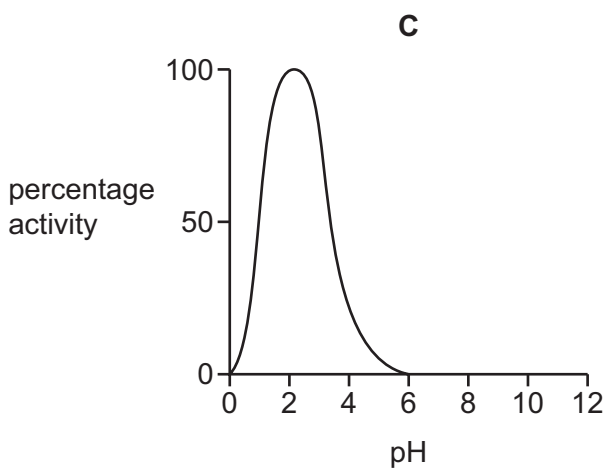
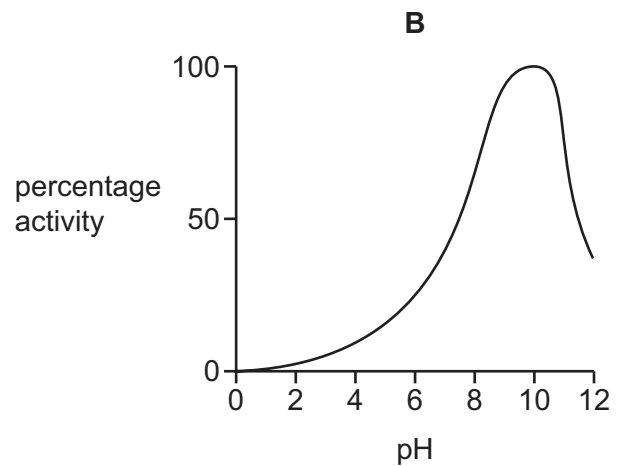
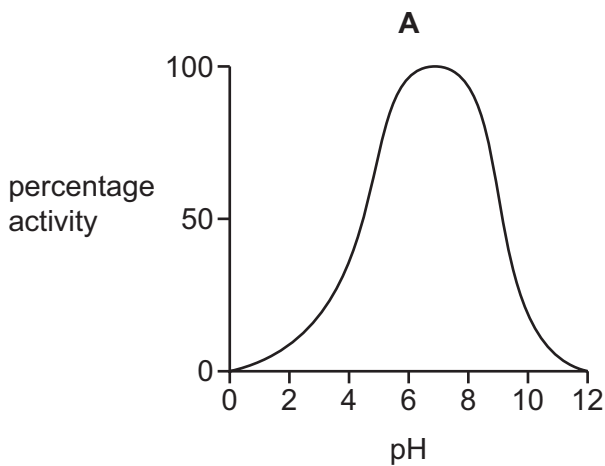
10 The diagram shows a cross-section of a molar tooth.

Which letter identifies the dentine?



11 The graphs show the effect of pH on the activity of four different digestive enzymes.

Which enzyme is most active in the stomach?



- 12 In which organ does physical digestion and chemical digestion occur?
- A gall bladder
 - B liver
 - C pancreas
 - D stomach
- 13 Which pathway is taken by water from the roots to the leaves?
- A root cortex cells → root hair cells → mesophyll cells → xylem
 - B root cortex cells → root hair cells → xylem → mesophyll cells
 - C root hair cells → root cortex cells → mesophyll cells → xylem
 - D root hair cells → root cortex cells → xylem → mesophyll cells
- 14 What is a function of red blood cells?
- A production of antibodies
 - B transport of glucose
 - C transport of hormones
 - D transport of oxygen
- 15 Which term can be used to describe all disease-causing organisms?
- A bacteria
 - B decomposers
 - C pathogens
 - D viruses
- 16 Which row shows how the composition of expired air differs from inspired air?

	carbon dioxide concentration	oxygen concentration
A	higher	higher
B	higher	lower
C	lower	higher
D	lower	lower

17 Which substance is produced in human muscles during anaerobic respiration?

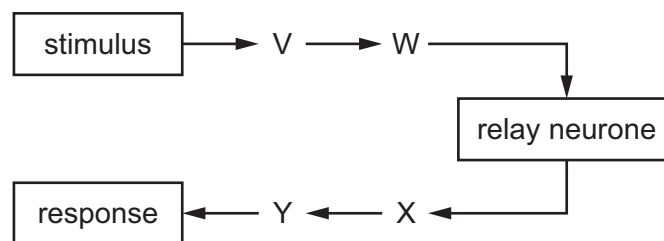
- A alcohol
- B carbon dioxide
- C lactic acid
- D urea

18 In humans, a ureter and a urethra are two structures involved in excretion.

Which comparison between a ureter and a urethra is correct?

- A Only fluid passing through a ureter contains excess mineral ions and water.
- B Only fluid passing through a urethra contains urea.
- C Only a ureter is directly attached to the bladder.
- D Only a ureter is directly attached to the kidney.

19 The diagram shows the pathway of a reflex arc.



Which row identifies the parts of the reflex arc?

	V	W	X	Y
A	effector	motor neurone	sensory neurone	receptor
B	effector	sensory neurone	motor neurone	receptor
C	receptor	motor neurone	sensory neurone	effector
D	receptor	sensory neurone	motor neurone	effector

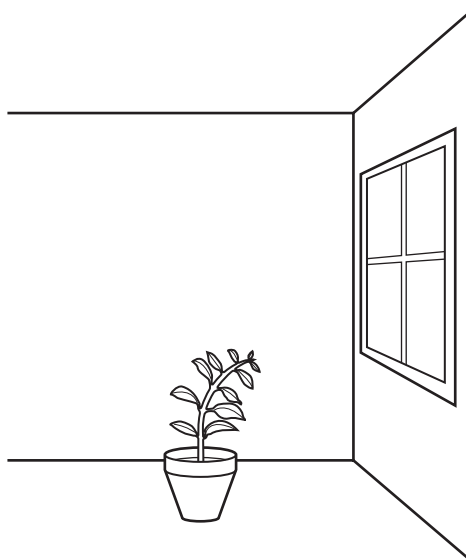
20 A person is being chased. This causes a hormone to be released.

Which row shows how this hormone affects the breathing rate, heart rate and pupil diameter?

	breathing rate	heart rate	pupil diameter
A	increases	increases	increases
B	increases	decreases	decreases
C	decreases	increases	increases
D	decreases	decreases	decreases

21 A plant is placed on the floor in a room, near a window that lets in a lot of light.

The shoot of the plant grows towards the window.



What could explain this response?

- 1 growth away from gravity
- 2 growth towards gravity
- 3 growth away from the light source
- 4 growth towards the light source

A 1 and 3

B 1 and 4

C 2 and 3

D 2 and 4

22 A person measures their reaction time five times before and after drinking a cup of coffee. The table shows the results.

	reaction time / s				
before drinking coffee	0.25	0.30	0.20	0.35	0.15
after drinking coffee	0.20	0.15	0.10	0.20	0.10

Which statement describes the results?

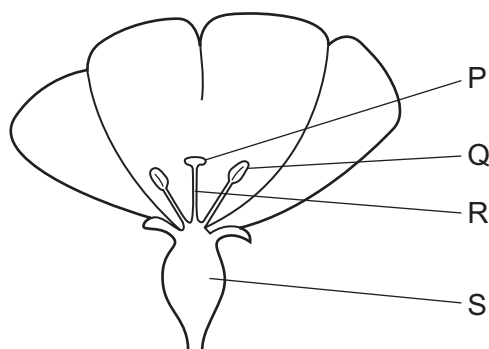
- A** The mean reaction time after drinking coffee is 0.25 s.
- B** The mean reaction time before drinking coffee is longer than after drinking coffee.
- C** The mean reaction time before drinking coffee is 0.15 s.
- D** The mean reaction time is **not** affected by drinking coffee.

23 Which set of conditions do most seeds require for germination?

	oxygen	warmth	water
A	✓	✗	✓
B	✓	✓	✗
C	✗	✓	✓
D	✓	✓	✓

key
 ✓ = required
 ✗ = **not** required

24 The diagram shows part of a flower.

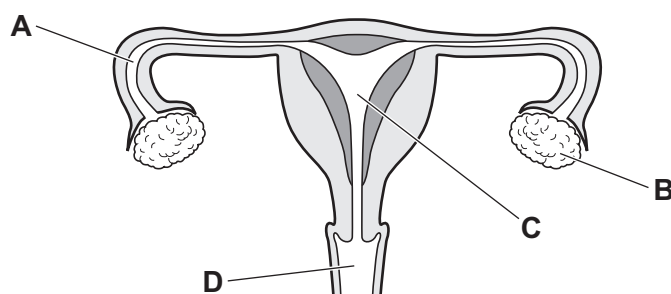


After pollination, where would pollen grains be found?

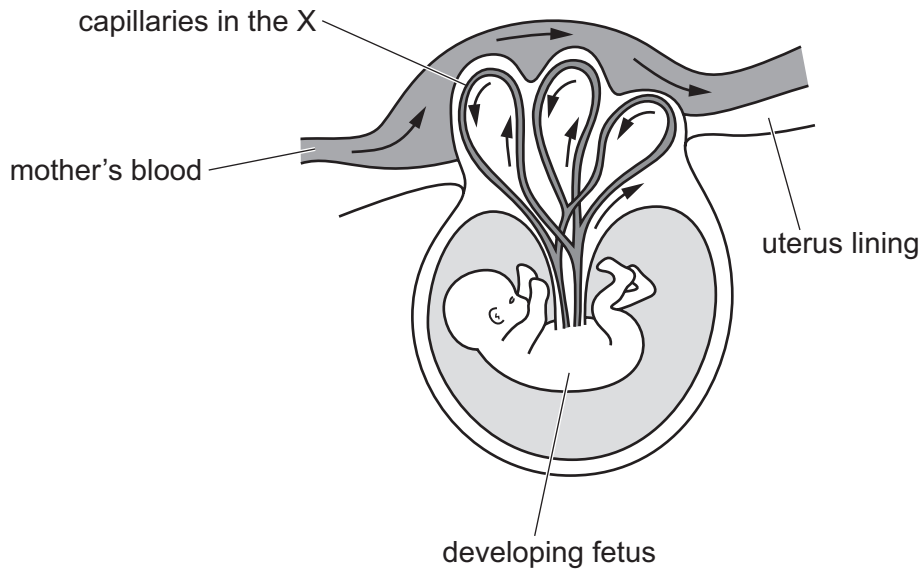
- A** P and Q **B** P and S **C** Q and R **D** R and S

25 The diagram shows the human female reproductive system.

Where does fertilisation normally take place?



26 The diagram shows a developing fetus.



Which structure is represented by X?

- A amniotic sac
- B cervix
- C placenta
- D umbilical cord

27 Which statements describe methods of transmission of HIV?

- 1 eating contaminated food
- 2 exchange of blood between individuals
- 3 from a mother to a fetus in pregnancy
- 4 insect bites
- 5 unprotected sex

- A 1, 2 and 5 B 1, 3 and 4 C 2, 3 and 5 D 5 only

28 What does a gene code for?

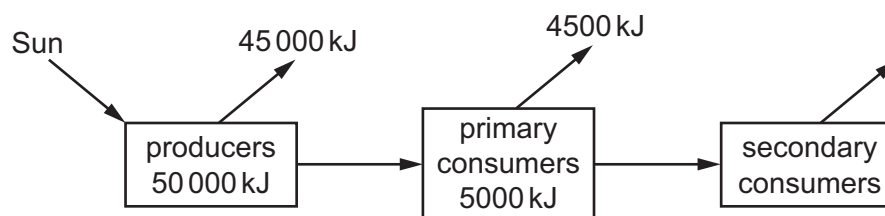
- A a carbohydrate
- B a fat
- C a protein
- D DNA

- 29** In a species of mammal, the allele for short hair, H, is dominant and the allele for long hair, h, is recessive.

A heterozygous individual breeds with a long-haired individual.

What is the expected ratio of the phenotypes in the offspring?

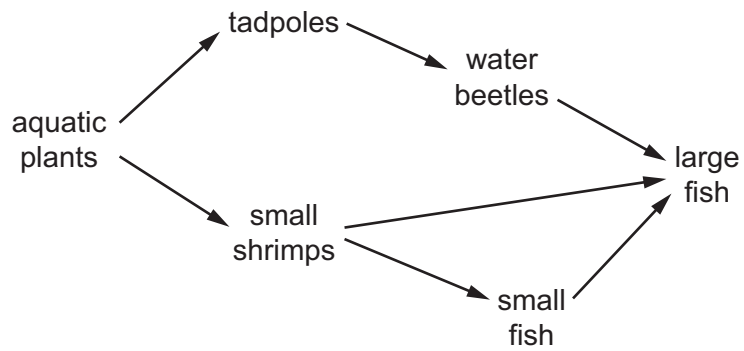
- A** 100% short-haired, 0% long-haired
 - B** 75% short-haired, 25% long-haired
 - C** 50% short-haired, 50% long-haired
 - D** 25% short-haired, 75% long-haired
- 30** Which statement about variation and example is correct?
- A** Continuous variation is caused by the environment and genes. An example is ABO blood groups.
 - B** Continuous variation is usually caused by genes only. An example is body mass.
 - C** Discontinuous variation is caused by both the environment and genes. An example is body mass.
 - D** Discontinuous variation is usually caused by genes only. An example is ABO blood groups.
- 31** The diagram shows the energy within the producers in an ecosystem and how much is transferred to primary consumers and eventually lost to the environment.



How much energy is transferred from primary consumers to secondary consumers?

- A** 5 kJ
- B** 50 kJ
- C** 500 kJ
- D** 5000 kJ

32 The diagram represents part of an aquatic food web.



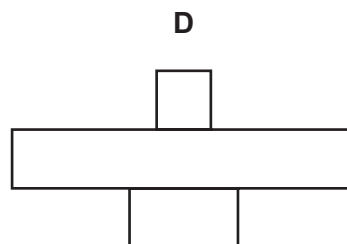
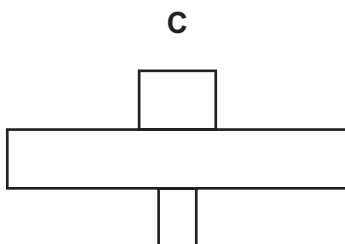
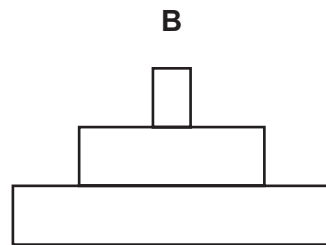
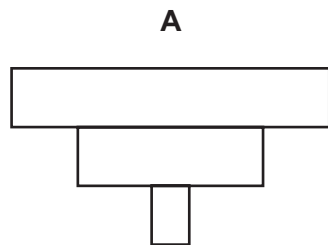
Which organisms are primary consumers in this food web?

- A small shrimps and tadpoles
- B tadpoles and small fish
- C water beetles and large fish
- D water beetles and small fish

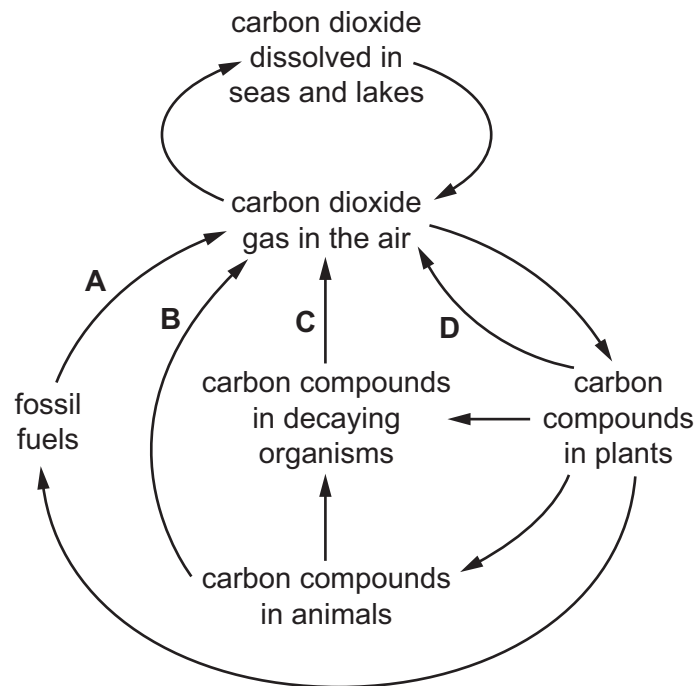
33 In a greenhouse there were many aphids feeding on 10 tomato plants.

A population of 20 ladybirds, which are a predator of aphids, was introduced into the greenhouse.

Which pyramid of numbers correctly represents this food chain?



34 Which arrow represents respiration in plants?

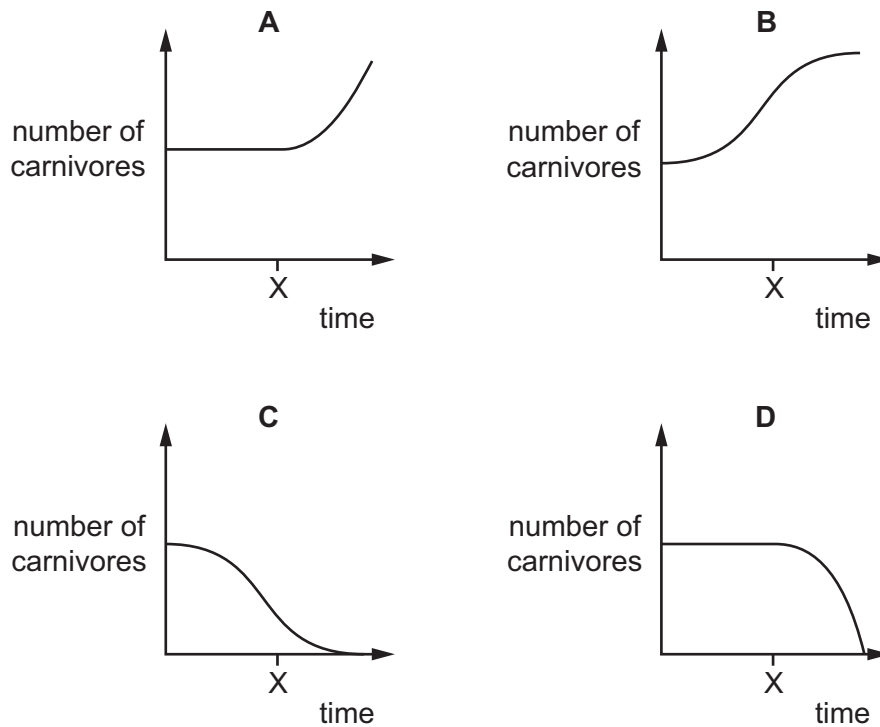


35 What is the correct sequence for the stages in a sigmoid population growth curve?

- A lag → log → stationary
- B log → lag → stationary
- C stationary → lag → log
- D stationary → log → lag

- 36** The graphs show the number of carnivores in a habitat over a period of time. At time X, the herbivores in the food chain are removed from the habitat.

Which graph shows the likely effect of this on the number of carnivores?



- 37** What is used to reduce competition from weeds when growing crops?

- A** deforestation
- B** fertiliser
- C** herbicide
- D** insecticide

- 38** Which statements describe the effects of deforestation?

- 1 Biodiversity increases.
- 2 The level of carbon dioxide in the atmosphere increases.
- 3 Soil is lost from the area.
- 4 There is a decrease in flooding.

- A** 1 and 2
- B** 1 and 4
- C** 2 and 3
- D** 3 and 4

39 Which substance is a greenhouse gas?

- A** herbicide
- B** insecticide
- C** methane
- D** oxygen

40 Biological washing powders contain enzymes.

Which enzyme removes an oil stain from clothes?

- A** amylase
- B** lipase
- C** pectinase
- D** protease

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Cambridge IGCSE™ (9–1)

BIOLOGY**0970/12**

Paper 1 Multiple Choice (Core)

October/November 2025**MARK SCHEME**Maximum Mark: 40

Published

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This document consists of **3** printed pages.

Question	Answer	Marks
1	B	1
2	D	1
3	C	1
4	B	1
5	C	1
6	A	1
7	A	1
8	B	1
9	B	1
10	B	1
11	C	1
12	D	1
13	D	1
14	D	1
15	C	1
16	B	1
17	C	1
18	D	1
19	D	1
20	A	1
21	B	1
22	B	1
23	D	1
24	A	1
25	A	1
26	C	1
27	C	1
28	C	1

Question	Answer	Marks
29	C	1
30	D	1
31	C	1
32	A	1
33	C	1
34	D	1
35	A	1
36	D	1
37	C	1
38	C	1
39	C	1
40	B	1